

Project

```
import pandas as pd
import numpy as np
```

```
df_app=pd.read_csv(r"Application_data - application_data.csv")
df_app
```

	SK_ID_CURR	TARGET	NAME_CONTRACT_TYPE	CODE_GENDER	FLAG_OWN_CAR
\					
0	100002	1	Cash loans	M	N
1	100003	0	Cash loans	F	N
2	100004	0	Revolving loans	M	Y
3	100006	0	Cash loans	F	N
4	100007	0	Cash loans	M	N
...	...	...	...	...	...
49994	157871	0	Cash loans	F	N
49995	157872	0	Cash loans	M	N
49996	157873	0	Cash loans	M	N
49997	157874	0	Cash loans	F	N
49998	157875	0	Cash loans	F	N

	FLAG_OWN_REALTY	CNT_CHILDREN	AMT_INCOME_TOTAL	AMT_CREDIT	\
0	Y	0	202500.0	406597.5	
1	N	0	270000.0	1293502.5	
2	Y	0	67500.0	135000.0	
3	Y	0	135000.0	312682.5	
4	Y	0	121500.0	513000.0	
...	...	...	...	...	
49994	N	0	180000.0	1206000.0	
49995	N	0	126000.0	1125000.0	
49996	N	1	112500.0	900000.0	
49997	Y	0	270000.0	820638.0	
49998	Y	0	117000.0	254700.0	

	AMT_ANNUITY	...	FLAG_DOCUMENT_18	FLAG_DOCUMENT_19
FLAG_DOCUMENT_20	\			
0	24700.5	...	0	0

0			
1	35698.5	...	0
0			
2	6750.0	...	0
0			
3	29686.5	...	0
0			
4	21865.5	...	0
0			
...	...	...	...
...			
49994	45936.0	...	0
0			
49995	47794.5	...	0
0			
49996	26316.0	...	0
0			
49997	34897.5	...	0
0			
49998	14751.0	...	0
0			

FLAG_DOCUMENT_21	AMT_REQ_CREDIT_BUREAU_HOUR
AMT_REQ_CREDIT_BUREAU_DAY \	

0	0	0.0
0.0		
1	0	0.0
0.0		
2	0	0.0
0.0		
3	0	NaN
NaN		
4	0	0.0
0.0		
...	...	...
...		
49994	0	0.0
0.0		
49995	0	0.0
0.0		
49996	0	0.0
0.0		
49997	0	0.0
0.0		
49998	0	0.0
0.0		

AMT_REQ_CREDIT_BUREAU_WEEK	AMT_REQ_CREDIT_BUREAU_MON \
0	0.0
	0.0

1	0.0	0.0
2	0.0	0.0
3	NaN	NaN
4	0.0	0.0
...	...	...
49994	0.0	0.0
49995	0.0	0.0
49996	0.0	0.0
49997	0.0	0.0
49998	0.0	0.0

	AMT_REQ_CREDIT_BUREAU_QRT	AMT_REQ_CREDIT_BUREAU_YEAR
0	0.0	1.0
1	0.0	0.0
2	0.0	0.0
3	NaN	NaN
4	0.0	0.0
...	...	...
49994	0.0	0.0
49995	0.0	0.0
49996	0.0	2.0
49997	2.0	4.0
49998	0.0	0.0

[49999 rows x 122 columns]

df_app.shape

(49999, 122)

df_app.size

6099878

df_app.index

RangeIndex(start=0, stop=49999, step=1)

df_app.columns

Index(['SK_ID_CURR', 'TARGET', 'NAME_CONTRACT_TYPE', 'CODE_GENDER',
'FLAG_OWN_CAR', 'FLAG_OWN_REALTY', 'CNT_CHILDREN',
'AMT_INCOME_TOTAL',
'AMT_CREDIT', 'AMT_ANNUITY',
...
'FLAG_DOCUMENT_18', 'FLAG_DOCUMENT_19', 'FLAG_DOCUMENT_20',
'FLAG_DOCUMENT_21', 'AMT_REQ_CREDIT_BUREAU_HOUR',
'AMT_REQ_CREDIT_BUREAU_DAY', 'AMT_REQ_CREDIT_BUREAU_WEEK',
'AMT_REQ_CREDIT_BUREAU_MON', 'AMT_REQ_CREDIT_BUREAU_QRT',
'AMT_REQ_CREDIT_BUREAU_YEAR'],
dtype='object', length=122)

```
df_app.columns.to_list()
```

```
['SK_ID_CURR',  
 'TARGET',  
 'NAME_CONTRACT_TYPE',  
 'CODE_GENDER',  
 'FLAG_OWN_CAR',  
 'FLAG_OWN_REALTY',  
 'CNT_CHILDREN',  
 'AMT_INCOME_TOTAL',  
 'AMT_CREDIT',  
 'AMT_ANNUITY',  
 'AMT_GOODS_PRICE',  
 'NAME_TYPE_SUITE',  
 'NAME_INCOME_TYPE',  
 'NAME_EDUCATION_TYPE',  
 'NAME_FAMILY_STATUS',  
 'NAME_HOUSING_TYPE',  
 'REGION_POPULATION_RELATIVE',  
 'DAYS_BIRTH',  
 'DAYS_EMPLOYED',  
 'DAYS_REGISTRATION',  
 'DAYS_ID_PUBLISH',  
 'OWN_CAR_AGE',  
 'FLAG_MOBIL',  
 'FLAG_EMP_PHONE',  
 'FLAG_WORK_PHONE',  
 'FLAG_CONT_MOBILE',  
 'FLAG_PHONE',  
 'FLAG_EMAIL',  
 'OCCUPATION_TYPE',  
 'CNT_FAM_MEMBERS',  
 'REGION_RATING_CLIENT',  
 'REGION_RATING_CLIENT_W_CITY',  
 'WEEKDAY_APPR_PROCESS_START',  
 'HOUR_APPR_PROCESS_START',  
 'REG_REGION_NOT_LIVE_REGION',  
 'REG_REGION_NOT_WORK_REGION',  
 'LIVE_REGION_NOT_WORK_REGION',  
 'REG_CITY_NOT_LIVE_CITY',  
 'REG_CITY_NOT_WORK_CITY',  
 'LIVE_CITY_NOT_WORK_CITY',  
 'ORGANIZATION_TYPE',  
 'EXT_SOURCE_1',  
 'EXT_SOURCE_2',  
 'EXT_SOURCE_3',  
 'APARTMENTS_AVG',  
 'BASEMENTAREA_AVG',  
 'YEARS_BEGINEXPLUATATION_AVG',  
 'YEARS_BUILD_AVG',
```

'COMMONAREA_AVG',
'ELEVATORS_AVG',
'ENTRANCES_AVG',
'FLOORSMAX_AVG',
'FLOORSMIN_AVG',
'LANDAREA_AVG',
'LIVINGAPARTMENTS_AVG',
'LIVINGAREA_AVG',
'NONLIVINGAPARTMENTS_AVG',
'NONLIVINGAREA_AVG',
'APARTMENTS_MODE',
'BASEMENTAREA_MODE',
'YEARS_BEGINEXPLUATATION_MODE',
'YEARS_BUILD_MODE',
'COMMONAREA_MODE',
'ELEVATORS_MODE',
'ENTRANCES_MODE',
'FLOORSMAX_MODE',
'FLOORSMIN_MODE',
'LANDAREA_MODE',
'LIVINGAPARTMENTS_MODE',
'LIVINGAREA_MODE',
'NONLIVINGAPARTMENTS_MODE',
'NONLIVINGAREA_MODE',
'APARTMENTS_MEDI',
'BASEMENTAREA_MEDI',
'YEARS_BEGINEXPLUATATION_MEDI',
'YEARS_BUILD_MEDI',
'COMMONAREA_MEDI',
'ELEVATORS_MEDI',
'ENTRANCES_MEDI',
'FLOORSMAX_MEDI',
'FLOORSMIN_MEDI',
'LANDAREA_MEDI',
'LIVINGAPARTMENTS_MEDI',
'LIVINGAREA_MEDI',
'NONLIVINGAPARTMENTS_MEDI',
'NONLIVINGAREA_MEDI',
'FONDKAPREMONT_MODE',
'HOUSETYPE_MODE',
'TOTALAREA_MODE',
'WALLSMATERIAL_MODE',
'EMERGENCYSTATE_MODE',
'OBS_30_CNT_SOCIAL_CIRCLE',
'DEF_30_CNT_SOCIAL_CIRCLE',
'OBS_60_CNT_SOCIAL_CIRCLE',
'DEF_60_CNT_SOCIAL_CIRCLE',
'DAYS_LAST_PHONE_CHANGE',
'FLAG_DOCUMENT_2',

```
'FLAG_DOCUMENT_3',
'FLAG_DOCUMENT_4',
'FLAG_DOCUMENT_5',
'FLAG_DOCUMENT_6',
'FLAG_DOCUMENT_7',
'FLAG_DOCUMENT_8',
'FLAG_DOCUMENT_9',
'FLAG_DOCUMENT_10',
'FLAG_DOCUMENT_11',
'FLAG_DOCUMENT_12',
'FLAG_DOCUMENT_13',
'FLAG_DOCUMENT_14',
'FLAG_DOCUMENT_15',
'FLAG_DOCUMENT_16',
'FLAG_DOCUMENT_17',
'FLAG_DOCUMENT_18',
'FLAG_DOCUMENT_19',
'FLAG_DOCUMENT_20',
'FLAG_DOCUMENT_21',
'AMT_REQ_CREDIT_BUREAU_HOUR',
'AMT_REQ_CREDIT_BUREAU_DAY',
'AMT_REQ_CREDIT_BUREAU_WEEK',
'AMT_REQ_CREDIT_BUREAU_MON',
'AMT_REQ_CREDIT_BUREAU_QRT',
'AMT_REQ_CREDIT_BUREAU_YEAR']
```

```
df_app.head()
```

	SK_ID_CURR	TARGET	NAME_CONTRACT_TYPE	CODE_GENDER	FLAG_OWN_CAR	\
0	100002	1	Cash loans	M	N	
1	100003	0	Cash loans	F	N	
2	100004	0	Revolving loans	M	Y	
3	100006	0	Cash loans	F	N	
4	100007	0	Cash loans	M	N	

	FLAG_OWN_REALTY	CNT_CHILDREN	AMT_INCOME_TOTAL	AMT_CREDIT	AMT_ANNUITY	\
0	Y	0	202500.0	406597.5	24700.5	
1	N	0	270000.0	1293502.5	35698.5	
2	Y	0	67500.0	135000.0	6750.0	
3	Y	0	135000.0	312682.5	29686.5	
4	Y	0	121500.0	513000.0	21865.5	

```
... FLAG_DOCUMENT_18 FLAG_DOCUMENT_19 FLAG_DOCUMENT_20
FLAG_DOCUMENT_21 \
```

0	...	0	0	0
0				
1	...	0	0	0
0				
2	...	0	0	0
0				
3	...	0	0	0
0				
4	...	0	0	0
0				

	AMT_REQ_CREDIT_BUREAU_HOUR	AMT_REQ_CREDIT_BUREAU_DAY	\
0	0.0	0.0	
1	0.0	0.0	
2	0.0	0.0	
3	NaN	NaN	
4	0.0	0.0	

	AMT_REQ_CREDIT_BUREAU_WEEK	AMT_REQ_CREDIT_BUREAU_MON	\
0	0.0	0.0	
1	0.0	0.0	
2	0.0	0.0	
3	NaN	NaN	
4	0.0	0.0	

	AMT_REQ_CREDIT_BUREAU_QRT	AMT_REQ_CREDIT_BUREAU_YEAR
0	0.0	1.0
1	0.0	0.0
2	0.0	0.0
3	NaN	NaN
4	0.0	0.0

[5 rows x 122 columns]

df_app.tail()

	SK_ID_CURR	TARGET	NAME_CONTRACT_TYPE	CODE_GENDER	FLAG_OWN_CAR
\					
49994	157871	0	Cash loans	F	N
49995	157872	0	Cash loans	M	N
49996	157873	0	Cash loans	M	N
49997	157874	0	Cash loans	F	N
49998	157875	0	Cash loans	F	N

	FLAG_OWN_REALTY	CNT_CHILDREN	AMT_INCOME_TOTAL	AMT_CREDIT	\
49994	N	0	180000.0	1206000.0	

49995	N	0	126000.0	1125000.0
49996	N	1	112500.0	900000.0
49997	Y	0	270000.0	820638.0
49998	Y	0	117000.0	254700.0

	AMT_ANNUITY	...	FLAG_DOCUMENT_18	FLAG_DOCUMENT_19
FLAG_DOCUMENT_20	\			
49994	45936.0	...	0	0
0				
49995	47794.5	...	0	0
0				
49996	26316.0	...	0	0
0				
49997	34897.5	...	0	0
0				
49998	14751.0	...	0	0
0				

	FLAG_DOCUMENT_21	AMT_REQ_CREDIT_BUREAU_HOUR
AMT_REQ_CREDIT_BUREAU_DAY	\	
49994	0	0.0
0.0		
49995	0	0.0
0.0		
49996	0	0.0
0.0		
49997	0	0.0
0.0		
49998	0	0.0
0.0		

	AMT_REQ_CREDIT_BUREAU_WEEK	AMT_REQ_CREDIT_BUREAU_MON	\
49994	0.0	0.0	
49995	0.0	0.0	
49996	0.0	0.0	
49997	0.0	0.0	
49998	0.0	0.0	

	AMT_REQ_CREDIT_BUREAU_QRT	AMT_REQ_CREDIT_BUREAU_YEAR
49994	0.0	0.0
49995	0.0	0.0
49996	0.0	2.0
49997	2.0	4.0
49998	0.0	0.0

[5 rows x 122 columns]

df_app.info()


```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 49999 entries, 0 to 49998
Columns: 122 entries, SK_ID_CURR to AMT_REQ_CREDIT_BUREAU_YEAR
dtypes: float64(64), int64(42), object(16)
memory usage: 46.5+ MB
```

```
df_app.info(verbose=True)
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 49999 entries, 0 to 49998
Data columns (total 122 columns):
#      Column                                         Dtype
---  -
0      SK_ID_CURR                                     int64
1      TARGET                                         int64
2      NAME_CONTRACT_TYPE                           object
3      CODE_GENDER                                   object
4      FLAG_OWN_CAR                                   object
5      FLAG_OWN_REALTY                               object
6      CNT_CHILDREN                                   int64
7      AMT_INCOME_TOTAL                             float64
8      AMT_CREDIT                                     float64
9      AMT_ANNUITY                                    float64
10     AMT_GOODS_PRICE                               float64
11     NAME_TYPE_SUITE                               object
12     NAME_INCOME_TYPE                             object
13     NAME_EDUCATION_TYPE                           object
14     NAME_FAMILY_STATUS                             object
15     NAME_HOUSING_TYPE                             object
16     REGION_POPULATION_RELATIVE                   float64
17     DAYS_BIRTH                                     int64
18     DAYS_EMPLOYED                                 int64
19     DAYS_REGISTRATION                             int64
20     DAYS_ID_PUBLISH                               int64
21     OWN_CAR_AGE                                   float64
22     FLAG_MOBIL                                     int64
23     FLAG_EMP_PHONE                               int64
24     FLAG_WORK_PHONE                              int64
25     FLAG_CONT_MOBILE                             int64
26     FLAG_PHONE                                    int64
27     FLAG_EMAIL                                    int64
28     OCCUPATION_TYPE                               object
29     CNT_FAM_MEMBERS                               float64
30     REGION_RATING_CLIENT                         int64
31     REGION_RATING_CLIENT_W_CITY                 int64
32     WEEKDAY_APPR_PROCESS_START                  object
33     HOUR_APPR_PROCESS_START                     int64
34     REG_REGION_NOT_LIVE_REGION                  int64
35     REG_REGION_NOT_WORK_REGION                  int64
36     LIVE_REGION_NOT_WORK_REGION                 int64
```

37	REG_CITY_NOT_LIVE_CITY	int64
38	REG_CITY_NOT_WORK_CITY	int64
39	LIVE_CITY_NOT_WORK_CITY	int64
40	ORGANIZATION_TYPE	object
41	EXT_SOURCE_1	float64
42	EXT_SOURCE_2	float64
43	EXT_SOURCE_3	float64
44	APARTMENTS_AVG	float64
45	BASEMENTAREA_AVG	float64
46	YEARS_BEGINEXPLUATATION_AVG	float64
47	YEARS_BUILD_AVG	float64
48	COMMONAREA_AVG	float64
49	ELEVATORS_AVG	float64
50	ENTRANCES_AVG	float64
51	FLOORSMAX_AVG	float64
52	FLOORSMIN_AVG	float64
53	LANDAREA_AVG	float64
54	LIVINGAPARTMENTS_AVG	float64
55	LIVINGAREA_AVG	float64
56	NONLIVINGAPARTMENTS_AVG	float64
57	NONLIVINGAREA_AVG	float64
58	APARTMENTS_MODE	float64
59	BASEMENTAREA_MODE	float64
60	YEARS_BEGINEXPLUATATION_MODE	float64
61	YEARS_BUILD_MODE	float64
62	COMMONAREA_MODE	float64
63	ELEVATORS_MODE	float64
64	ENTRANCES_MODE	float64
65	FLOORSMAX_MODE	float64
66	FLOORSMIN_MODE	float64
67	LANDAREA_MODE	float64
68	LIVINGAPARTMENTS_MODE	float64
69	LIVINGAREA_MODE	float64
70	NONLIVINGAPARTMENTS_MODE	float64
71	NONLIVINGAREA_MODE	float64
72	APARTMENTS_MEDI	float64
73	BASEMENTAREA_MEDI	float64
74	YEARS_BEGINEXPLUATATION_MEDI	float64
75	YEARS_BUILD_MEDI	float64
76	COMMONAREA_MEDI	float64
77	ELEVATORS_MEDI	float64
78	ENTRANCES_MEDI	float64
79	FLOORSMAX_MEDI	float64
80	FLOORSMIN_MEDI	float64
81	LANDAREA_MEDI	float64
82	LIVINGAPARTMENTS_MEDI	float64
83	LIVINGAREA_MEDI	float64
84	NONLIVINGAPARTMENTS_MEDI	float64
85	NONLIVINGAREA_MEDI	float64

```

86  FONDKAPREMONT_MODE      object
87  HOUSETYPE_MODE         object
88  TOTALAREA_MODE         float64
89  WALLSMATERIAL_MODE     object
90  EMERGENCYSTATE_MODE    object
91  OBS_30_CNT_SOCIAL_CIRCLE float64
92  DEF_30_CNT_SOCIAL_CIRCLE float64
93  OBS_60_CNT_SOCIAL_CIRCLE float64
94  DEF_60_CNT_SOCIAL_CIRCLE float64
95  DAYS_LAST_PHONE_CHANGE float64
96  FLAG_DOCUMENT_2        int64
97  FLAG_DOCUMENT_3        int64
98  FLAG_DOCUMENT_4        int64
99  FLAG_DOCUMENT_5        int64
100 FLAG_DOCUMENT_6        int64
101 FLAG_DOCUMENT_7        int64
102 FLAG_DOCUMENT_8        int64
103 FLAG_DOCUMENT_9        int64
104 FLAG_DOCUMENT_10       int64
105 FLAG_DOCUMENT_11       int64
106 FLAG_DOCUMENT_12       int64
107 FLAG_DOCUMENT_13       int64
108 FLAG_DOCUMENT_14       int64
109 FLAG_DOCUMENT_15       int64
110 FLAG_DOCUMENT_16       int64
111 FLAG_DOCUMENT_17       int64
112 FLAG_DOCUMENT_18       int64
113 FLAG_DOCUMENT_19       int64
114 FLAG_DOCUMENT_20       int64
115 FLAG_DOCUMENT_21       int64
116 AMT_REQ_CREDIT_BUREAU_HOUR float64
117 AMT_REQ_CREDIT_BUREAU_DAY float64
118 AMT_REQ_CREDIT_BUREAU_WEEK float64
119 AMT_REQ_CREDIT_BUREAU_MON float64
120 AMT_REQ_CREDIT_BUREAU_QRT float64
121 AMT_REQ_CREDIT_BUREAU_YEAR float64
dtypes: float64(64), int64(42), object(16)
memory usage: 46.5+ MB

df_app.info(verbose=True,show_counts=True)

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 49999 entries, 0 to 49998
Data columns (total 122 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   SK_ID_CURR                           49999 non-null  int64
1   TARGET                               49999 non-null  int64
2   NAME_CONTRACT_TYPE                   49999 non-null  object
3   CODE_GENDER                          49999 non-null  object

```

4	FLAG_OWN_CAR	49999	non-null	object
5	FLAG_OWN_REALTY	49999	non-null	object
6	CNT_CHILDREN	49999	non-null	int64
7	AMT_INCOME_TOTAL	49999	non-null	float64
8	AMT_CREDIT	49999	non-null	float64
9	AMT_ANNUITY	49998	non-null	float64
10	AMT_GOODS_PRICE	49961	non-null	float64
11	NAME_TYPE_SUITE	49807	non-null	object
12	NAME_INCOME_TYPE	49999	non-null	object
13	NAME_EDUCATION_TYPE	49999	non-null	object
14	NAME_FAMILY_STATUS	49999	non-null	object
15	NAME_HOUSING_TYPE	49999	non-null	object
16	REGION_POPULATION_RELATIVE	49999	non-null	float64
17	DAYS_BIRTH	49999	non-null	int64
18	DAYS_EMPLOYED	49999	non-null	int64
19	DAYS_REGISTRATION	49999	non-null	int64
20	DAYS_ID_PUBLISH	49999	non-null	int64
21	OWN_CAR_AGE	17049	non-null	float64
22	FLAG_MOBIL	49999	non-null	int64
23	FLAG_EMP_PHONE	49999	non-null	int64
24	FLAG_WORK_PHONE	49999	non-null	int64
25	FLAG_CONT_MOBILE	49999	non-null	int64
26	FLAG_PHONE	49999	non-null	int64
27	FLAG_EMAIL	49999	non-null	int64
28	OCCUPATION_TYPE	34345	non-null	object
29	CNT_FAM_MEMBERS	49998	non-null	float64
30	REGION_RATING_CLIENT	49999	non-null	int64
31	REGION_RATING_CLIENT_W_CITY	49999	non-null	int64
32	WEEKDAY_APPR_PROCESS_START	49999	non-null	object
33	HOUR_APPR_PROCESS_START	49999	non-null	int64
34	REG_REGION_NOT_LIVE_REGION	49999	non-null	int64
35	REG_REGION_NOT_WORK_REGION	49999	non-null	int64
36	LIVE_REGION_NOT_WORK_REGION	49999	non-null	int64
37	REG_CITY_NOT_LIVE_CITY	49999	non-null	int64
38	REG_CITY_NOT_WORK_CITY	49999	non-null	int64
39	LIVE_CITY_NOT_WORK_CITY	49999	non-null	int64
40	ORGANIZATION_TYPE	49999	non-null	object
41	EXT_SOURCE_1	21827	non-null	float64
42	EXT_SOURCE_2	49873	non-null	float64
43	EXT_SOURCE_3	40055	non-null	float64
44	APARTMENTS_AVG	24614	non-null	float64
45	BASEMENTAREA_AVG	20800	non-null	float64
46	YEARS_BEGINEXPLUATATION_AVG	25605	non-null	float64
47	YEARS_BUILD_AVG	16760	non-null	float64
48	COMMONAREA_AVG	15039	non-null	float64
49	ELEVATORS_AVG	23348	non-null	float64
50	ENTRANCES_AVG	24804	non-null	float64
51	FLOORSMAX_AVG	25124	non-null	float64
52	FLOORSMIN_AVG	16105	non-null	float64

53	LANDAREA_AVG	20278	non-null	float64
54	LIVINGAPARTMENTS_AVG	15773	non-null	float64
55	LIVINGAREA_AVG	24862	non-null	float64
56	NONLIVINGAPARTMENTS_AVG	15285	non-null	float64
57	NONLIVINGAREA_AVG	22427	non-null	float64
58	APARTMENTS_MODE	24614	non-null	float64
59	BASEMENTAREA_MODE	20800	non-null	float64
60	YEARS_BEGINEXPLUATATION_MODE	25605	non-null	float64
61	YEARS_BUILD_MODE	16760	non-null	float64
62	COMMONAREA_MODE	15039	non-null	float64
63	ELEVATORS_MODE	23348	non-null	float64
64	ENTRANCES_MODE	24804	non-null	float64
65	FLOORSMAX_MODE	25124	non-null	float64
66	FLOORSMIN_MODE	16105	non-null	float64
67	LANDAREA_MODE	20278	non-null	float64
68	LIVINGAPARTMENTS_MODE	15773	non-null	float64
69	LIVINGAREA_MODE	24862	non-null	float64
70	NONLIVINGAPARTMENTS_MODE	15285	non-null	float64
71	NONLIVINGAREA_MODE	22427	non-null	float64
72	APARTMENTS_MEDI	24614	non-null	float64
73	BASEMENTAREA_MEDI	20800	non-null	float64
74	YEARS_BEGINEXPLUATATION_MEDI	25605	non-null	float64
75	YEARS_BUILD_MEDI	16760	non-null	float64
76	COMMONAREA_MEDI	15039	non-null	float64
77	ELEVATORS_MEDI	23348	non-null	float64
78	ENTRANCES_MEDI	24804	non-null	float64
79	FLOORSMAX_MEDI	25124	non-null	float64
80	FLOORSMIN_MEDI	16105	non-null	float64
81	LANDAREA_MEDI	20278	non-null	float64
82	LIVINGAPARTMENTS_MEDI	15773	non-null	float64
83	LIVINGAREA_MEDI	24862	non-null	float64
84	NONLIVINGAPARTMENTS_MEDI	15285	non-null	float64
85	NONLIVINGAREA_MEDI	22427	non-null	float64
86	FONDKAPREMONT_MODE	15808	non-null	object
87	HOUSETYPE_MODE	24924	non-null	object
88	TOTALAREA_MODE	25851	non-null	float64
89	WALLSMATERIAL_MODE	24540	non-null	object
90	EMERGENCYSTATE_MODE	26301	non-null	object
91	OBS_30_CNT_SOCIAL_CIRCLE	49831	non-null	float64
92	DEF_30_CNT_SOCIAL_CIRCLE	49831	non-null	float64
93	OBS_60_CNT_SOCIAL_CIRCLE	49831	non-null	float64
94	DEF_60_CNT_SOCIAL_CIRCLE	49831	non-null	float64
95	DAYS_LAST_PHONE_CHANGE	49998	non-null	float64
96	FLAG_DOCUMENT_2	49999	non-null	int64
97	FLAG_DOCUMENT_3	49999	non-null	int64
98	FLAG_DOCUMENT_4	49999	non-null	int64
99	FLAG_DOCUMENT_5	49999	non-null	int64
100	FLAG_DOCUMENT_6	49999	non-null	int64
101	FLAG_DOCUMENT_7	49999	non-null	int64

```

102 FLAG_DOCUMENT_8          49999 non-null int64
103 FLAG_DOCUMENT_9          49999 non-null int64
104 FLAG_DOCUMENT_10         49999 non-null int64
105 FLAG_DOCUMENT_11         49999 non-null int64
106 FLAG_DOCUMENT_12         49999 non-null int64
107 FLAG_DOCUMENT_13         49999 non-null int64
108 FLAG_DOCUMENT_14         49999 non-null int64
109 FLAG_DOCUMENT_15         49999 non-null int64
110 FLAG_DOCUMENT_16         49999 non-null int64
111 FLAG_DOCUMENT_17         49999 non-null int64
112 FLAG_DOCUMENT_18         49999 non-null int64
113 FLAG_DOCUMENT_19         49999 non-null int64
114 FLAG_DOCUMENT_20         49999 non-null int64
115 FLAG_DOCUMENT_21         49999 non-null int64
116 AMT_REQ_CREDIT_BUREAU_HOUR 43265 non-null float64
117 AMT_REQ_CREDIT_BUREAU_DAY  43265 non-null float64
118 AMT_REQ_CREDIT_BUREAU_WEEK 43265 non-null float64
119 AMT_REQ_CREDIT_BUREAU_MON  43265 non-null float64
120 AMT_REQ_CREDIT_BUREAU_QRT  43265 non-null float64
121 AMT_REQ_CREDIT_BUREAU_YEAR 43265 non-null float64

```

dtypes: float64(64), int64(42), object(16)

memory usage: 46.5+ MB

df_app.describe()

	SK_ID_CURR	TARGET	CNT_CHILDREN	AMT_INCOME_TOTAL \
count	49999.000000	49999.000000	49999.000000	4.999900e+04
mean	129013.210584	0.080522	0.419848	1.707676e+05
std	16690.512048	0.272102	0.724039	5.318191e+05
min	100002.000000	0.000000	0.000000	2.565000e+04
25%	114570.500000	0.000000	0.000000	1.125000e+05
50%	129076.000000	0.000000	0.000000	1.458000e+05
75%	143438.500000	0.000000	1.000000	2.025000e+05
max	157875.000000	1.000000	11.000000	1.170000e+08

	AMT_CREDIT	AMT_ANNUITY	AMT_GOODS_PRICE \
count	4.999900e+04	49998.000000	4.996100e+04
mean	5.997006e+05	27107.377355	5.390600e+05
std	4.024154e+05	14562.944435	3.698533e+05
min	4.500000e+04	2052.000000	4.500000e+04
25%	2.700000e+05	16456.500000	2.385000e+05
50%	5.147775e+05	24939.000000	4.500000e+05
75%	8.086500e+05	34596.000000	6.795000e+05
max	4.050000e+06	258025.500000	4.050000e+06

	REGION_POPULATION_RELATIVE	DAYS_BIRTH	DAYS_EMPLOYED	...	\
count	49999.000000	49999.000000	49999.000000	...	
mean	0.020798	-16022.042081	63219.424488	...	
std	0.013761	4361.400270	140794.605668	...	
min	0.000533	-25184.000000	-17531.000000	...	

25%	0.010006	-19644.000000	-2786.000000	...
50%	0.018850	-15731.000000	-1221.000000	...
75%	0.028663	-12378.500000	-292.000000	...
max	0.072508	-7680.000000	365243.000000	...

	FLAG_DOCUMENT_18	FLAG_DOCUMENT_19	FLAG_DOCUMENT_20
FLAG_DOCUMENT_21 \			
count	49999.000000	49999.000000	49999.000000
49999.000000			
mean	0.008500	0.000700	0.000520
0.00038			
std	0.091805	0.026449	0.022798
0.01949			
min	0.000000	0.000000	0.000000
0.00000			
25%	0.000000	0.000000	0.000000
0.00000			
50%	0.000000	0.000000	0.000000
0.00000			
75%	0.000000	0.000000	0.000000
0.00000			
max	1.000000	1.000000	1.000000
1.00000			

	AMT_REQ_CREDIT_BUREAU_HOUR	AMT_REQ_CREDIT_BUREAU_DAY \
count	43265.000000	43265.000000
mean	0.007096	0.007512
std	0.087709	0.107992
min	0.000000	0.000000
25%	0.000000	0.000000
50%	0.000000	0.000000
75%	0.000000	0.000000
max	3.000000	6.000000

	AMT_REQ_CREDIT_BUREAU_WEEK	AMT_REQ_CREDIT_BUREAU_MON \
count	43265.000000	43265.000000
mean	0.032382	0.270288
std	0.194080	0.928560
min	0.000000	0.000000
25%	0.000000	0.000000
50%	0.000000	0.000000
75%	0.000000	0.000000
max	6.000000	24.000000

	AMT_REQ_CREDIT_BUREAU_QRT	AMT_REQ_CREDIT_BUREAU_YEAR
count	43265.000000	43265.000000
mean	0.260973	1.881035
std	0.606996	1.865054
min	0.000000	0.000000
25%	0.000000	0.000000

50%	0.000000	1.000000
75%	0.000000	3.000000
max	8.000000	25.000000

[8 rows x 106 columns]

df_app.describe().columns

```
Index(['SK_ID_CURR', 'TARGET', 'CNT_CHILDREN', 'AMT_INCOME_TOTAL',
      'AMT_CREDIT', 'AMT_ANNUITY', 'AMT_GOODS_PRICE',
      'REGION_POPULATION_RELATIVE', 'DAYS_BIRTH', 'DAYS_EMPLOYED',
      ...,
      'FLAG_DOCUMENT_18', 'FLAG_DOCUMENT_19', 'FLAG_DOCUMENT_20',
      'FLAG_DOCUMENT_21', 'AMT_REQ_CREDIT_BUREAU_HOUR',
      'AMT_REQ_CREDIT_BUREAU_DAY', 'AMT_REQ_CREDIT_BUREAU_WEEK',
      'AMT_REQ_CREDIT_BUREAU_MON', 'AMT_REQ_CREDIT_BUREAU_QRT',
      'AMT_REQ_CREDIT_BUREAU_YEAR'],
      dtype='object', length=106)
```

```
list(set(df_app.columns)-set(df_app.describe().columns))
```

```
['NAME_TYPE_SUITE',
 'NAME_FAMILY_STATUS',
 'NAME_EDUCATION_TYPE',
 'NAME_CONTRACT_TYPE',
 'WALLSMATERIAL_MODE',
 'WEEKDAY_APPR_PROCESS_START',
 'EMERGENCYSTATE_MODE',
 'FLAG_OWN_REALTY',
 'ORGANIZATION_TYPE',
 'OCCUPATION_TYPE',
 'NAME_HOUSING_TYPE',
 'FLAG_OWN_CAR',
 'HOUSETYPE_MODE',
 'NAME_INCOME_TYPE',
 'FONDKAPREMONT_MODE',
 'CODE_GENDER']
```

```
import pandas as pd
import numpy as np
```

```
df_app["CODE_GENDER"]
```

0	M
1	F
2	M
3	F
4	M
...	
49994	F
49995	M


```

49996      M
49997      F
49998      F
Name: CODE_GENDER, Length: 49999, dtype: object

```

```
df_app.CODE_GENDER
```

```

0      M
1      F
2      M
3      F
4      M
...
49994    F
49995    M
49996    M
49997    F
49998    F
Name: CODE_GENDER, Length: 49999, dtype: object

```

```
df_app[["CODE_GENDER", "AMT_ANNUITY"]]
```

	CODE_GENDER	AMT_ANNUITY
0	M	24700.5
1	F	35698.5
2	M	6750.0
3	F	29686.5
4	M	21865.5
...	...	...
49994	F	45936.0
49995	M	47794.5
49996	M	26316.0
49997	F	34897.5
49998	F	14751.0

```
[49999 rows x 2 columns]
```

```

#unique value count return karta hai (nunique)
# nunique() is a dataframe method that returns the unique value count
of each column
df_app.nunique()

```

SK_ID_CURR	49999
TARGET	2
NAME_CONTRACT_TYPE	2
CODE_GENDER	3
FLAG_OWN_CAR	2
...	...
AMT_REQ_CREDIT_BUREAU_DAY	7
AMT_REQ_CREDIT_BUREAU_WEEK	7
AMT_REQ_CREDIT_BUREAU_MON	20

```

AMT_REQ_CREDIT_BUREAU_QRT      9
AMT_REQ_CREDIT_BUREAU_YEAR    19
Length: 122, dtype: int64

```

```
df_app["CODE_GENDER"].unique() # is a return of values
```

```
array(['M', 'F', 'XNA'], dtype=object)
```

#isnull() is a method of dataframe and it returns the n ull value in boolean..

#True -- if null exists.

```
df_app.isnull()
```

	SK_ID_CURR	TARGET	NAME_CONTRACT_TYPE	CODE_GENDER
FLAG_OWN_CAR \				
0	False	False	False	False
False				
1	False	False	False	False
False				
2	False	False	False	False
False				
3	False	False	False	False
False				
4	False	False	False	False
False				
...	...	...	...	...
..				
49994	False	False	False	False
False				
49995	False	False	False	False
False				
49996	False	False	False	False
False				
49997	False	False	False	False
False				
49998	False	False	False	False
False				

	FLAG_OWN_REALTY	CNT_CHILDREN	AMT_INCOME_TOTAL	AMT_CREDIT \
0	False	False	False	False
1	False	False	False	False
2	False	False	False	False
3	False	False	False	False
4	False	False	False	False
...	...	...	...	...
49994	False	False	False	False
49995	False	False	False	False
49996	False	False	False	False
49997	False	False	False	False
49998	False	False	False	False

	AMT_ANNUITY	...	FLAG_DOCUMENT_18	FLAG_DOCUMENT_19
FLAG_DOCUMENT_20 \				
0	False	...	False	False
False				
1	False	...	False	False
False				
2	False	...	False	False
False				
3	False	...	False	False
False				
4	False	...	False	False
False				
...	...	...	...	...
...				
49994	False	...	False	False
False				
49995	False	...	False	False
False				
49996	False	...	False	False
False				
49997	False	...	False	False
False				
49998	False	...	False	False
False				

	FLAG_DOCUMENT_21	AMT_REQ_CREDIT_BUREAU_HOUR \
0	False	False
1	False	False
2	False	False
3	False	True
4	False	False
...	...	...
49994	False	False
49995	False	False
49996	False	False
49997	False	False
49998	False	False

	AMT_REQ_CREDIT_BUREAU_DAY	AMT_REQ_CREDIT_BUREAU_WEEK \
0	False	False
1	False	False
2	False	False
3	True	True
4	False	False
...	...	...
49994	False	False
49995	False	False
49996	False	False
49997	False	False

49998	False	False
	AMT_REQ_CREDIT_BUREAU_MON	AMT_REQ_CREDIT_BUREAU_QRT \
0	False	False
1	False	False
2	False	False
3	True	True
4	False	False
...	...	...
49994	False	False
49995	False	False
49996	False	False
49997	False	False
49998	False	False

	AMT_REQ_CREDIT_BUREAU_YEAR
0	False
1	False
2	False
3	True
4	False
...	...
49994	False
49995	False
49996	False
49997	False
49998	False

[49999 rows x 122 columns]

#isna() -- predefined function of pandas for null values.
pd.isna(df_app)

	SK_ID_CURR	TARGET	NAME_CONTRACT_TYPE	CODE_GENDER
FLAG_OWN_CAR \				
0	False	False	False	False
False				
1	False	False	False	False
False				
2	False	False	False	False
False				
3	False	False	False	False
False				
4	False	False	False	False
False				
...	...	...	...	...
..				
49994	False	False	False	False
False				
49995	False	False	False	False

False					
49996	False	False	False	False	False
False					
49997	False	False	False	False	False
False					
49998	False	False	False	False	False
False					
	FLAG_OWN_REALTY	CNT_CHILDREN	AMT_INCOME_TOTAL	AMT_CREDIT	\
0	False	False	False	False	
1	False	False	False	False	
2	False	False	False	False	
3	False	False	False	False	
4	False	False	False	False	
...	...	...	...	...	
49994	False	False	False	False	
49995	False	False	False	False	
49996	False	False	False	False	
49997	False	False	False	False	
49998	False	False	False	False	
	AMT_ANNUITY	...	FLAG_DOCUMENT_18	FLAG_DOCUMENT_19	
FLAG_DOCUMENT_20	\				
0	False	...	False	False	
False					
1	False	...	False	False	
False					
2	False	...	False	False	
False					
3	False	...	False	False	
False					
4	False	...	False	False	
False					
...	...	...	...	...	
...					
49994	False	...	False	False	
False					
49995	False	...	False	False	
False					
49996	False	...	False	False	
False					
49997	False	...	False	False	
False					
49998	False	...	False	False	
False					
	FLAG_DOCUMENT_21	AMT_REQ_CREDIT_BUREAU_HOUR	\		
0	False	False	False		
1	False	False	False		
2	False	False	False		

3	False	True
4	False	False
...	...	...
49994	False	False
49995	False	False
49996	False	False
49997	False	False
49998	False	False

	AMT_REQ_CREDIT_BUREAU_DAY	AMT_REQ_CREDIT_BUREAU_WEEK	\
0	False	False	
1	False	False	
2	False	False	
3	True	True	
4	False	False	
...	...	...	
49994	False	False	
49995	False	False	
49996	False	False	
49997	False	False	
49998	False	False	

	AMT_REQ_CREDIT_BUREAU_MON	AMT_REQ_CREDIT_BUREAU_QRT	\
0	False	False	
1	False	False	
2	False	False	
3	True	True	
4	False	False	
...	...	...	
49994	False	False	
49995	False	False	
49996	False	False	
49997	False	False	
49998	False	False	

	AMT_REQ_CREDIT_BUREAU_YEAR
0	False
1	False
2	False
3	True
4	False
...	...
49994	False
49995	False
49996	False
49997	False
49998	False

[49999 rows x 122 columns]

```
df_app.isnull().sum()
```

SK_ID_CURR	0
TARGET	0
NAME_CONTRACT_TYPE	0
CODE_GENDER	0
FLAG_OWN_CAR	0
...	
AMT_REQ_CREDIT_BUREAU_DAY	6734
AMT_REQ_CREDIT_BUREAU_WEEK	6734
AMT_REQ_CREDIT_BUREAU_MON	6734
AMT_REQ_CREDIT_BUREAU_QRT	6734
AMT_REQ_CREDIT_BUREAU_YEAR	6734

Length: 122, dtype: int64

```
df_app.isnull().mean()
```

SK_ID_CURR	0.000000
TARGET	0.000000
NAME_CONTRACT_TYPE	0.000000
CODE_GENDER	0.000000
FLAG_OWN_CAR	0.000000
...	
AMT_REQ_CREDIT_BUREAU_DAY	0.134683
AMT_REQ_CREDIT_BUREAU_WEEK	0.134683
AMT_REQ_CREDIT_BUREAU_MON	0.134683
AMT_REQ_CREDIT_BUREAU_QRT	0.134683
AMT_REQ_CREDIT_BUREAU_YEAR	0.134683

Length: 122, dtype: float64

```
df_app.isnull().mean()*100
```

SK_ID_CURR	0.000000
TARGET	0.000000
NAME_CONTRACT_TYPE	0.000000
CODE_GENDER	0.000000
FLAG_OWN_CAR	0.000000
...	
AMT_REQ_CREDIT_BUREAU_DAY	13.468269
AMT_REQ_CREDIT_BUREAU_WEEK	13.468269
AMT_REQ_CREDIT_BUREAU_MON	13.468269
AMT_REQ_CREDIT_BUREAU_QRT	13.468269
AMT_REQ_CREDIT_BUREAU_YEAR	13.468269

Length: 122, dtype: float64

```
round(df_app.isnull().mean()*100,2)
```

SK_ID_CURR	0.00
TARGET	0.00
NAME_CONTRACT_TYPE	0.00
CODE_GENDER	0.00

```

FLAG_OWN_CAR          0.00
...
AMT_REQ_CREDIT_BUREAU_DAY  13.47
AMT_REQ_CREDIT_BUREAU_WEEK  13.47
AMT_REQ_CREDIT_BUREAU_MON  13.47
AMT_REQ_CREDIT_BUREAU_QRT  13.47
AMT_REQ_CREDIT_BUREAU_YEAR  13.47
Length: 122, dtype: float64

```

#sort values is method used for sorting the records..

```

round(df_app.isnull().mean()*100,2).sort_values(ascending=False)

```

```

COMMONAREA_MEDI      69.92
COMMONAREA_AVG        69.92
COMMONAREA_MODE       69.92
NONLIVINGAPARTMENTS_MODE  69.43
NONLIVINGAPARTMENTS_AVG  69.43
...
NAME_FAMILY_STATUS    0.00
NAME_EDUCATION_TYPE   0.00
NAME_INCOME_TYPE      0.00
AMT_ANNUITY           0.00
SK_ID_CURR            0.00
Length: 122, dtype: float64

```

```

def missing_value(Dataframe):
    return

```

```

round(Dataframe.isnull().mean()*100,2).sort_values(ascending=False)

```

```

missing_value(df_app)

```

```

COMMONAREA_MEDI      69.92
COMMONAREA_AVG        69.92
COMMONAREA_MODE       69.92
NONLIVINGAPARTMENTS_MODE  69.43
NONLIVINGAPARTMENTS_AVG  69.43
...
NAME_FAMILY_STATUS    0.00
NAME_EDUCATION_TYPE   0.00
NAME_INCOME_TYPE      0.00
AMT_ANNUITY           0.00
SK_ID_CURR            0.00
Length: 122, dtype: float64

```

```

null=missing_value(df_app)[missing_value(df_app)>50]

```

```

null

```

```

COMMONAREA_MEDI      69.92
COMMONAREA_AVG        69.92
COMMONAREA_MODE       69.92

```


NONLIVINGAPARTMENTS_MODE	69.43
NONLIVINGAPARTMENTS_AVG	69.43
NONLIVINGAPARTMENTS_MEDI	69.43
LIVINGAPARTMENTS_MODE	68.45
LIVINGAPARTMENTS_AVG	68.45
LIVINGAPARTMENTS_MEDI	68.45
FONDKAPREMONT_MODE	68.38
FLOORSMIN_AVG	67.79
FLOORSMIN_MODE	67.79
FLOORSMIN_MEDI	67.79
YEARS_BUILD_MEDI	66.48
YEARS_BUILD_MODE	66.48
YEARS_BUILD_AVG	66.48
OWN_CAR_AGE	65.90
LANDAREA_MEDI	59.44
LANDAREA_MODE	59.44
LANDAREA_AVG	59.44
BASEMENTAREA_MEDI	58.40
BASEMENTAREA_AVG	58.40
BASEMENTAREA_MODE	58.40
EXT_SOURCE_1	56.35
NONLIVINGAREA_MODE	55.15
NONLIVINGAREA_AVG	55.15
NONLIVINGAREA_MEDI	55.15
ELEVATORS_MEDI	53.30
ELEVATORS_AVG	53.30
ELEVATORS_MODE	53.30
WALLSMATERIAL_MODE	50.92
APARTMENTS_MEDI	50.77
APARTMENTS_AVG	50.77
APARTMENTS_MODE	50.77
ENTRANCES_MEDI	50.39
ENTRANCES_AVG	50.39
ENTRANCES_MODE	50.39
LIVINGAREA_AVG	50.28
LIVINGAREA_MODE	50.28
LIVINGAREA_MEDI	50.28
HOUSETYPE_MODE	50.15

dtype: float64

null.index

```
Index(['COMMONAREA_MEDI', 'COMMONAREA_AVG', 'COMMONAREA_MODE',
      'NONLIVINGAPARTMENTS_MODE', 'NONLIVINGAPARTMENTS_AVG',
      'NONLIVINGAPARTMENTS_MEDI', 'LIVINGAPARTMENTS_MODE',
      'LIVINGAPARTMENTS_AVG', 'LIVINGAPARTMENTS_MEDI',
      'FONDKAPREMONT_MODE',
      'FLOORSMIN_AVG', 'FLOORSMIN_MODE', 'FLOORSMIN_MEDI',
      'YEARS_BUILD_MEDI',
      'YEARS_BUILD_MODE', 'YEARS_BUILD_AVG', 'OWN_CAR_AGE',
```

```
'LANDAREA_MEDI',
    'LANDAREA_MODE', 'LANDAREA_AVG', 'BASEMENTAREA_MEDI',
    'BASEMENTAREA_AVG', 'BASEMENTAREA_MODE', 'EXT_SOURCE_1',
    'NONLIVINGAREA_MODE', 'NONLIVINGAREA_AVG',
'NONLIVINGAREA_MEDI',
    'ELEVATORS_MEDI', 'ELEVATORS_AVG', 'ELEVATORS_MODE',
    'WALLSMATERIAL_MODE', 'APARTMENTS_MEDI', 'APARTMENTS_AVG',
    'APARTMENTS_MODE', 'ENTRANCES_MEDI', 'ENTRANCES_AVG',
'ENTRANCES_MODE',
    'LIVINGAREA_AVG', 'LIVINGAREA_MODE', 'LIVINGAREA_MEDI',
    'HOUSETYPE_MODE'],
    dtype='object')
```

```
# drop() method
```

```
# DataFrame.drop(columns,inplace)
```

```
df_app.drop(columns=null.index,inplace=True)
```

```
df_app
```

	SK_ID_CURR	TARGET	NAME_CONTRACT_TYPE	CODE_GENDER	FLAG_OWN_CAR
0	100002	1	Cash loans	M	N
1	100003	0	Cash loans	F	N
2	100004	0	Revolving loans	M	Y
3	100006	0	Cash loans	F	N
4	100007	0	Cash loans	M	N
...	...	...	...	...	...
49994	157871	0	Cash loans	F	N
49995	157872	0	Cash loans	M	N
49996	157873	0	Cash loans	M	N
49997	157874	0	Cash loans	F	N
49998	157875	0	Cash loans	F	N

	FLAG_OWN_REALTY	CNT_CHILDREN	AMT_INCOME_TOTAL	AMT_CREDIT	\
0	Y	0	202500.0	406597.5	
1	N	0	270000.0	1293502.5	
2	Y	0	67500.0	135000.0	
3	Y	0	135000.0	312682.5	
4	Y	0	121500.0	513000.0	
...	...	...	...	...	
49994	N	0	180000.0	1206000.0	

49995	N	0	126000.0	1125000.0
49996	N	1	112500.0	900000.0
49997	Y	0	270000.0	820638.0
49998	Y	0	117000.0	254700.0

	AMT_ANNUITY	...	FLAG_DOCUMENT_18	FLAG_DOCUMENT_19
FLAG_DOCUMENT_20 \				
0	24700.5	...	0	0
0				
1	35698.5	...	0	0
0				
2	6750.0	...	0	0
0				
3	29686.5	...	0	0
0				
4	21865.5	...	0	0
0				

...	...	...	...	...
...				
49994	45936.0	...	0	0
0				
49995	47794.5	...	0	0
0				
49996	26316.0	...	0	0
0				
49997	34897.5	...	0	0
0				
49998	14751.0	...	0	0
0				

	FLAG_DOCUMENT_21	AMT_REQ_CREDIT_BUREAU_HOUR
AMT_REQ_CREDIT_BUREAU_DAY \		
0	0	0.0
0.0		
1	0	0.0
0.0		
2	0	0.0
0.0		
3	0	NaN
NaN		
4	0	0.0
0.0		
...	...	...
...		
49994	0	0.0
0.0		
49995	0	0.0
0.0		
49996	0	0.0

0.0		
49997	0	0.0
0.0		
49998	0	0.0
0.0		

	AMT_REQ_CREDIT_BUREAU_WEEK	AMT_REQ_CREDIT_BUREAU_MON \
0	0.0	0.0
1	0.0	0.0
2	0.0	0.0
3	NaN	NaN
4	0.0	0.0
...	...	...
49994	0.0	0.0
49995	0.0	0.0
49996	0.0	0.0
49997	0.0	0.0
49998	0.0	0.0

	AMT_REQ_CREDIT_BUREAU_QRT	AMT_REQ_CREDIT_BUREAU_YEAR
0	0.0	1.0
1	0.0	0.0
2	0.0	0.0
3	NaN	NaN
4	0.0	0.0
...	...	...
49994	0.0	0.0
49995	0.0	0.0
49996	0.0	2.0
49997	2.0	4.0
49998	0.0	0.0

[49999 rows x 81 columns]

```
null1=missing_value(df_app)[missing_value(df_app)>40]
null1
```

FLOORSMAX_AVG	49.75
FLOORSMAX_MODE	49.75
FLOORSMAX_MEDI	49.75
YEARS_BEGINEXPLUATATION_AVG	48.79
YEARS_BEGINEXPLUATATION_MODE	48.79
YEARS_BEGINEXPLUATATION_MEDI	48.79
TOTALAREA_MODE	48.30
EMERGENCYSTATE_MODE	47.40

dtype: float64

null1.index

```
Index(['FLOORSMAX_AVG', 'FLOORSMAX_MODE', 'FLOORSMAX_MEDI',
      'YEARS_BEGINEXPLUATATION_AVG', 'YEARS_BEGINEXPLUATATION_MODE',
      'YEARS_BEGINEXPLUATATION_MEDI', 'TOTALAREA_MODE',
      'EMERGENCYSTATE_MODE'],
      dtype='object')
```

```
df_app.drop(columns=null1.index,inplace=True)
df_app
```

	SK_ID_CURR	TARGET	NAME_CONTRACT_TYPE	CODE_GENDER	FLAG_OWN_CAR
\					
0	100002	1	Cash loans	M	N
1	100003	0	Cash loans	F	N
2	100004	0	Revolving loans	M	Y
3	100006	0	Cash loans	F	N
4	100007	0	Cash loans	M	N
...	...	...	...	...	...
49994	157871	0	Cash loans	F	N
49995	157872	0	Cash loans	M	N
49996	157873	0	Cash loans	M	N
49997	157874	0	Cash loans	F	N
49998	157875	0	Cash loans	F	N

	FLAG_OWN_REALTY	CNT_CHILDREN	AMT_INCOME_TOTAL	AMT_CREDIT	\
0	Y	0	202500.0	406597.5	
1	N	0	270000.0	1293502.5	
2	Y	0	67500.0	135000.0	
3	Y	0	135000.0	312682.5	
4	Y	0	121500.0	513000.0	
...	...	...	...	...	
49994	N	0	180000.0	1206000.0	
49995	N	0	126000.0	1125000.0	
49996	N	1	112500.0	900000.0	
49997	Y	0	270000.0	820638.0	
49998	Y	0	117000.0	254700.0	

	AMT_ANNUITY	...	FLAG_DOCUMENT_18	FLAG_DOCUMENT_19
FLAG_DOCUMENT_20	\			
0	24700.5	...	0	0
0				

1	35698.5	...	0	0
0				
2	6750.0	...	0	0
0				
3	29686.5	...	0	0
0				
4	21865.5	...	0	0
0				
...	...	...	...	...
...				
49994	45936.0	...	0	0
0				
49995	47794.5	...	0	0
0				
49996	26316.0	...	0	0
0				
49997	34897.5	...	0	0
0				
49998	14751.0	...	0	0
0				

FLAG_DOCUMENT_21	AMT_REQ_CREDIT_BUREAU_HOUR
AMT_REQ_CREDIT_BUREAU_DAY \	

0	0	0.0
0.0		
1	0	0.0
0.0		
2	0	0.0
0.0		
3	0	NaN
NaN		
4	0	0.0
0.0		
...	...	...
...		
49994	0	0.0
0.0		
49995	0	0.0
0.0		
49996	0	0.0
0.0		
49997	0	0.0
0.0		
49998	0	0.0
0.0		

AMT_REQ_CREDIT_BUREAU_WEEK	AMT_REQ_CREDIT_BUREAU_MON \
0	0.0 0.0
1	0.0 0.0

2	0.0	0.0
3	NaN	NaN
4	0.0	0.0
...	...	...
49994	0.0	0.0
49995	0.0	0.0
49996	0.0	0.0
49997	0.0	0.0
49998	0.0	0.0

	AMT_REQ_CREDIT_BUREAU_QRT	AMT_REQ_CREDIT_BUREAU_YEAR
0	0.0	1.0
1	0.0	0.0
2	0.0	0.0
3	NaN	NaN
4	0.0	0.0
...	...	...
49994	0.0	0.0
49995	0.0	0.0
49996	0.0	2.0
49997	2.0	4.0
49998	0.0	0.0

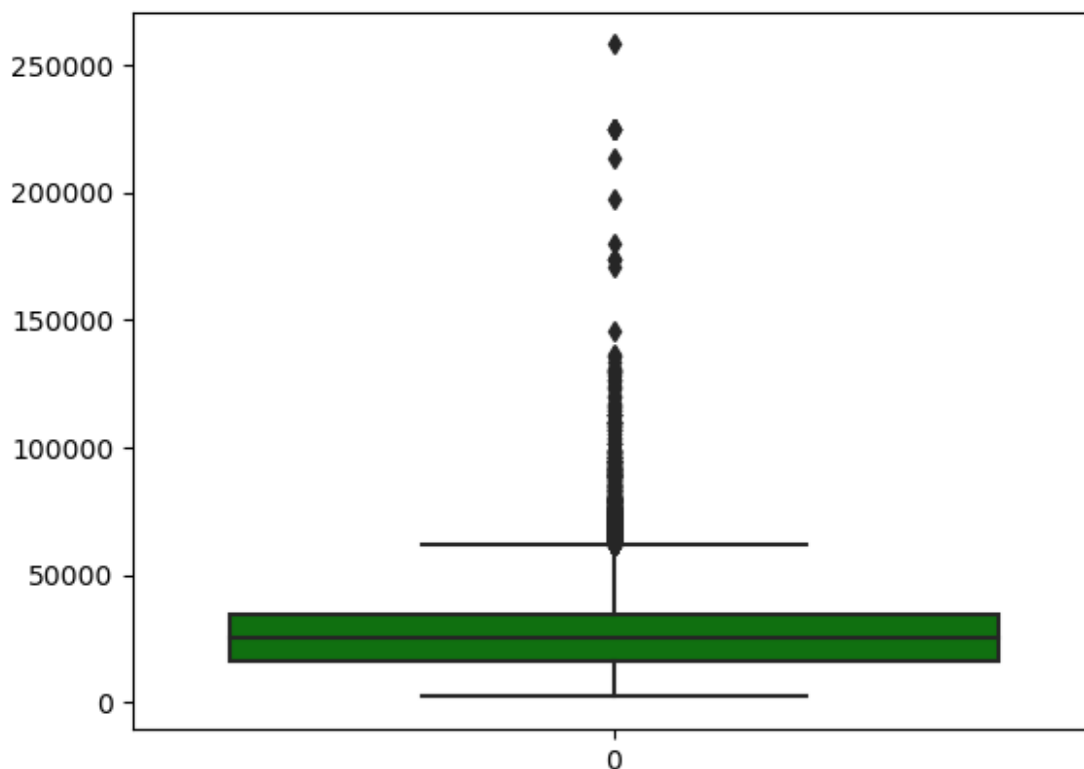
[49999 rows x 73 columns]

```
df_app["AMT_ANNUITY"].isnull().sum()
```

1

```
import matplotlib.pyplot as plt
import seaborn as sns
```

```
sns.boxplot(df_app["AMT_ANNUITY"],color="green")
plt.show()
```



```
df_app["AMT_ANNUIITY"].describe()
```

```
count      49998.000000
mean       27107.377355
std        14562.944435
min         2052.000000
25%        16456.500000
50%        24939.000000
75%        34596.000000
max       258025.500000
Name: AMT_ANNUIITY, dtype: float64
```

```
q1=np.quantile(df_app["AMT_ANNUIITY"].dropna(),0.25)
q1
```

```
16456.5
```

```
q2=np.quantile(df_app["AMT_ANNUIITY"].dropna(),0.50)
q2
```

```
24939.0
```

```
q3=np.quantile(df_app["AMT_ANNUIITY"].dropna(),0.75)
q3
```

```
34596.0
```



```
IQR=q3-q1
```

```
IQR
```

```
18139.5
```

```
upper_fence=q3+1.5*(IQR)
```

```
lower_fence=q1-1.5*(IQR)
```

```
l=[]
```

```
for i in df_app["AMT_ANNUITY"]:
```

```
    if i>upper_fence or i<lower_fence:
```

```
        l.append(i)
```

```
print(l)
```

```
[86989.5, 64107.0, 73611.0, 83515.5, 77494.5, 66262.5, 65956.5,
62698.5, 67500.0, 68643.0, 67500.0, 67500.0, 62019.0, 116266.5,
72778.5, 79065.0, 62019.0, 68643.0, 74191.5, 63549.0, 72477.0,
68490.0, 62572.5, 81108.0, 84694.5, 66019.5, 63549.0, 67500.0,
69651.0, 64998.0, 65988.0, 67203.0, 73732.5, 116505.0, 63684.0,
107356.5, 68440.5, 73341.0, 96174.0, 65088.0, 83034.0, 68089.5,
99045.0, 69754.5, 70033.5, 67891.5, 102658.5, 63463.5, 67221.0,
72607.5, 69781.5, 68512.5, 63504.0, 83241.0, 79020.0, 135936.0,
62698.5, 79708.5, 69754.5, 63243.0, 68350.5, 66618.0, 72607.5,
62820.0, 71374.5, 61915.5, 68814.0, 67500.0, 62770.5, 72607.5,
72607.5, 69295.5, 62010.0, 65475.0, 62613.0, 94509.0, 93928.5,
119731.5, 83214.0, 65556.0, 69754.5, 76104.0, 90918.0, 109053.0,
63567.0, 62698.5, 112909.5, 72607.5, 66712.5, 66757.5, 71590.5,
62019.0, 66928.5, 62568.0, 80239.5, 62505.0, 76473.0, 62613.0,
62613.0, 70083.0, 88789.5, 61879.5, 61911.0, 91570.5, 69754.5,
66721.5, 68350.5, 65907.0, 85585.5, 91642.5, 71928.0, 69214.5,
88245.0, 71028.0, 62613.0, 78903.0, 65826.0, 82152.0, 63976.5,
77364.0, 68512.5, 90058.5, 66402.0, 62847.0, 64264.5, 62613.0,
94639.5, 69354.0, 68166.0, 62613.0, 62613.0, 65061.0, 62613.0,
62028.0, 68512.5, 62203.5, 95935.5, 64264.5, 68512.5, 62311.5,
74416.5, 71928.0, 68512.5, 63549.0, 80010.0, 67513.5, 62770.5,
82620.0, 98379.0, 89928.0, 64120.5, 73125.0, 63351.0, 67500.0,
67500.0, 109597.5, 76531.5, 80662.5, 93487.5, 76729.5, 66627.0,
75640.5, 62698.5, 64818.0, 66487.5, 74772.0, 109809.0, 69660.0,
61942.5, 74254.5, 110488.5, 63351.0, 64516.5, 65889.0, 90117.0,
75307.5, 72337.5, 62311.5, 69781.5, 71928.0, 67500.0, 69007.5,
72607.5, 69781.5, 63549.0, 72076.5, 67248.0, 94639.5, 73944.0,
67500.0, 62572.5, 67500.0, 61875.0, 78903.0, 93123.0, 70132.5,
61875.0, 66532.5, 77382.0, 91570.5, 62446.5, 71460.0, 79155.0,
69048.0, 63549.0, 71599.5, 62568.0, 103374.0, 64908.0, 62874.0,
68625.0, 98640.0, 63549.0, 72036.0, 105511.5, 123480.0, 74641.5,
88164.0, 62568.0, 73786.5, 63549.0, 84267.0, 72607.5, 68643.0,
62653.5, 62950.5, 69354.0, 71928.0, 62613.0, 77089.5, 67500.0,
69417.0, 77332.5, 67203.0, 79753.5, 67500.0, 67500.0, 62568.0,
72274.5, 67891.5, 62820.0, 62698.5, 64107.0, 70209.0, 67500.0,
```

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67203.0, 73957.5, 78669.0, 62964.0, 69781.5, 67500.0, 62568.0,
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62964.0, 68490.0, 72036.0, 108805.5, 62698.5, 61906.5, 64647.0,
75343.5, 62698.5, 68643.0, 130779.0, 62698.5, 71590.5, 66843.0,
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71478.0, 65358.0, 74565.0, 62707.5, 62698.5, 63427.5, 91147.5,
225000.0, 68512.5, 61875.0, 70434.0, 64444.5, 68512.5, 62698.5,
61906.5, 67203.0, 62698.5, 110047.5, 63936.0, 88834.5, 76234.5,
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62568.0, 76234.5, 74470.5, 65758.5, 67500.0, 73566.0, 65133.0,
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63958.5, 65137.5, 197230.5, 68512.5, 62820.0, 64147.5, 74817.0,
79065.0, 76234.5, 68440.5, 71590.5, 146002.5, 86580.0, 76626.0,
81684.0, 72607.5, 67923.0, 70879.5, 61906.5, 90009.0, 76837.5,
66015.0, 62311.5, 71590.5, 78903.0, 62568.0, 225000.0, 72274.5,
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95782.5, 62568.0, 91219.5, 63774.0, 96345.0, 71590.5, 76842.0,
67500.0, 61879.5, 97177.5, 89536.5, 74353.5, 76095.0, 73732.5,
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62950.5, 62568.0, 78979.5, 66757.5, 61875.0, 98338.5, 85428.0,
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225000.0, 65065.5, 66915.0, 117369.0, 68980.5, 62964.0, 213291.0,
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62311.5, 61875.0, 68098.5, 62658.0, 61875.0, 130180.5, 76234.5,
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66532.5, 61911.0, 88969.5, 62698.5, 65866.5, 61875.0, 71028.0,
75667.5, 67540.5, 68512.5, 171040.5, 62302.5, 225000.0, 62010.0,
78367.5, 88969.5, 68719.5, 80194.5, 74254.5, 66402.0, 67500.0,

62316.0, 66244.5, 108481.5, 69781.5, 71959.5, 70173.0, 64147.5,
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72040.5, 71905.5, 72454.5, 61875.0, 68062.5, 72576.0, 62698.5,
66226.5, 101250.0, 63738.0, 97762.5, 62698.5, 80995.5, 69538.5,
62613.0, 65052.0, 68512.5, 67248.0, 87808.5, 67203.0, 78052.5,
92326.5, 62860.5, 72607.5, 68314.5, 67500.0, 67203.0, 66325.5,
68643.0, 130018.5, 68643.0, 80149.5, 75123.0, 67500.0, 78736.5,
62613.0, 62892.0, 130018.5, 67500.0, 62527.5, 68022.0, 71253.0,
76909.5, 67203.0, 64395.0, 66757.5, 79708.5, 65065.5, 62568.0,
63499.5, 93100.5, 62568.0, 72828.0, 64188.0, 101250.0, 72036.0,
62698.5, 69912.0, 62005.5, 73444.5, 69799.5, 70006.5, 79155.0,
66712.5, 69354.0, 88839.0, 62203.5, 78916.5, 126594.0, 87957.0,
73818.0, 75352.5, 64107.0, 110047.5, 66708.0, 64539.0, 79708.5,
71905.5, 76531.5, 65313.0, 64948.5, 88942.5, 69295.5, 62568.0,
66532.5, 90000.0, 78903.0, 68512.5, 75375.0, 62613.0, 62820.0,
69165.0, 69624.0, 112500.0, 67500.0, 86274.0, 66712.5, 67500.0,
74749.5, 77089.5, 62698.5, 82152.0, 78651.0, 68022.0, 62568.0,
62797.5, 92223.0, 64683.0, 92191.5, 63549.0, 69961.5, 112500.0,
85500.0, 67500.0, 95913.0, 89374.5, 70857.0, 62482.5, 65997.0,
68643.0, 65889.0, 62698.5, 63706.5, 98811.0, 69538.5, 71010.0,
79758.0, 71883.0, 69772.5, 92596.5, 72292.5, 68314.5, 62442.0,
68643.0, 109597.5, 67500.0, 67203.0, 66015.0, 69007.5, 67923.0,
94657.5, 111969.0, 65601.0, 66532.5, 62568.0, 63184.5, 61875.0,
68220.0, 62365.5, 69354.0, 62653.5, 68643.0, 69295.5, 62698.5,
80653.5, 69912.0, 62568.0, 75285.0, 73980.0, 71599.5, 71095.5,
64647.0, 130306.5, 62568.0, 76473.0, 88965.0, 68490.0, 77674.5,
101497.5, 79632.0, 225000.0, 74124.0, 66532.5, 71761.5, 71397.0,
91278.0, 66294.0, 62311.5, 67486.5, 67500.0, 73125.0, 124141.5,
70348.5, 72058.5, 80032.5, 69754.5, 62311.5, 68350.5, 78903.0,
72085.5, 64107.0, 62698.5, 136687.5, 68643.0, 78772.5, 70218.0,
70299.0, 83407.5, 62010.0, 62721.0, 66712.5, 80527.5, 62203.5,
72058.5, 80239.5, 66487.5, 66942.0, 74781.0, 62613.0, 64372.5,
68359.5, 66595.5, 68944.5, 66712.5, 65299.5, 79578.0, 86710.5,
90544.5, 64309.5, 173704.5, 67108.5, 61906.5, 93478.5, 89928.0,
98640.0, 89482.5, 66753.0, 72607.5, 63549.0, 95800.5, 67500.0,
71280.0, 74830.5, 67792.5, 66033.0, 135724.5, 64309.5, 62698.5,
64683.0, 64395.0, 66595.5, 68350.5, 94509.0, 62991.0, 72607.5,
68643.0, 64147.5, 62311.5, 70713.0, 85585.5, 79065.0, 62185.5,
96214.5, 68980.5, 62829.0, 62041.5, 62613.0, 97762.5, 62527.5,
69354.0, 69795.0, 94752.0, 93960.0, 70605.0, 67500.0, 68980.5,
97528.5, 80239.5, 61879.5, 74110.5, 70209.0, 66532.5, 73579.5,
71815.5, 62095.5, 66069.0, 84334.5, 84937.5, 64107.0, 87291.0,
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61888.5, 91728.0, 131535.0, 82476.0, 68710.5, 65151.0, 84442.5,
68764.5, 85018.5, 130018.5, 120798.0, 62203.5, 122832.0, 62698.5,
112909.5, 68490.0, 65835.0, 68643.0, 68643.0, 66357.0, 63675.0,
78903.0, 66757.5, 68643.0, 62005.5, 64107.0, 65128.5, 89325.0,

```
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65835.0, 62613.0, 98982.0, 72036.0, 70357.5]
```

```
df_app["AMT_ANNUITY"]=df_app["AMT_ANNUITY"].fillna(df_app["AMT_ANNUITY"]
"].median())
```

```
df_app["AMT_ANNUITY"].isnull().sum()
```

```
0
```

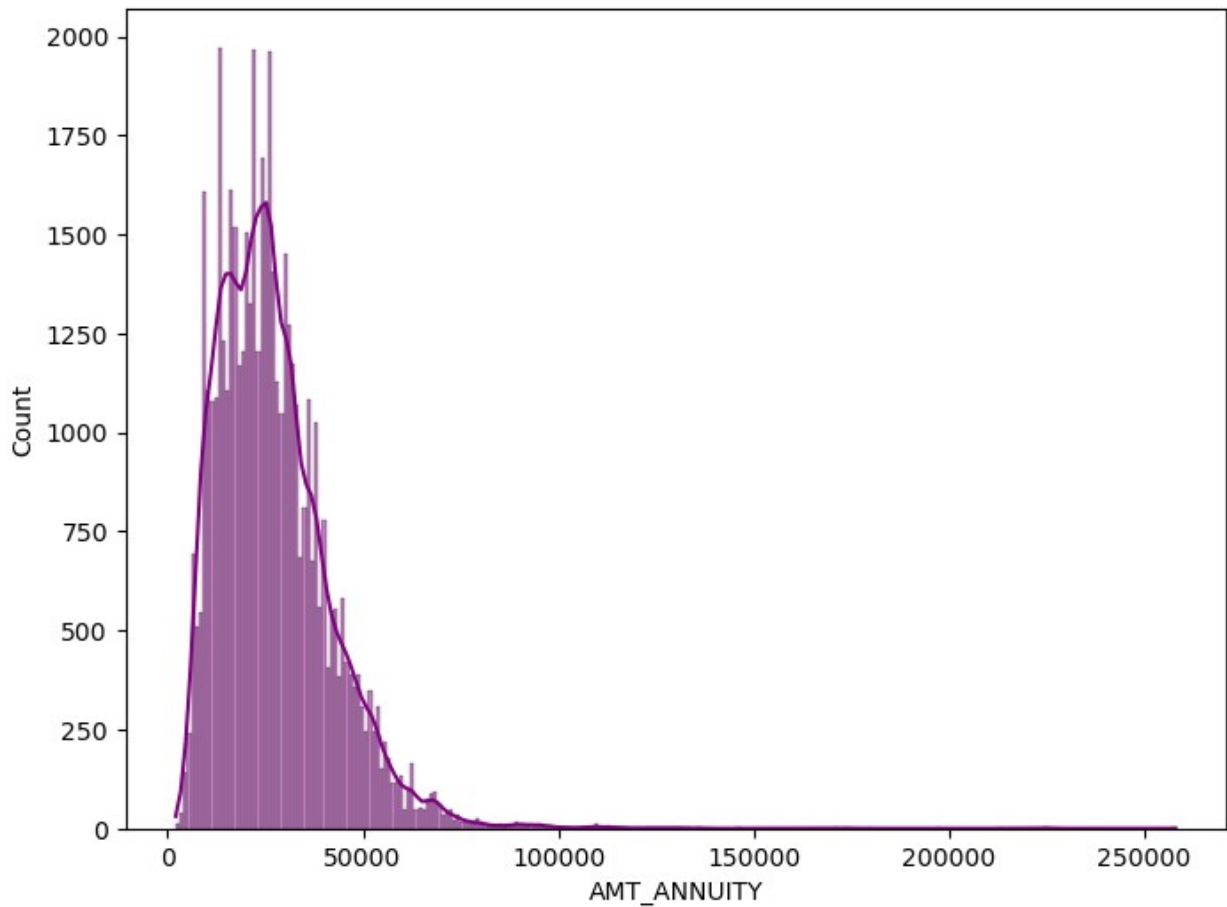
```
#kde= kurnel density estimator
```

```
#histogram plot
```

```
plt.figure(figsize=(8,6))
```

```
sns.histplot(df_app["AMT_ANNUITY"],kde=True,color="purple")
```

```
plt.show()
```



```
df_app["AMT_GOODS_PRICE"].isnull().sum()
```

```
38
```

```
df_app["AMT_GOODS_PRICE"].describe()
```

```
count    4.996100e+04
```

```
mean      5.390600e+05
```

```
std       3.698533e+05
```

```
min       4.500000e+04
```

```
25%       2.385000e+05
```

```
50%       4.500000e+05
```

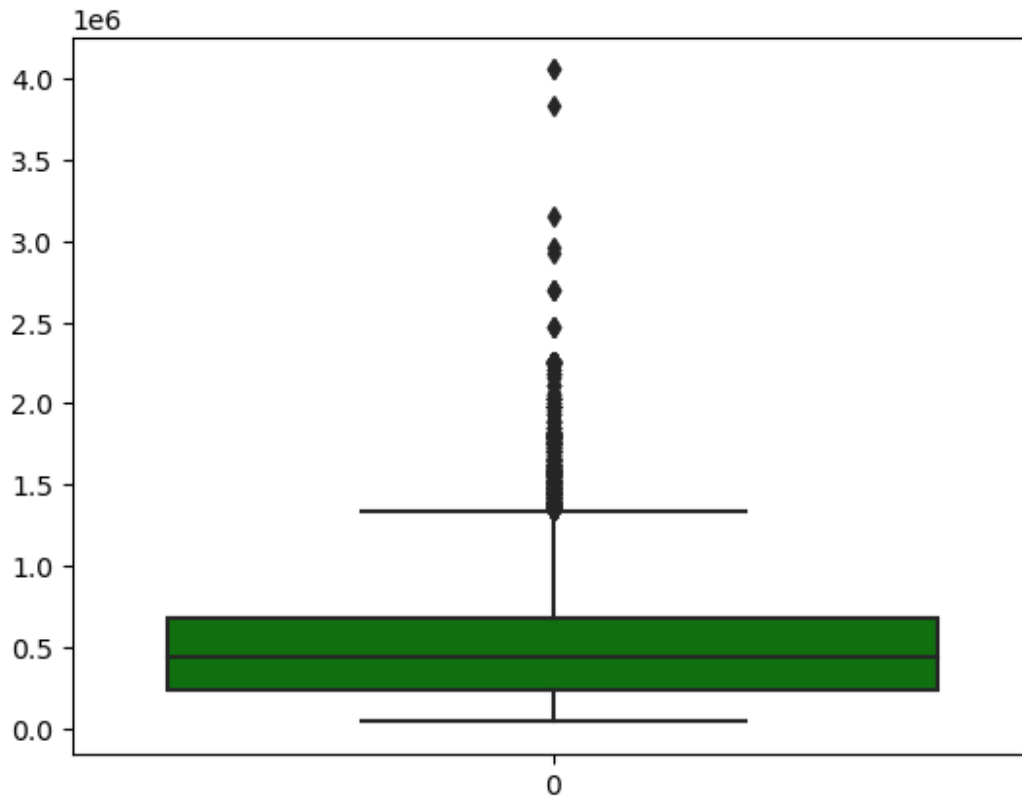
```
75%       6.795000e+05
```

```
max       4.050000e+06
```

```
Name: AMT_GOODS_PRICE, dtype: float64
```

```
sns.boxplot(df_app["AMT_GOODS_PRICE"],color="green")
```

```
plt.show()
```



```

q1=np.quantile(df_app["AMT_GOODS_PRICE"].dropna(),0.25)
q1
238500.0

q2=np.quantile(df_app["AMT_GOODS_PRICE"].dropna(),0.50)
q2
450000.0

q3=np.quantile(df_app["AMT_GOODS_PRICE"].dropna(),0.75)
q3
679500.0

IQR=q3-q1
IQR
441000.0

upper_fence=q3+1.5*IQR
lower_fence=q1-1.5*IQR

l=[]
for i in df_app["AMT_GOODS_PRICE"]:
    if i>upper_fence or i<lower_fence:

```

```
l.append(i)
print(l)
```

```
[1395000.0, 1530000.0, 1350000.0, 1350000.0, 1588500.0, 1435500.0,
1755000.0, 1354500.0, 2250000.0, 1354500.0, 1575000.0, 1710000.0,
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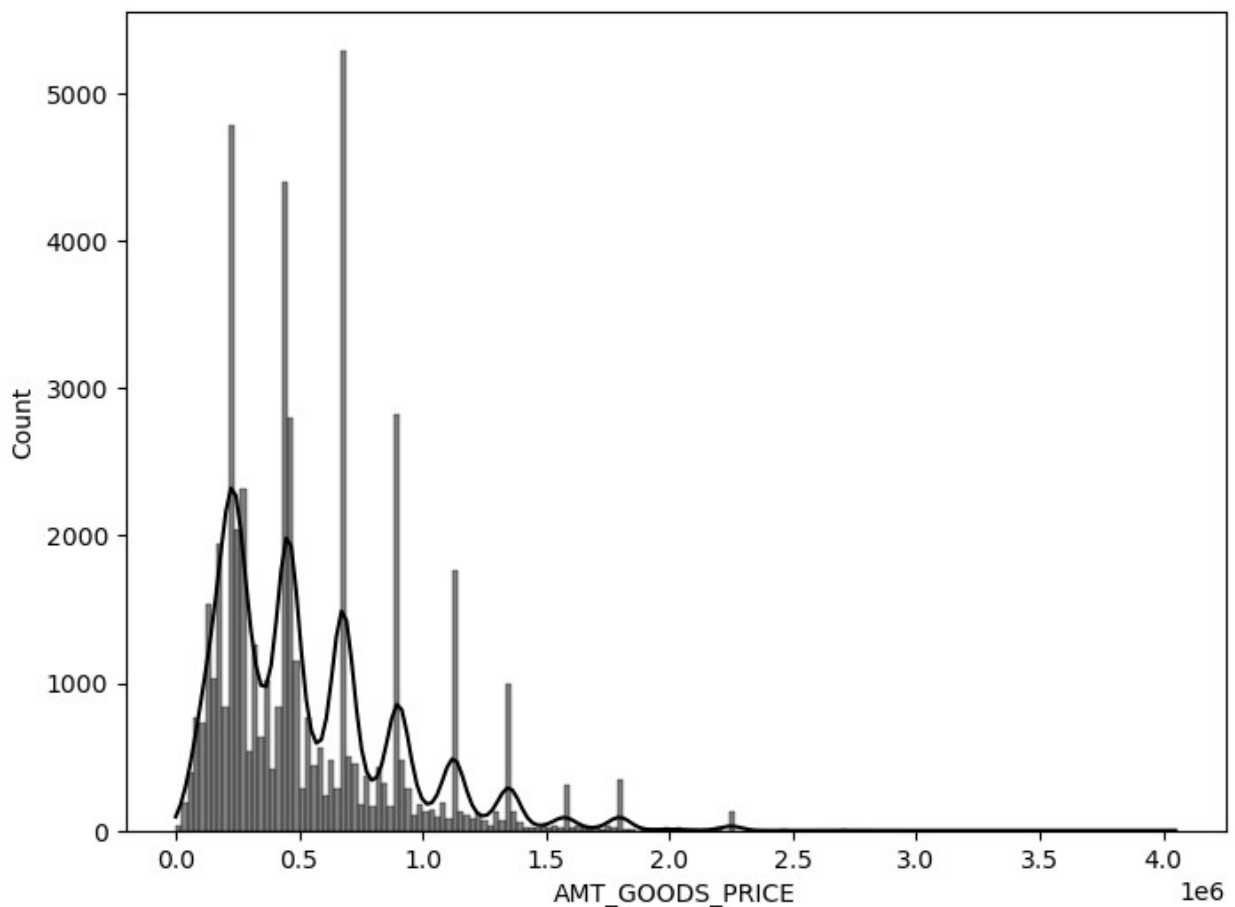
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```

```
df_app["AMT_GOODS_PRICE"] = df_app["AMT_GOODS_PRICE"].fillna(df_app["AMT_GOODS_PRICE"] == df_app["AMT_CREDIT"])
```

```
df_app["AMT_GOODS_PRICE"].isnull().sum()
```

```
0
```

```
plt.figure(figsize=(8,6))
sns.histplot(df_app["AMT_GOODS_PRICE"], kde=True, color="black")
plt.show()
```



```
df_app["OCCUPATION_TYPE"]
```

```

0      Laborers
1      Core staff
2      Laborers
3      Laborers
4      Core staff
...
49994   Waiters/barmen staff
49995      Laborers
49996      NaN
49997      Cleaning staff
49998      NaN
Name: OCCUPATION_TYPE, Length: 49999, dtype: object

```

```

df_app["OCCUPATION_TYPE"].isnull().sum()

15654

```

```

#value_counts()--records--frequency.
df_app["OCCUPATION_TYPE"].value_counts()

```

```

OCCUPATION_TYPE
Laborers      8952
Sales staff   5160
Core staff    4434
Managers      3489
Drivers       3044
High skill tech staff  1852
Accountants   1621
Medicine staff  1403
Security staff  1140
Cooking staff   963
Cleaning staff   739
Private service staff  447
Low-skill Laborers  357
Waiters/barmen staff  228
Secretaries     212
Realty agents   123
HR staff        101
IT staff         80
Name: count, dtype: int64

```

```

df_app["OCCUPATION_TYPE"].value_counts(normalize=True)

```

```

OCCUPATION_TYPE
Laborers      0.260649
Sales staff    0.150240
Core staff     0.129102
Managers       0.101587
Drivers        0.088630
High skill tech staff  0.053923
Accountants    0.047198

```


Medicine staff	0.040850
Security staff	0.033193
Cooking staff	0.028039
Cleaning staff	0.021517
Private service staff	0.013015
Low-skill Laborers	0.010395
Waiters/barmen staff	0.006639
Secretaries	0.006173
Realty agents	0.003581
HR staff	0.002941
IT staff	0.002329

Name: proportion, dtype: float64

```
df_app["OCCUPATION_TYPE"]=df_app["OCCUPATION_TYPE"].fillna("Unknown")
```

```
df_app["OCCUPATION_TYPE"].isnull().sum()
```

0

```
occupation_type=df_app["OCCUPATION_TYPE"].value_counts()
occupation_type
```

OCCUPATION_TYPE	
Unknown	15654
Laborers	8952
Sales staff	5160
Core staff	4434
Managers	3489
Drivers	3044
High skill tech staff	1852
Accountants	1621
Medicine staff	1403
Security staff	1140
Cooking staff	963
Cleaning staff	739
Private service staff	447
Low-skill Laborers	357
Waiters/barmen staff	228
Secretaries	212
Realty agents	123
HR staff	101
IT staff	80

Name: count, dtype: int64

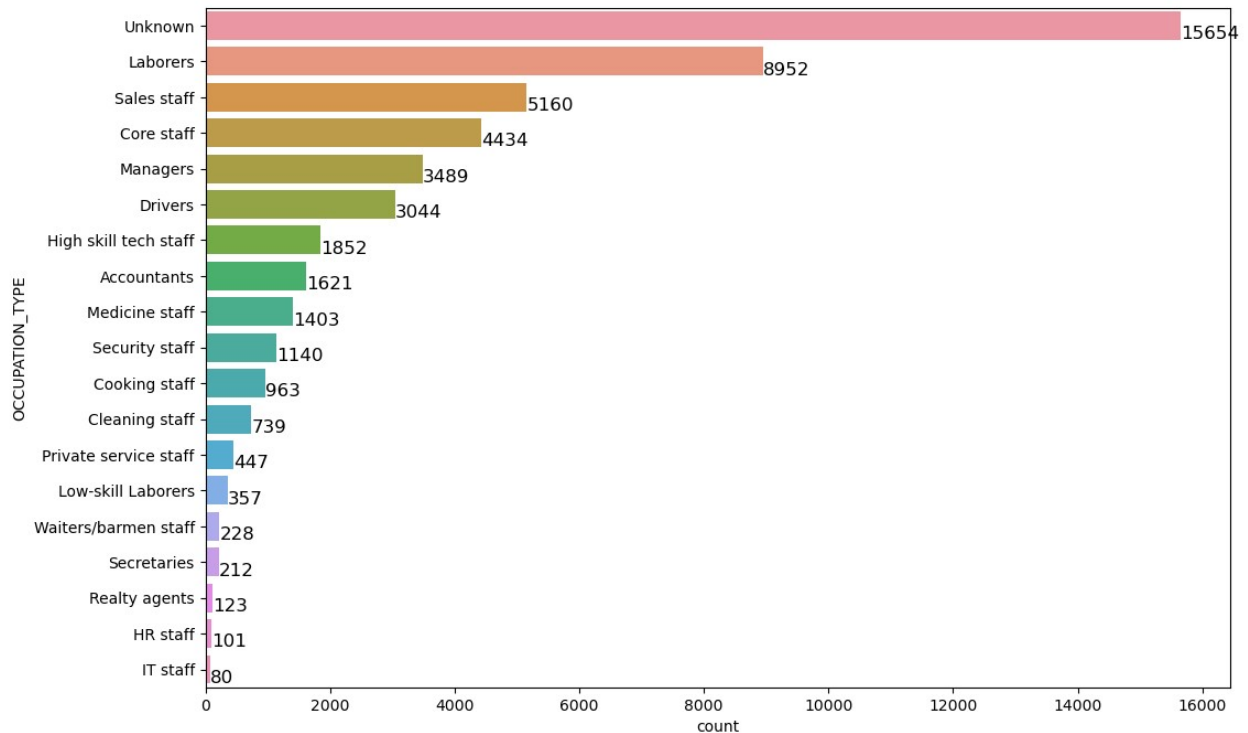
```
plt.figure(figsize=(12,8))
```

```
sns.barplot(x=occupation_type,y=occupation_type.index)
```

```
for index,value in enumerate(occupation_type):
```

```
plt.text(value,index,f"{value}",ha="left",va="top",color="black",fontsize=12)
```

```
plt.show()
```



```
df_app["NAME_TYPE_SUITE"]=df_app["NAME_TYPE_SUITE"].fillna("Unacompanied")
```

```
df_app["NAME_TYPE_SUITE"].isnull().sum()
```

```
0
```

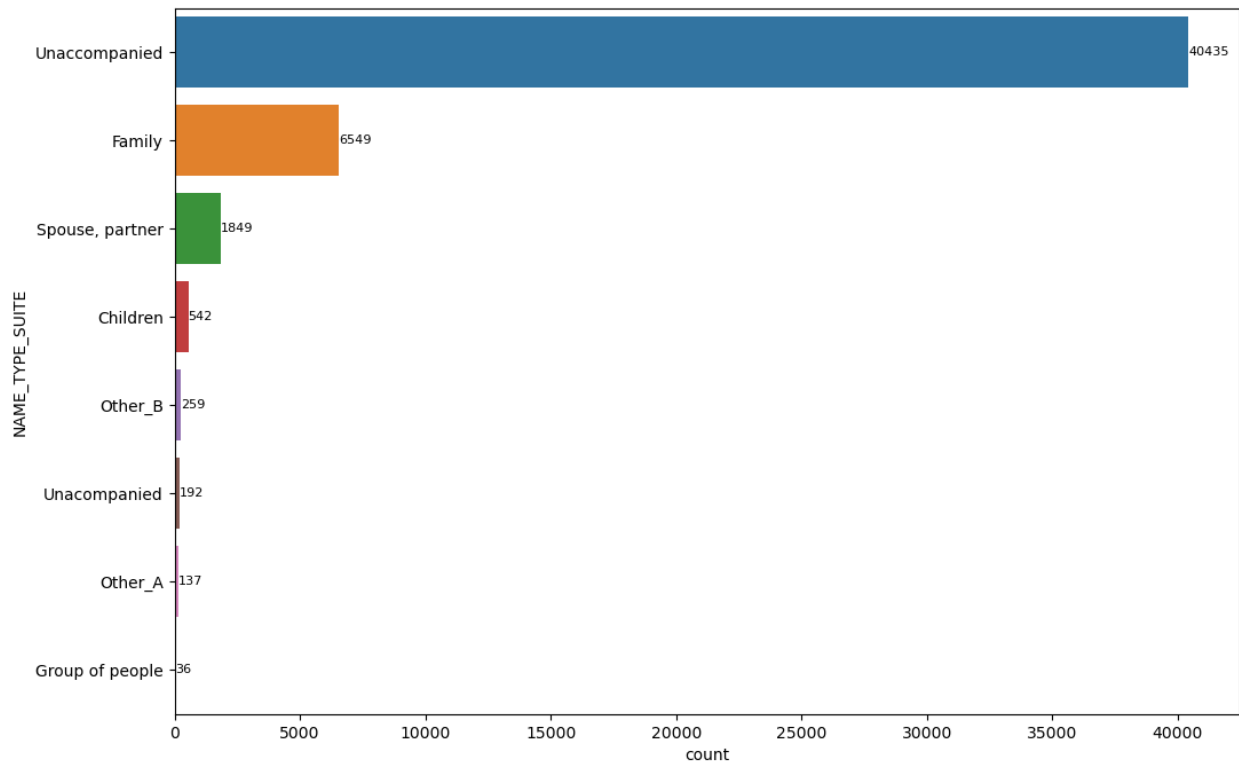
```
X=df_app["NAME_TYPE_SUITE"].value_counts()
```

```
X
```

```
NAME_TYPE_SUITE
Unaccompanied    40435
Family           6549
Spouse, partner   1849
Children          542
Other_B           259
Unacompanied      192
Other_A           137
Group of people    36
Name: count, dtype: int64
```

```
plt.figure(figsize=(12,8))
sns.barplot(x=X,y=X.index)
for index,value in enumerate(X):
```

```
plt.text(value,index,f"{value}",ha="left",va="center",color="black",fontsize=8)
plt.show()
```



```
df_app["CNT_FAM_MEMBERS"]
```

```
0      1.0
1      2.0
2      1.0
3      2.0
4      1.0
```

```
...
```

```
49994    1.0
49995    2.0
49996    3.0
49997    2.0
49998    2.0
```

```
Name: CNT_FAM_MEMBERS, Length: 49999, dtype: float64
```

```
df_app["CNT_FAM_MEMBERS"].isnull().sum()
```

```
1
```

```
df_app["CNT_FAM_MEMBERS"].describe()
```

```
count    49998.000000
mean      2.158946
std       0.911332
min       1.000000
25%       2.000000
50%       2.000000
```

```

75%          3.000000
max          13.000000
Name: CNT_FAM_MEMBERS, dtype: float64

df_app["CNT_FAM_MEMBERS"]=df_app["CNT_FAM_MEMBERS"].fillna(df_app["CNT_FAM_MEMBERS"].median())

df_app["CNT_FAM_MEMBERS"].isnull().sum()

0

df_app.columns

Index(['SK_ID_CURR', 'TARGET', 'NAME_CONTRACT_TYPE', 'CODE_GENDER',
      'FLAG_OWN_CAR', 'FLAG_OWN_REALTY', 'CNT_CHILDREN',
      'AMT_INCOME_TOTAL',
      'AMT_CREDIT', 'AMT_ANNUITY', 'AMT_GOODS_PRICE',
      'NAME_TYPE_SUITE',
      'NAME_INCOME_TYPE', 'NAME_EDUCATION_TYPE',
      'NAME_FAMILY_STATUS',
      'NAME_HOUSING_TYPE', 'REGION_POPULATION_RELATIVE',
      'DAYS_BIRTH',
      'DAYS_EMPLOYED', 'DAYS_REGISTRATION', 'DAYS_ID_PUBLISH',
      'FLAG_MOBIL',
      'FLAG_EMP_PHONE', 'FLAG_WORK_PHONE', 'FLAG_CONT_MOBILE',
      'FLAG_PHONE',
      'FLAG_EMAIL', 'OCCUPATION_TYPE', 'CNT_FAM_MEMBERS',
      'REGION_RATING_CLIENT', 'REGION_RATING_CLIENT_W_CITY',
      'WEEKDAY_APPR_PROCESS_START', 'HOUR_APPR_PROCESS_START',
      'REG_REGION_NOT_LIVE_REGION', 'REG_REGION_NOT_WORK_REGION',
      'LIVE_REGION_NOT_WORK_REGION', 'REG_CITY_NOT_LIVE_CITY',
      'REG_CITY_NOT_WORK_CITY', 'LIVE_CITY_NOT_WORK_CITY',
      'ORGANIZATION_TYPE', 'EXT_SOURCE_2', 'EXT_SOURCE_3',
      'OBS_30_CNT_SOCIAL_CIRCLE', 'DEF_30_CNT_SOCIAL_CIRCLE',
      'OBS_60_CNT_SOCIAL_CIRCLE', 'DEF_60_CNT_SOCIAL_CIRCLE',
      'DAYS_LAST_PHONE_CHANGE', 'FLAG_DOCUMENT_2', 'FLAG_DOCUMENT_3',
      'FLAG_DOCUMENT_4', 'FLAG_DOCUMENT_5', 'FLAG_DOCUMENT_6',
      'FLAG_DOCUMENT_7', 'FLAG_DOCUMENT_8', 'FLAG_DOCUMENT_9',
      'FLAG_DOCUMENT_10', 'FLAG_DOCUMENT_11', 'FLAG_DOCUMENT_12',
      'FLAG_DOCUMENT_13', 'FLAG_DOCUMENT_14', 'FLAG_DOCUMENT_15',
      'FLAG_DOCUMENT_16', 'FLAG_DOCUMENT_17', 'FLAG_DOCUMENT_18',
      'FLAG_DOCUMENT_19', 'FLAG_DOCUMENT_20', 'FLAG_DOCUMENT_21',
      'AMT_REQ_CREDIT_BUREAU_HOUR', 'AMT_REQ_CREDIT_BUREAU_DAY',
      'AMT_REQ_CREDIT_BUREAU_WEEK', 'AMT_REQ_CREDIT_BUREAU_MON',
      'AMT_REQ_CREDIT_BUREAU_QRT', 'AMT_REQ_CREDIT_BUREAU_YEAR'],
      dtype='object')

df_app[['AMT_REQ_CREDIT_BUREAU_HOUR', 'AMT_REQ_CREDIT_BUREAU_DAY',
      'AMT_REQ_CREDIT_BUREAU_WEEK', 'AMT_REQ_CREDIT_BUREAU_MON',
      'AMT_REQ_CREDIT_BUREAU_QRT', 'AMT_REQ_CREDIT_BUREAU_YEAR']]

```

	AMT_REQ_CREDIT_BUREAU_HOUR	AMT_REQ_CREDIT_BUREAU_DAY \
0	0.0	0.0
1	0.0	0.0
2	0.0	0.0
3	NaN	NaN
4	0.0	0.0
...	...	...
49994	0.0	0.0
49995	0.0	0.0
49996	0.0	0.0
49997	0.0	0.0
49998	0.0	0.0

	AMT_REQ_CREDIT_BUREAU_WEEK	AMT_REQ_CREDIT_BUREAU_MON \
0	0.0	0.0
1	0.0	0.0
2	0.0	0.0
3	NaN	NaN
4	0.0	0.0
...	...	...
49994	0.0	0.0
49995	0.0	0.0
49996	0.0	0.0
49997	0.0	0.0
49998	0.0	0.0

	AMT_REQ_CREDIT_BUREAU_QRT	AMT_REQ_CREDIT_BUREAU_YEAR
0	0.0	1.0
1	0.0	0.0
2	0.0	0.0
3	NaN	NaN
4	0.0	0.0
...	...	...
49994	0.0	0.0
49995	0.0	0.0
49996	0.0	2.0
49997	2.0	4.0
49998	0.0	0.0

[49999 rows x 6 columns]

```
AMT_REQ_CREDIT=['AMT_REQ_CREDIT_BUREAU_HOUR',
'AMT_REQ_CREDIT_BUREAU_DAY',
'AMT_REQ_CREDIT_BUREAU_WEEK', 'AMT_REQ_CREDIT_BUREAU_MON',
'AMT_REQ_CREDIT_BUREAU_QRT', 'AMT_REQ_CREDIT_BUREAU_YEAR']
```

```
df_app[AMT_REQ_CREDIT]
```

	AMT_REQ_CREDIT_BUREAU_HOUR	AMT_REQ_CREDIT_BUREAU_DAY \
0	0.0	0.0

1	0.0	0.0
2	0.0	0.0
3	NaN	NaN
4	0.0	0.0
...	...	...
49994	0.0	0.0
49995	0.0	0.0
49996	0.0	0.0
49997	0.0	0.0
49998	0.0	0.0

	AMT_REQ_CREDIT_BUREAU_WEEK	AMT_REQ_CREDIT_BUREAU_MON \
0	0.0	0.0
1	0.0	0.0
2	0.0	0.0
3	NaN	NaN
4	0.0	0.0
...	...	...
49994	0.0	0.0
49995	0.0	0.0
49996	0.0	0.0
49997	0.0	0.0
49998	0.0	0.0

	AMT_REQ_CREDIT_BUREAU_QRT	AMT_REQ_CREDIT_BUREAU_YEAR
0	0.0	1.0
1	0.0	0.0
2	0.0	0.0
3	NaN	NaN
4	0.0	0.0
...	...	...
49994	0.0	0.0
49995	0.0	0.0
49996	0.0	2.0
49997	2.0	4.0
49998	0.0	0.0

[49999 rows x 6 columns]

```
df_app[AMT_REQ_CREDIT].isnull().sum()
```

```
AMT_REQ_CREDIT_BUREAU_HOUR    6734
AMT_REQ_CREDIT_BUREAU_DAY      6734
AMT_REQ_CREDIT_BUREAU_WEEK    6734
AMT_REQ_CREDIT_BUREAU_MON      6734
AMT_REQ_CREDIT_BUREAU_QRT      6734
AMT_REQ_CREDIT_BUREAU_YEAR    6734
dtype: int64
```

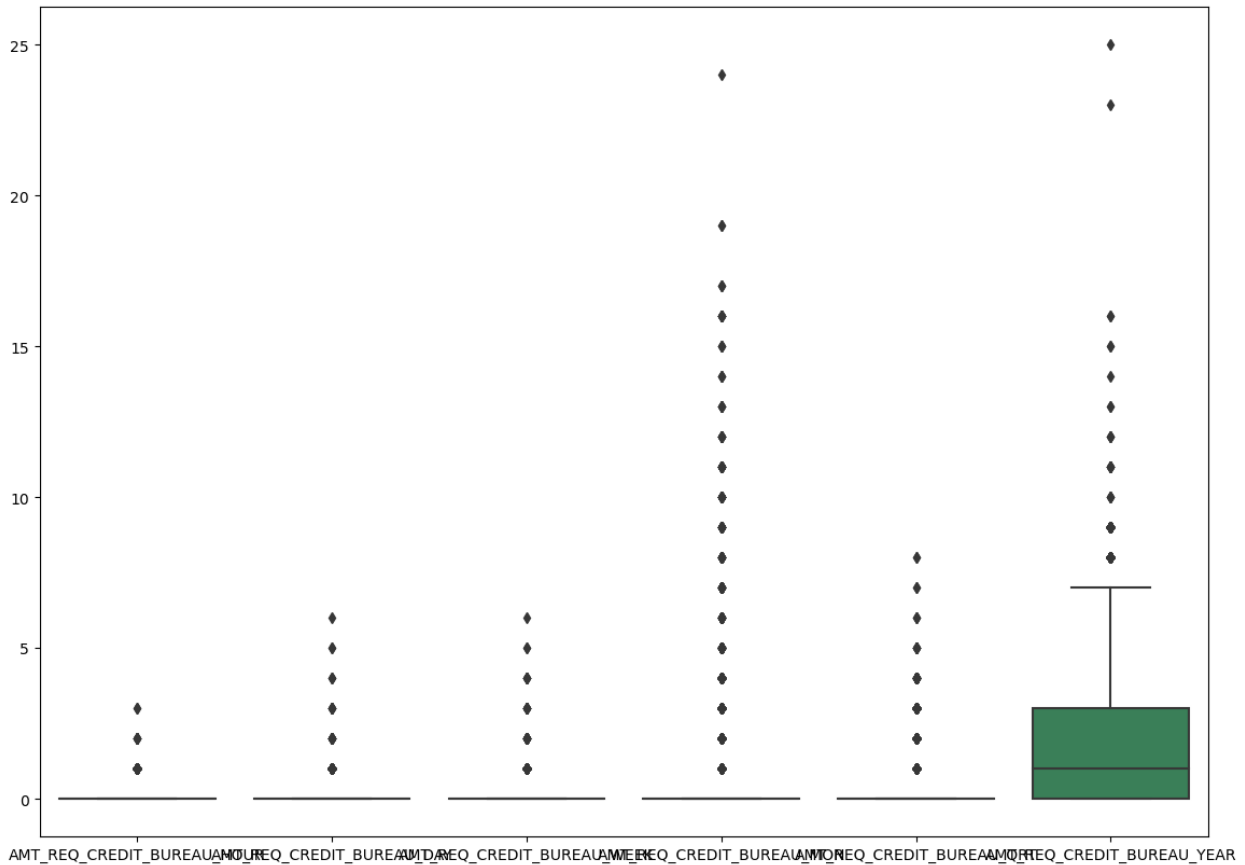
```
df_app[AMT_REQ_CREDIT].describe()
```

	AMT_REQ_CREDIT_BUREAU_HOUR	AMT_REQ_CREDIT_BUREAU_DAY \
count	43265.000000	43265.000000
mean	0.007096	0.007512
std	0.087709	0.107992
min	0.000000	0.000000
25%	0.000000	0.000000
50%	0.000000	0.000000
75%	0.000000	0.000000
max	3.000000	6.000000

	AMT_REQ_CREDIT_BUREAU_WEEK	AMT_REQ_CREDIT_BUREAU_MON \
count	43265.000000	43265.000000
mean	0.032382	0.270288
std	0.194080	0.928560
min	0.000000	0.000000
25%	0.000000	0.000000
50%	0.000000	0.000000
75%	0.000000	0.000000
max	6.000000	24.000000

	AMT_REQ_CREDIT_BUREAU_QRT	AMT_REQ_CREDIT_BUREAU_YEAR
count	43265.000000	43265.000000
mean	0.260973	1.881035
std	0.606996	1.865054
min	0.000000	0.000000
25%	0.000000	0.000000
50%	0.000000	1.000000
75%	0.000000	3.000000
max	8.000000	25.000000

```
plt.figure(figsize=(14,10))
sns.boxplot(df_app[AMT_REQ_CREDIT],color="seagreen")
plt.show()
```



```
df_app[AMT_REQ_CREDIT]=df_app[AMT_REQ_CREDIT].fillna(df_app[AMT_REQ_CREDIT].median())
```

```
df_app[AMT_REQ_CREDIT].isnull().sum()
```

```
AMT_REQ_CREDIT_BUREAU_HOUR    0
AMT_REQ_CREDIT_BUREAU_DAY    0
AMT_REQ_CREDIT_BUREAU_WEEK    0
AMT_REQ_CREDIT_BUREAU_MON    0
AMT_REQ_CREDIT_BUREAU_QRT    0
AMT_REQ_CREDIT_BUREAU_YEAR    0
dtype: int64
```

```
df_app.columns
```

```
Index(['SK_ID_CURR', 'TARGET', 'NAME_CONTRACT_TYPE', 'CODE_GENDER',
      'FLAG_OWN_CAR', 'FLAG_OWN_REALTY', 'CNT_CHILDREN',
      'AMT_INCOME_TOTAL',
      'AMT_CREDIT', 'AMT_ANNUITY', 'AMT_GOODS_PRICE',
      'NAME_TYPE_SUITE',
      'NAME_INCOME_TYPE', 'NAME_EDUCATION_TYPE',
      'NAME_FAMILY_STATUS',
      'NAME_HOUSING_TYPE', 'REGION_POPULATION_RELATIVE',
      'DAYS_BIRTH',
```



```

'DAYS_EMPLOYED', 'DAYS_REGISTRATION', 'DAYS_ID_PUBLISH',
'FLAG_MOBIL',
'FLAG_EMP_PHONE', 'FLAG_WORK_PHONE', 'FLAG_CONT_MOBILE',
'FLAG_PHONE',
'FLAG_EMAIL', 'OCCUPATION_TYPE', 'CNT_FAM_MEMBERS',
'REGION_RATING_CLIENT', 'REGION_RATING_CLIENT_W_CITY',
'WEEKDAY_APPR_PROCESS_START', 'HOUR_APPR_PROCESS_START',
'REG_REGION_NOT_LIVE_REGION', 'REG_REGION_NOT_WORK_REGION',
'LIVE_REGION_NOT_WORK_REGION', 'REG_CITY_NOT_LIVE_CITY',
'REG_CITY_NOT_WORK_CITY', 'LIVE_CITY_NOT_WORK_CITY',
'ORGANIZATION_TYPE', 'EXT_SOURCE_2', 'EXT_SOURCE_3',
'OBS_30_CNT_SOCIAL_CIRCLE', 'DEF_30_CNT_SOCIAL_CIRCLE',
'OBS_60_CNT_SOCIAL_CIRCLE', 'DEF_60_CNT_SOCIAL_CIRCLE',
'DAYS_LAST_PHONE_CHANGE', 'FLAG_DOCUMENT_2', 'FLAG_DOCUMENT_3',
'FLAG_DOCUMENT_4', 'FLAG_DOCUMENT_5', 'FLAG_DOCUMENT_6',
'FLAG_DOCUMENT_7', 'FLAG_DOCUMENT_8', 'FLAG_DOCUMENT_9',
'FLAG_DOCUMENT_10', 'FLAG_DOCUMENT_11', 'FLAG_DOCUMENT_12',
'FLAG_DOCUMENT_13', 'FLAG_DOCUMENT_14', 'FLAG_DOCUMENT_15',
'FLAG_DOCUMENT_16', 'FLAG_DOCUMENT_17', 'FLAG_DOCUMENT_18',
'FLAG_DOCUMENT_19', 'FLAG_DOCUMENT_20', 'FLAG_DOCUMENT_21',
'AMT_REQ_CREDIT_BUREAU_HOUR', 'AMT_REQ_CREDIT_BUREAU_DAY',
'AMT_REQ_CREDIT_BUREAU_WEEK', 'AMT_REQ_CREDIT_BUREAU_MON',
'AMT_REQ_CREDIT_BUREAU_QRT', 'AMT_REQ_CREDIT_BUREAU_YEAR'],
dtype='object')

```

```

df_app[['OBS_30_CNT_SOCIAL_CIRCLE', 'DEF_30_CNT_SOCIAL_CIRCLE',
'OBS_60_CNT_SOCIAL_CIRCLE', 'DEF_60_CNT_SOCIAL_CIRCLE']]

```

	OBS_30_CNT_SOCIAL_CIRCLE	DEF_30_CNT_SOCIAL_CIRCLE	\
0	2.0	2.0	
1	1.0	0.0	
2	0.0	0.0	
3	2.0	0.0	
4	0.0	0.0	
...	...	...	
49994	0.0	0.0	
49995	1.0	0.0	
49996	0.0	0.0	
49997	1.0	0.0	
49998	1.0	0.0	

	OBS_60_CNT_SOCIAL_CIRCLE	DEF_60_CNT_SOCIAL_CIRCLE
0	2.0	2.0
1	1.0	0.0
2	0.0	0.0
3	2.0	0.0
4	0.0	0.0
...	...	...
49994	0.0	0.0
49995	1.0	0.0

49996	0.0	0.0
49997	1.0	0.0
49998	1.0	0.0

[49999 rows x 4 columns]

```
SOCIAL_CIRCLE=['OBS_30_CNT_SOCIAL_CIRCLE', 'DEF_30_CNT_SOCIAL_CIRCLE',
               'OBS_60_CNT_SOCIAL_CIRCLE', 'DEF_60_CNT_SOCIAL_CIRCLE']
df_app[SOCIAL_CIRCLE]
```

	OBS_30_CNT_SOCIAL_CIRCLE	DEF_30_CNT_SOCIAL_CIRCLE \
0	2.0	2.0
1	1.0	0.0
2	0.0	0.0
3	2.0	0.0
4	0.0	0.0
...	...	...
49994	0.0	0.0
49995	1.0	0.0
49996	0.0	0.0
49997	1.0	0.0
49998	1.0	0.0

	OBS_60_CNT_SOCIAL_CIRCLE	DEF_60_CNT_SOCIAL_CIRCLE
0	2.0	2.0
1	1.0	0.0
2	0.0	0.0
3	2.0	0.0
4	0.0	0.0
...	...	...
49994	0.0	0.0
49995	1.0	0.0
49996	0.0	0.0
49997	1.0	0.0
49998	1.0	0.0

[49999 rows x 4 columns]

```
df_app[SOCIAL_CIRCLE].isnull().sum()
```

```
OBS_30_CNT_SOCIAL_CIRCLE    168
DEF_30_CNT_SOCIAL_CIRCLE    168
OBS_60_CNT_SOCIAL_CIRCLE    168
DEF_60_CNT_SOCIAL_CIRCLE    168
dtype: int64
```

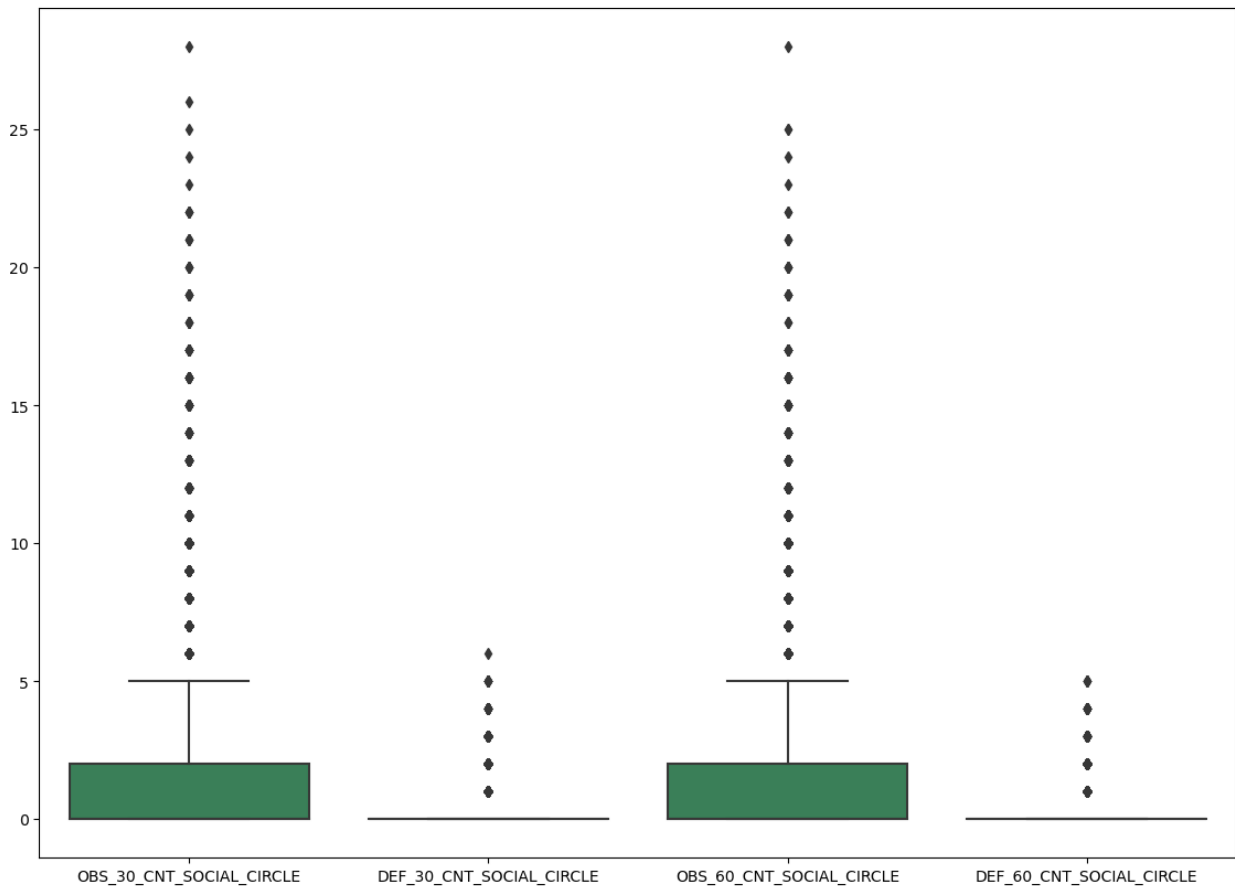
```
df_app[SOCIAL_CIRCLE].describe()
```

	OBS_30_CNT_SOCIAL_CIRCLE	DEF_30_CNT_SOCIAL_CIRCLE \
count	49831.000000	49831.000000
mean	1.420782	0.141819

std	2.302086	0.440540
min	0.000000	0.000000
25%	0.000000	0.000000
50%	0.000000	0.000000
75%	2.000000	0.000000
max	28.000000	6.000000

	OBS_60_CNT_SOCIAL_CIRCLE	DEF_60_CNT_SOCIAL_CIRCLE
count	49831.000000	49831.000000
mean	1.403664	0.098332
std	2.281782	0.357264
min	0.000000	0.000000
25%	0.000000	0.000000
50%	0.000000	0.000000
75%	2.000000	0.000000
max	28.000000	5.000000

```
plt.figure(figsize=(14,10))
sns.boxplot(df_app[SOCIAL_CIRCLE],color="seagreen")
plt.show()
```



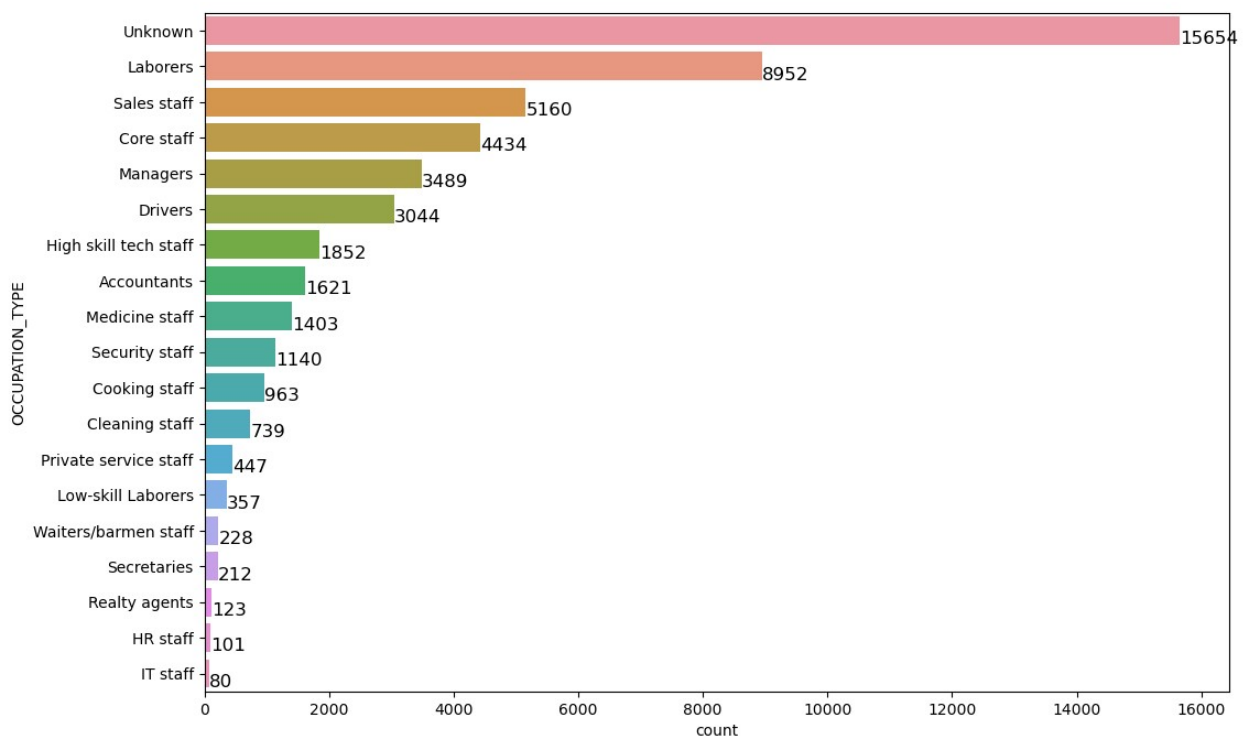
```
df_app[SOCIAL_CIRCLE]=df_app[SOCIAL_CIRCLE].fillna(df_app[SOCIAL_CIRCLE].median())
```

```
df_app[SOCIAL_CIRCLE].isnull().sum()
```

```
OBS_30_CNT_SOCIAL_CIRCLE    0
DEF_30_CNT_SOCIAL_CIRCLE    0
OBS_60_CNT_SOCIAL_CIRCLE    0
DEF_60_CNT_SOCIAL_CIRCLE    0
dtype: int64
```

```
plt.figure(figsize=(12,8))
sns.barplot(x=occupation_type,y=occupation_type.index)
for index,value in enumerate(occupation_type):
```

```
plt.text(value,index,f"{value}",ha="left",va="top",color="black",fontsize=12)
plt.show()
```



```
df_app["CODE_GENDER"]
```

```
0      M
1      F
2      M
3      F
4      M
..
```

```

49994    F
49995    M
49996    M
49997    F
49998    F
Name: CODE_GENDER, Length: 49999, dtype: object

df_app["CODE_GENDER"].isnull().sum()

0

df_app["CODE_GENDER"].value_counts()

CODE_GENDER
F      32823
M      17174
XNA         2
Name: count, dtype: int64

df_app["CODE_GENDER"].value_counts(normalize=True)

CODE_GENDER
F      0.656473
M      0.343487
XNA     0.000040
Name: proportion, dtype: float64

df_app["CODE_GENDER"] = df_app["CODE_GENDER"].fillna("F")

df_app["CODE_GENDER"].isnull().sum()

0

code_gender = df_app["CODE_GENDER"].value_counts()
code_gender

CODE_GENDER
F      32823
M      17174
XNA         2
Name: count, dtype: int64

code_gender.index

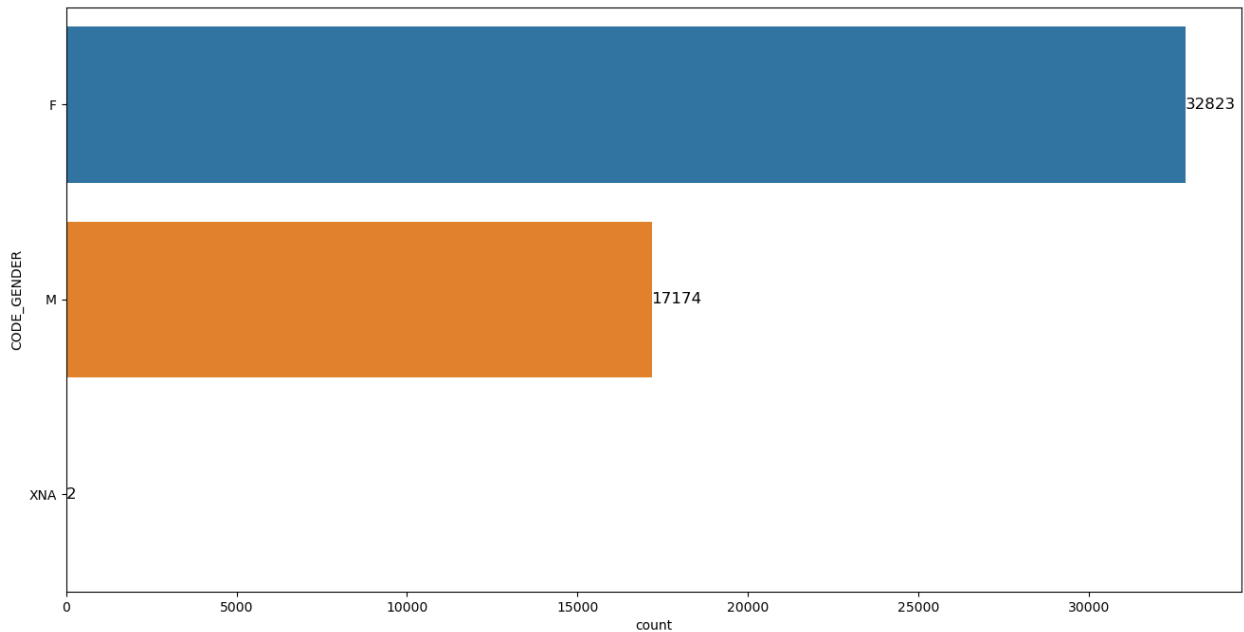
Index(['F', 'M', 'XNA'], dtype='object', name='CODE_GENDER')

plt.figure(figsize=(16,8))
sns.barplot(x=code_gender, y=code_gender.index)
for index, value in enumerate(code_gender):

plt.text(value, index, f"{value}", ha="left", va="center", color="black", fo

```

```
ntsize=12)
plt.show()
```



#loc -- loc is a labeled based and meaning that we use labels(column_name,index_name to select aor axis data)

#loc(rows,columns]

stop condition included

```
df_app.loc[0:10:2,
["SK_ID_CURR","CODE_GENDER","AMT_ANNUITY","DAYS_BIRTH"]]
```

	SK_ID_CURR	CODE_GENDER	AMT_ANNUITY	DAYS_BIRTH
0	100002	M	24700.5	-9461
2	100004	M	6750.0	-19046
4	100007	M	21865.5	-19932
6	100009	F	41301.0	-13778
8	100011	F	33826.5	-20099
10	100014	F	21177.0	-10197

#ILOC-- INTEGER BASED # NOT A INCLUDED STOP(N-1)

```
df_app.iloc[0:10:2,0:10:2]
```

	SK_ID_CURR	NAME_CONTRACT_TYPE	FLAG_OWN_CAR	CNT_CHILDREN
0	100002	Cash loans	N	0
2	100004	Revolving loans	Y	0
4	100007	Cash loans	N	0
6	100009	Cash loans	Y	1

1560726.0

8 100011 Cash loans N 0

1019610.0

df_app[df_app["AMT_ANNUITY"]>50000]

	SK_ID_CURR	TARGET	NAME_CONTRACT_TYPE	CODE_GENDER	FLAG_OWN_CAR
\					
28	100033	0	Cash loans	M	Y
60	100071	0	Cash loans	F	N
94	100112	1	Cash loans	M	Y
112	100132	0	Cash loans	F	N
135	100157	0	Cash loans	M	N
...	...	...	...	...	...
49895	157754	0	Cash loans	F	N
49933	157797	0	Cash loans	F	N
49938	157806	0	Cash loans	M	Y
49945	157813	0	Cash loans	M	Y
49966	157838	0	Cash loans	F	N

	FLAG_OWN_REALTY	CNT_CHILDREN	AMT_INCOME_TOTAL	AMT_CREDIT	\
28	Y	0	270000.0	790830.0	
60	Y	0	180000.0	1663987.5	
94	Y	0	315000.0	953460.0	
112	Y	0	202500.0	1136439.0	
135	Y	0	180000.0	1755000.0	
...	...	...	...	...	
49895	N	0	270000.0	1909107.0	
49933	N	0	337500.0	2169913.5	
49938	N	0	441000.0	701730.0	
49945	Y	1	225000.0	1540588.5	
49966	N	1	180000.0	1701729.0	

	AMT_ANNUITY	...	FLAG_DOCUMENT_18	FLAG_DOCUMENT_19
FLAG_DOCUMENT_20				
\				
28	57676.5	...	0	0
0				
60	86989.5	...	0	0
0				
94	64107.0	...	0	0

0				
112	73611.0	...	0	0
0				
135	61132.5	...	0	0
0				
...	...	...	...	...
...				
49895	98982.0	...	0	0
0				
49933	59800.5	...	0	0
0				
49938	72036.0	...	0	0
0				
49945	58810.5	...	0	0
0				
49966	70357.5	...	0	0
0				

FLAG_DOCUMENT_21 AMT_REQ_CREDIT_BUREAU_HOUR

AMT_REQ_CREDIT_BUREAU_DAY \

28	0	0.0
0.0		
60	0	0.0
0.0		
94	0	0.0
0.0		
112	0	0.0
0.0		
135	0	0.0
0.0		
...	...	...
...		
49895	0	0.0
0.0		
49933	0	0.0
0.0		
49938	0	0.0
0.0		
49945	0	0.0
0.0		
49966	0	0.0
0.0		

	AMT_REQ_CREDIT_BUREAU_WEEK	AMT_REQ_CREDIT_BUREAU_MON \
28	0.0	1.0
60	0.0	2.0
94	0.0	0.0
112	0.0	0.0
135	0.0	1.0

...	...	...
49895	0.0	0.0
49933	0.0	0.0
49938	0.0	0.0
49945	0.0	1.0
49966	0.0	0.0

	AMT_REQ_CREDIT_BUREAU_QRT	AMT_REQ_CREDIT_BUREAU_YEAR
28	0.0	1.0
60	0.0	1.0
94	0.0	2.0
112	0.0	1.0
135	0.0	2.0
...	...	...
49895	0.0	0.0
49933	1.0	3.0
49938	0.0	5.0
49945	0.0	1.0
49966	0.0	1.0

[3437 rows x 73 columns]

df_app[df_app["CODE_GENDER"]=="F"]

	SK_ID_CURR	TARGET	NAME_CONTRACT_TYPE	CODE_GENDER	FLAG_OWN_CAR
\					
1	100003	0	Cash loans	F	N
3	100006	0	Cash loans	F	N
6	100009	0	Cash loans	F	Y
8	100011	0	Cash loans	F	N
10	100014	0	Cash loans	F	N
...	...	...	...	...	...
49992	157869	0	Revolving loans	F	N
49993	157870	0	Revolving loans	F	N
49994	157871	0	Cash loans	F	N
49997	157874	0	Cash loans	F	N
49998	157875	0	Cash loans	F	N
1	FLAG_OWN_REALTY	CNT_CHILDREN	AMT_INCOME_TOTAL	AMT_CREDIT	\
	N	0	270000.0	1293502.5	

3	Y	0	135000.0	312682.5
6	Y	1	171000.0	1560726.0
8	Y	0	112500.0	1019610.0
10	Y	1	112500.0	652500.0
...	...	...	...	...
49992	Y	3	58500.0	157500.0
49993	N	0	202500.0	450000.0
49994	N	0	180000.0	1206000.0
49997	Y	0	270000.0	820638.0
49998	Y	0	117000.0	254700.0

	AMT_ANNUITY	...	FLAG_DOCUMENT_18	FLAG_DOCUMENT_19
FLAG_DOCUMENT_20 \				
1	35698.5	...	0	0
0				
3	29686.5	...	0	0
0				
6	41301.0	...	0	0
0				
8	33826.5	...	0	0
0				
10	21177.0	...	0	0
0				
...	...	...	...	...
...				
49992	7875.0	...	1	0
0				
49993	22500.0	...	0	0
0				
49994	45936.0	...	0	0
0				
49997	34897.5	...	0	0
0				
49998	14751.0	...	0	0
0				

	FLAG_DOCUMENT_21	AMT_REQ_CREDIT_BUREAU_HOUR
AMT_REQ_CREDIT_BUREAU_DAY \		
1	0	0.0
0.0		
3	0	0.0
0.0		
6	0	0.0
0.0		
8	0	0.0
0.0		
10	0	0.0
0.0		
...	...	...

```

...
49992          0          0.0
0.0
49993          0          0.0
0.0
49994          0          0.0
0.0
49997          0          0.0
0.0
49998          0          0.0
0.0

```

```

          AMT_REQ_CREDIT_BUREAU_WEEK  AMT_REQ_CREDIT_BUREAU_MON  \
1                0.0                0.0
3                0.0                0.0
6                0.0                1.0
8                0.0                0.0
10               0.0                1.0
...
49992            0.0            0.0
49993            0.0            0.0
49994            0.0            0.0
49997            0.0            0.0
49998            0.0            0.0

```

```

          AMT_REQ_CREDIT_BUREAU_QRT  AMT_REQ_CREDIT_BUREAU_YEAR
1                0.0                0.0
3                0.0                1.0
6                1.0                2.0
8                0.0                1.0
10               0.0                0.0
...
49992            0.0                1.0
49993            2.0                5.0
49994            0.0                0.0
49997            2.0                4.0
49998            0.0                0.0

```

[32823 rows x 73 columns]

```
df_app[(df_app["AMT_ANNUITY"]>50000) & (df_app["CODE_GENDER"]=="F")]
```

```

          SK_ID_CURR  TARGET  NAME_CONTRACT_TYPE  CODE_GENDER  FLAG_OWN_CAR
\
60          100071      0      Cash loans          F          N
112         100132      0      Cash loans          F          N
152         100179      0      Cash loans          F          Y

```

159	100188	0	Cash loans	F	N
191	100221	0	Cash loans	F	N
...	...	...	...	...	...
49802	157655	0	Cash loans	F	N
49882	157741	0	Cash loans	F	N
49895	157754	0	Cash loans	F	N
49933	157797	0	Cash loans	F	N
49966	157838	0	Cash loans	F	N
	FLAG_OWN_REALTY	CNT_CHILDREN	AMT_INCOME_TOTAL	AMT_CREDIT	\
60	Y	0	180000.0	1663987.5	
112	Y	0	202500.0	1136439.0	
152	N	0	202500.0	675000.0	
159	N	1	225000.0	1325475.0	
191	Y	0	225000.0	684054.0	
...	...	...	...	...	
49802	N	0	180000.0	1125000.0	
49882	N	0	180000.0	1649376.0	
49895	N	0	270000.0	1909107.0	
49933	N	0	337500.0	2169913.5	
49966	N	1	180000.0	1701729.0	
	AMT_ANNUITY	...	FLAG_DOCUMENT_18	FLAG_DOCUMENT_19	
FLAG_DOCUMENT_20	\				
60	86989.5	...	0	0	
0					
112	73611.0	...	0	0	
0					
152	53329.5	...	0	0	
0					
159	56290.5	...	0	0	
0					
191	77494.5	...	0	0	
0					
...	...	...	...	...	
...					
49802	50967.0	...	0	0	
0					
49882	51916.5	...	0	0	
0					
49895	98982.0	...	0	0	
0					

49933	59800.5	...	0	0
0				
49966	70357.5	...	0	0
0				

FLAG_DOCUMENT_21	AMT_REQ_CREDIT_BUREAU_HOUR
AMT_REQ_CREDIT_BUREAU_DAY	\

60	0	0.0
0.0		
112	0	0.0
0.0		
152	0	0.0
0.0		
159	0	0.0
0.0		
191	0	0.0
0.0		
...	...	...
...		
49802	0	0.0
0.0		
49882	0	0.0
0.0		
49895	0	0.0
0.0		
49933	0	0.0
0.0		
49966	0	0.0
0.0		

AMT_REQ_CREDIT_BUREAU_WEEK	AMT_REQ_CREDIT_BUREAU_MON	\
60	0.0	2.0
112	0.0	0.0
152	0.0	1.0
159	0.0	1.0
191	0.0	0.0
...	...	...
49802	0.0	4.0
49882	0.0	0.0
49895	0.0	0.0
49933	0.0	0.0
49966	0.0	0.0

AMT_REQ_CREDIT_BUREAU_QRT	AMT_REQ_CREDIT_BUREAU_YEAR
60	0.0
112	0.0
152	0.0
159	0.0
191	1.0
...	...

49802	0.0	1.0
49882	0.0	1.0
49895	0.0	0.0
49933	1.0	3.0
49966	0.0	1.0

[1995 rows x 73 columns]

df_app.columns

```
Index(['SK_ID_CURR', 'TARGET', 'NAME_CONTRACT_TYPE', 'CODE_GENDER',
      'FLAG_OWN_CAR', 'FLAG_OWN_REALTY', 'CNT_CHILDREN',
      'AMT_INCOME_TOTAL',
      'AMT_CREDIT', 'AMT_ANNUITY', 'AMT_GOODS_PRICE',
      'NAME_TYPE_SUITE',
      'NAME_INCOME_TYPE', 'NAME_EDUCATION_TYPE',
      'NAME_FAMILY_STATUS',
      'NAME_HOUSING_TYPE', 'REGION_POPULATION_RELATIVE',
      'DAYS_BIRTH',
      'DAYS_EMPLOYED', 'DAYS_REGISTRATION', 'DAYS_ID_PUBLISH',
      'FLAG_MOBIL',
      'FLAG_EMP_PHONE', 'FLAG_WORK_PHONE', 'FLAG_CONT_MOBILE',
      'FLAG_PHONE',
      'FLAG_EMAIL', 'OCCUPATION_TYPE', 'CNT_FAM_MEMBERS',
      'REGION_RATING_CLIENT', 'REGION_RATING_CLIENT_W_CITY',
      'WEEKDAY_APPR_PROCESS_START', 'HOUR_APPR_PROCESS_START',
      'REG_REGION_NOT_LIVE_REGION', 'REG_REGION_NOT_WORK_REGION',
      'LIVE_REGION_NOT_WORK_REGION', 'REG_CITY_NOT_LIVE_CITY',
      'REG_CITY_NOT_WORK_CITY', 'LIVE_CITY_NOT_WORK_CITY',
      'ORGANIZATION_TYPE', 'EXT_SOURCE_2', 'EXT_SOURCE_3',
      'OBS_30_CNT_SOCIAL_CIRCLE', 'DEF_30_CNT_SOCIAL_CIRCLE',
      'OBS_60_CNT_SOCIAL_CIRCLE', 'DEF_60_CNT_SOCIAL_CIRCLE',
      'DAYS_LAST_PHONE_CHANGE', 'FLAG_DOCUMENT_2', 'FLAG_DOCUMENT_3',
      'FLAG_DOCUMENT_4', 'FLAG_DOCUMENT_5', 'FLAG_DOCUMENT_6',
      'FLAG_DOCUMENT_7', 'FLAG_DOCUMENT_8', 'FLAG_DOCUMENT_9',
      'FLAG_DOCUMENT_10', 'FLAG_DOCUMENT_11', 'FLAG_DOCUMENT_12',
      'FLAG_DOCUMENT_13', 'FLAG_DOCUMENT_14', 'FLAG_DOCUMENT_15',
      'FLAG_DOCUMENT_16', 'FLAG_DOCUMENT_17', 'FLAG_DOCUMENT_18',
      'FLAG_DOCUMENT_19', 'FLAG_DOCUMENT_20', 'FLAG_DOCUMENT_21',
      'AMT_REQ_CREDIT_BUREAU_HOUR', 'AMT_REQ_CREDIT_BUREAU_DAY',
      'AMT_REQ_CREDIT_BUREAU_WEEK', 'AMT_REQ_CREDIT_BUREAU_MON',
      'AMT_REQ_CREDIT_BUREAU_QRT', 'AMT_REQ_CREDIT_BUREAU_YEAR'],
      dtype='object')
```

```
df_app[['DAYS_BIRTH', 'DAYS_EMPLOYED', 'DAYS_REGISTRATION',
        'DAYS_ID_PUBLISH', 'DAYS_LAST_PHONE_CHANGE']]
```

	DAYS_BIRTH	DAYS_EMPLOYED	DAYS_REGISTRATION	DAYS_ID_PUBLISH \
0	-9461	-637	-3648	-2120

1	-16765	-1188	-1186	-291
2	-19046	-225	-4260	-2531
3	-19005	-3039	-9833	-2437
4	-19932	-3038	-4311	-3458
...	...	...	...	...
49994	-10667	-285	-2521	-3333
49995	-20211	-4651	-11281	-3722
49996	-10280	-1158	-8620	-2604
49997	-23485	-2181	-2662	-4200
49998	-19251	365243	-12934	-2783

	DAYS_LAST_PHONE_CHANGE
0	-1134.0
1	-828.0
2	-815.0
3	-617.0
4	-1106.0
...	...
49994	-14.0
49995	-1930.0
49996	-712.0
49997	-1989.0
49998	-295.0

[49999 rows x 5 columns]

```
#absolute fn== (-) change in (+)
df_app[['DAYS_BIRTH', 'DAYS_EMPLOYED', 'DAYS_REGISTRATION',
'DAYS_ID_PUBLISH', 'DAYS_LAST_PHONE_CHANGE']] = abs(df_app[['DAYS_BIRTH',
'DAYS_EMPLOYED', 'DAYS_REGISTRATION',
'DAYS_ID_PUBLISH', 'DAYS_LAST_PHONE_CHANGE']])
df_app[['DAYS_BIRTH', 'DAYS_EMPLOYED', 'DAYS_REGISTRATION',
'DAYS_ID_PUBLISH', 'DAYS_LAST_PHONE_CHANGE']]
```

	DAYS_BIRTH	DAYS_EMPLOYED	DAYS_REGISTRATION
DAYS_ID_PUBLISH \			
0	9461	637	3648
			2120

1	16765	1188	1186	291
2	19046	225	4260	2531
3	19005	3039	9833	2437
4	19932	3038	4311	3458
...	...	...	...	...
49994	10667	285	2521	3333
49995	20211	4651	11281	3722
49996	10280	1158	8620	2604
49997	23485	2181	2662	4200
49998	19251	365243	12934	2783

	DAYS_LAST_PHONE_CHANGE
0	1134.0
1	828.0
2	815.0
3	617.0
4	1106.0
...	...
49994	14.0
49995	1930.0
49996	712.0
49997	1989.0
49998	295.0

[49999 rows x 5 columns]

```
df_app["DAYS_BIRTH_IN_YEAR"]=round(df_app["DAYS_BIRTH"]/365,2)
df_app["DAYS_BIRTH_IN_YEAR"]
```

0	25.92
1	45.93
2	52.18
3	52.07
4	54.61
...	...
49994	29.22
49995	55.37
49996	28.16
49997	64.34
49998	52.74

Name: DAYS_BIRTH_IN_YEAR, Length: 49999, dtype: float64


```
df_app["DAYS_EMPLOYED_IN_YEAR"]=round(df_app["DAYS_EMPLOYED"]/365,2)
df_app["DAYS_EMPLOYED_IN_YEAR"]
```

```
0      1.75
1      3.25
2      0.62
3      8.33
4      8.32
```

```
...
49994    0.78
49995   12.74
49996    3.17
49997    5.98
49998  1000.67
```

```
Name: DAYS_EMPLOYED_IN_YEAR, Length: 49999, dtype: float64
```

```
df_app["DAYS_EMPLOYED_IN_YEAR"].max()
```

```
1000.67
```

```
df_app["DAYS_EMPLOYED_IN_YEAR"].describe()
```

```
count    49999.000000
mean      184.001660
std       380.708187
min        0.000000
25%        2.560000
50%        6.070000
75%       15.670000
max      1000.670000
```

```
Name: DAYS_EMPLOYED_IN_YEAR, dtype: float64
```

```
df_app["DAYS_EMPLOYED_IN_YEAR"]=df_app["DAYS_EMPLOYED_IN_YEAR"].replac
e(df_app["DAYS_EMPLOYED_IN_YEAR"].max(),np.nan)
```

```
df_app["DAYS_EMPLOYED_IN_YEAR"].max()
```

```
48.03
```

```
df_app["DAYS_BIRTH_RANGE"]=pd.cut(df_app["DAYS_EMPLOYED_IN_YEAR"],bins
=[20,25,30,35,40,45,50,55,60,65,70],labels=["20-25","25-30","30-
35","35-40","40-45","45-50","50-55","55-60","60-65","65&moore"])
```

```
df_app["DAYS_BIRTH_RANGE"]
```

```
0      NaN
1      NaN
2      NaN
3      NaN
4      NaN
```

```
...
```

```

49994    NaN
49995    NaN
49996    NaN
49997    NaN
49998    NaN
Name: DAYS_BIRTH_RANGE, Length: 49999, dtype: category
Categories (10, object): ['20-25' < '25-30' < '30-35' < '35-40' ...
'50-55' < '55-60' < '60-65' < '65&moore']

```

```
df_app["DAYS_BIRTH_RANGE"].describe()
```

```

count      2119
unique         6
top      20-25
freq      1085
Name: DAYS_BIRTH_RANGE, dtype: object

```

```
df_app[["AMT_CREDIT", "AMT_INCOME_TOTAL"]]
```

	AMT_CREDIT	AMT_INCOME_TOTAL
0	406597.5	202500.0
1	1293502.5	270000.0
2	135000.0	67500.0
3	312682.5	135000.0
4	513000.0	121500.0
...	...	...
49994	1206000.0	180000.0
49995	1125000.0	126000.0
49996	900000.0	112500.0
49997	820638.0	270000.0
49998	254700.0	117000.0

```
[49999 rows x 2 columns]
```

```
df_app[["AMT_CREDIT_LAC", "AMT_INCOME_TOTAL_LAC"]]=round(df_app[["AMT_CREDIT", "AMT_INCOME_TOTAL"]]/100000,2)
```

```
df_app[["AMT_CREDIT_LAC", "AMT_INCOME_TOTAL_LAC"]]
```

	AMT_CREDIT_LAC	AMT_INCOME_TOTAL_LAC
0	4.07	2.02
1	12.94	2.70
2	1.35	0.68
3	3.13	1.35
4	5.13	1.22
...	...	...
49994	12.06	1.80
49995	11.25	1.26
49996	9.00	1.12
49997	8.21	2.70
49998	2.55	1.17

```
[49999 rows x 2 columns]
```

```
df_app[["AMT_CREDIT_LAC"]].describe()
```

	AMT_CREDIT_LAC
count	49999.000000
mean	5.996987
std	4.023996
min	0.450000
25%	2.700000
50%	5.150000
75%	8.090000
max	40.500000

```
df_app["AMT_CREDIT_LAC_RANGE"]=pd.cut(df_app["AMT_CREDIT_LAC"],bins=[0,5,10,15,20,25,30,35,40,45],labels=["0-5l","5-10l","10-15l","15-20l","20-25l","25-30l","30-35l","35-40l","45&above"])
```

```
df_app["AMT_CREDIT_LAC_RANGE"]
```

0	0-5l
1	10-15l
2	0-5l
3	0-5l
4	5-10l

	...
49994	10-15l
49995	10-15l
49996	5-10l
49997	5-10l
49998	0-5l

Name: AMT_CREDIT_LAC_RANGE, Length: 49999, dtype: category

Categories (9, object): ['0-5l' < '5-10l' < '10-15l' < '15-20l' ... '25-30l' < '30-35l' < '35-40l' < '45&above']

```
df_app["AMT_CREDIT_LAC_RANGE"].value_counts()
```

AMT_CREDIT_LAC_RANGE	
0-5l	24335
5-10l	17518
10-15l	6412
15-20l	1411
20-25l	271
25-30l	48
45&above	2
30-35l	1
35-40l	1

Name: count, dtype: int64

```
df_app["AMT_INCOME_TOTAL_LAC"].describe()
```

```
count    49999.000000
mean      1.707480
std       5.318205
min       0.260000
25%       1.120000
50%       1.460000
75%       2.020000
max       1170.000000
```

```
Name: AMT_INCOME_TOTAL_LAC, dtype: float64
```

```
df_app["AMT_INCOME_TOTAL_LAC_RANGE"]=pd.cut(df_app["AMT_INCOME_TOTAL_LAC"],bins=[0,1,2,3,4,5,6,7,8,9,10,100],labels=["0-1l","1-2l","2-3l","3-4l","4-5l","5-6l","6-7l","7-8l","8-9l","9-10l","100&above"])
```

```
df_app["AMT_INCOME_TOTAL_LAC_RANGE"]
```

```
0      2-3l
1      2-3l
2      0-1l
3      1-2l
4      1-2l
```

```
...
49994   1-2l
49995   1-2l
49996   1-2l
49997   2-3l
49998   1-2l
```

```
Name: AMT_INCOME_TOTAL_LAC_RANGE, Length: 49999, dtype: category
```

```
Categories (11, object): ['0-1l' < '1-2l' < '2-3l' < '3-4l' ... '7-8l' < '8-9l' < '9-10l' < '100&above']
```

```
df_app["AMT_INCOME_TOTAL_LAC_RANGE"].value_counts()
```

```
AMT_INCOME_TOTAL_LAC_RANGE
```

```
1-2l      25261
2-3l      10603
0-1l      10395
3-4l       2443
4-5l       843
5-6l       167
6-7l       157
8-9l        55
100&above   39
7-8l        33
9-10l        2
```

```
Name: count, dtype: int64
```

```
df_pre=pd.read_csv(r"previous_application - previous_application.csv")
df_pre
```

	SK_ID_PREV	SK_ID_CURR	NAME_CONTRACT	TYPE	AMT_ANNUITY	\
0	2030495	271877	Consumer	loans	1730.430	
1	2802425	108129	Cash	loans	25188.615	
2	2523466	122040	Cash	loans	15060.735	
3	2819243	176158	Cash	loans	47041.335	
4	1784265	202054	Cash	loans	31924.395	
...	...	...				
49994	1171956	339569	Cash	loans	NaN	
49995	1904808	363980	Cash	loans	NaN	
49996	2331005	231295	Cash	loans	22176.405	
49997	1960897	346691	Cash	loans	NaN	
49998	1979352	363244	Cash	loans	24909.390	
	AMT_APPLICATION	AMT_CREDIT	AMT_DOWN_PAYMENT	AMT_GOODS_PRICE		
\						
0	17145.0	17145.0	0.0	17145.0		
1	607500.0	679671.0	NaN	607500.0		
2	112500.0	136444.5	NaN	112500.0		
3	450000.0	470790.0	NaN	450000.0		
4	337500.0	404055.0	NaN	337500.0		
...	...	...	...	...		
49994	0.0	0.0	NaN	NaN		
49995	0.0	0.0	NaN	NaN		
49996	180000.0	216418.5	NaN	180000.0		
49997	0.0	0.0	NaN	NaN		
49998	360000.0	409896.0	NaN	360000.0		
	WEEKDAY_APPR_PROCESS_START	hour_APPR_PROCESS_START	...	\		
0	SATURDAY	15	...			
1	THURSDAY	11	...			
2	TUESDAY	11	...			
3	MONDAY	7	...			
4	THURSDAY	9	...			
...	...	...	...			
49994	THURSDAY	11	...			
49995	FRIDAY	10	...			
49996	TUESDAY	12	...			
49997	WEDNESDAY	16	...			
49998	FRIDAY	18	...			

	NAME_SELLER_INDUSTRY	CNT_PAYMENT	NAME_YIELD_GROUP	\
0	Connectivity	12.0	middle	
1	XNA	36.0	low_action	
2	XNA	12.0	high	
3	XNA	12.0	middle	
4	XNA	24.0	high	
...	...	...	...	
49994	XNA	NaN	XNA	
49995	XNA	NaN	XNA	
49996	XNA	12.0	middle	
49997	XNA	NaN	XNA	
49998	XNA	36.0	high	

	PRODUCT_COMBINATION	DAYS_FIRST_DRAWING	DAYS_FIRST_DUE	\
0	POS mobile with interest	365243.0	-42.0	
1	Cash X-Sell: low	365243.0	-134.0	
2	Cash X-Sell: high	365243.0	-271.0	
3	Cash X-Sell: middle	365243.0	-482.0	
4	Cash Street: high	NaN	NaN	
...	...	...	...	
49994	Cash	NaN	NaN	
49995	Cash	NaN	NaN	
49996	Cash X-Sell: middle	365243.0	-670.0	
49997	Cash	NaN	NaN	
49998	Cash X-Sell: high	NaN	NaN	

	DAYS_LAST_DUE_1ST_VERSION	DAYS_LAST_DUE	DAYS_TERMINATION	\
0	300.0	-42.0	-37.0	
1	916.0	365243.0	365243.0	
2	59.0	365243.0	365243.0	
3	-152.0	-182.0	-177.0	
4	NaN	NaN	NaN	
...	...	...	...	
49994	NaN	NaN	NaN	
49995	NaN	NaN	NaN	
49996	-340.0	-340.0	-338.0	
49997	NaN	NaN	NaN	
49998	NaN	NaN	NaN	

	NFLAG_INSURED_ON_APPROVAL
0	0.0
1	1.0
2	1.0
3	1.0
4	NaN
...	...
49994	NaN
49995	NaN
49996	1.0
49997	NaN

49998

NaN

[49999 rows x 37 columns]

df_pre.head()

	SK_ID_PREV	SK_ID_CURR	NAME_CONTRACT_TYPE	AMT_ANNUITY
AMT_APPLICATION \				
0	2030495	271877	Consumer loans	1730.430
17145.0				
1	2802425	108129	Cash loans	25188.615
607500.0				
2	2523466	122040	Cash loans	15060.735
112500.0				
3	2819243	176158	Cash loans	47041.335
450000.0				
4	1784265	202054	Cash loans	31924.395
337500.0				

	AMT_CREDIT	AMT_DOWN_PAYMENT	AMT_GOODS_PRICE
WEEKDAY_APPR_PROCESS_START \			
0	17145.0	0.0	17145.0
SATURDAY			
1	679671.0	NaN	607500.0
THURSDAY			
2	136444.5	NaN	112500.0
TUESDAY			
3	470790.0	NaN	450000.0
MONDAY			
4	404055.0	NaN	337500.0
THURSDAY			

	HOUR_APPR_PROCESS_START	...	NAME_SELLER_INDUSTRY	CNT_PAYMENT	\
0	15	...	Connectivity	12.0	
1	11	...	XNA	36.0	
2	11	...	XNA	12.0	
3	7	...	XNA	12.0	
4	9	...	XNA	24.0	

	NAME_YIELD_GROUP	PRODUCT_COMBINATION	DAYS_FIRST_DRAWING	\
0	middle	POS mobile with interest	365243.0	
1	low_action	Cash X-Sell: low	365243.0	
2	high	Cash X-Sell: high	365243.0	
3	middle	Cash X-Sell: middle	365243.0	
4	high	Cash Street: high	NaN	

	DAYS_FIRST_DUE	DAYS_LAST_DUE_1ST_VERSION	DAYS_LAST_DUE	
DAYS_TERMINATION \				
0	-42.0	300.0	-42.0	-
37.0				

1	-134.0	916.0	365243.0
365243.0			
2	-271.0	59.0	365243.0
365243.0			
3	-482.0	-152.0	-182.0
177.0			-
4	NaN	NaN	NaN
NaN			

NFLAG_INSURED_ON_APPROVAL	
0	0.0
1	1.0
2	1.0
3	1.0
4	NaN

[5 rows x 37 columns]

df_pre.tail()

	SK_ID_PREV	SK_ID_CURR	NAME_CONTRACT_TYPE	AMT_ANNUITY	\
49994	1171956	339569	Cash loans	NaN	
49995	1904808	363980	Cash loans	NaN	
49996	2331005	231295	Cash loans	22176.405	
49997	1960897	346691	Cash loans	NaN	
49998	1979352	363244	Cash loans	24909.390	

	AMT_APPLICATION	AMT_CREDIT	AMT_DOWN_PAYMENT	AMT_GOODS_PRICE	\
49994	0.0	0.0	NaN	NaN	
49995	0.0	0.0	NaN	NaN	
49996	180000.0	216418.5	NaN	180000.0	
49997	0.0	0.0	NaN	NaN	
49998	360000.0	409896.0	NaN	360000.0	

	WEEKDAY_APPR_PROCESS_START	HOUR_APPR_PROCESS_START	...	\
49994	THURSDAY	11	...	
49995	FRIDAY	10	...	
49996	TUESDAY	12	...	
49997	WEDNESDAY	16	...	
49998	FRIDAY	18	...	

	NAME_SELLER_INDUSTRY	CNT_PAYMENT	NAME_YIELD_GROUP	\
49994	XNA	NaN	XNA	
49995	XNA	NaN	XNA	
49996	XNA	12.0	middle	

49997	XNA	NaN	XNA
49998	XNA	36.0	high
	PRODUCT_COMBINATION	DAYS_FIRST_DRAWING	DAYS_FIRST_DUE \
49994	Cash	NaN	NaN
49995	Cash	NaN	NaN
49996	Cash X-Sell: middle	365243.0	-670.0
49997	Cash	NaN	NaN
49998	Cash X-Sell: high	NaN	NaN
	DAYS_LAST_DUE_1ST_VERSION	DAYS_LAST_DUE	DAYS_TERMINATION \
49994	NaN	NaN	NaN
49995	NaN	NaN	NaN
49996	-340.0	-340.0	-338.0
49997	NaN	NaN	NaN
49998	NaN	NaN	NaN
	NFLAG_INSURED_ON_APPROVAL		
49994	NaN		
49995	NaN		
49996	1.0		
49997	NaN		
49998	NaN		

[5 rows x 37 columns]

df_pre.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 49999 entries, 0 to 49998
Data columns (total 37 columns):
```

#	Column	Non-Null Count	Dtype
---	-----	-----	-----
0	SK_ID_PREV	49999 non-null	int64
1	SK_ID_CURR	49999 non-null	int64
2	NAME_CONTRACT_TYPE	49999 non-null	object
3	AMT_ANNUITY	39407 non-null	float64
4	AMT_APPLICATION	49999 non-null	float64
5	AMT_CREDIT	49999 non-null	float64
6	AMT_DOWN_PAYMENT	24801 non-null	float64
7	AMT_GOODS_PRICE	39255 non-null	float64
8	WEEKDAY_APPR_PROCESS_START	49999 non-null	object
9	HOUR_APPR_PROCESS_START	49999 non-null	int64
10	FLAG_LAST_APPL_PER_CONTRACT	49999 non-null	object
11	NFLAG_LAST_APPL_IN_DAY	49999 non-null	int64
12	RATE_DOWN_PAYMENT	24801 non-null	float64
13	RATE_INTEREST_PRIMARY	165 non-null	float64
14	RATE_INTEREST_PRIVILEGED	165 non-null	float64
15	NAME_CASH_LOAN_PURPOSE	49999 non-null	object
16	NAME_CONTRACT_STATUS	49999 non-null	object

```

17 DAYS_DECISION 49999 non-null int64
18 NAME_PAYMENT_TYPE 49999 non-null object
19 CODE_REJECT_REASON 49999 non-null object
20 NAME_TYPE_SUITE 25756 non-null object
21 NAME_CLIENT_TYPE 49999 non-null object
22 NAME_GOODS_CATEGORY 49999 non-null object
23 NAME_PORTFOLIO 49999 non-null object
24 NAME_PRODUCT_TYPE 49999 non-null object
25 CHANNEL_TYPE 49999 non-null object
26 SELLERPLACE_AREA 49999 non-null int64
27 NAME_SELLER_INDUSTRY 49999 non-null object
28 CNT_PAYMENT 39407 non-null float64
29 NAME_YIELD_GROUP 49999 non-null object
30 PRODUCT_COMBINATION 49991 non-null object
31 DAYS_FIRST_DRAWING 30839 non-null float64
32 DAYS_FIRST_DUE 30839 non-null float64
33 DAYS_LAST_DUE_1ST_VERSION 30839 non-null float64
34 DAYS_LAST_DUE 30839 non-null float64
35 DAYS_TERMINATION 30839 non-null float64
36 NFLAG_INSURED_ON_APPROVAL 30839 non-null float64
dtypes: float64(15), int64(6), object(16)
memory usage: 14.1+ MB

```

```
df_pre.describe()
```

	SK_ID_PREV	SK_ID_CURR	AMT_ANNUITY	AMT_APPLICATION \
count	4.999900e+04	49999.000000	39407.000000	4.999900e+04
mean	1.922254e+06	278983.187604	15482.596847	1.688925e+05
std	5.351980e+05	102780.124434	14530.971854	2.822035e+05
min	1.000001e+06	100007.000000	0.000000	0.000000e+00
25%	1.457920e+06	189919.500000	6122.835000	2.204550e+04
50%	1.920889e+06	279264.000000	10879.920000	7.155000e+04
75%	2.388632e+06	368527.500000	19669.140000	1.800000e+05
max	2.845367e+06	456254.000000	234478.395000	3.826372e+06

	AMT_CREDIT	AMT_DOWN_PAYMENT	AMT_GOODS_PRICE \
count	4.999900e+04	2.480100e+04	3.925500e+04
mean	1.885429e+05	6.557571e+03	2.151414e+05
std	3.084736e+05	1.744458e+04	3.024993e+05
min	0.000000e+00	0.000000e+00	0.000000e+00
25%	2.605500e+04	0.000000e+00	4.941000e+04
50%	7.890750e+04	1.566000e+03	1.040175e+05
75%	1.981058e+05	7.875000e+03	2.250000e+05
max	4.104351e+06	1.035000e+06	3.826372e+06

	HOUR_APPR_PROCESS_START	NFLAG_LAST_APPL_IN_DAY
RATE_DOWN_PAYMENT \		
count	49999.000000	49999.000000
24801.000000		
mean	12.478330	0.996500

0.079083		
std	3.333012	0.059058
0.107658		
min	0.000000	0.000000
0.000000		
25%	10.000000	1.000000
0.000000		
50%	12.000000	1.000000
0.049732		
75%	15.000000	1.000000
0.108909		
max	23.000000	1.000000
0.944776		

	...	RATE_INTEREST_PRIVILEGED	DAYS_DECISION	SELLERPLACE_AREA
\				
count	...	165.000000	49999.000000	4.999900e+04
mean	...	0.787674	-900.112622	4.016558e+02
std	...	0.091985	786.531303	1.793772e+04
min	...	0.424419	-2922.000000	-1.000000e+00
25%	...	0.715645	-1335.000000	-1.000000e+00
50%	...	0.835095	-599.000000	1.000000e+01
75%	...	0.852537	-292.000000	1.000000e+02
max	...	0.867336	-2.000000	4.000000e+06

	CNT_PAYMENT	DAYS_FIRST_DRAWING	DAYS_FIRST_DUE	\
count	39407.000000	30839.000000	30839.000000	
mean	15.555891	344485.142806	14217.240150	
std	13.985174	84683.650627	73348.984383	
min	0.000000	-2910.000000	-2891.000000	
25%	6.000000	365243.000000	-1642.000000	
50%	12.000000	365243.000000	-822.000000	
75%	18.000000	365243.000000	-404.000000	
max	60.000000	365243.000000	365243.000000	

	DAYS_LAST_DUE_1ST_VERSION	DAYS_LAST_DUE	DAYS_TERMINATION	\
count	30839.000000	30839.000000	30839.000000	
mean	31528.148611	76724.982101	81666.162586	
std	103691.881189	149757.893751	153101.159809	
min	-2800.000000	-2850.000000	-2844.000000	
25%	-1270.000000	-1337.000000	-1293.000000	
50%	-366.000000	-536.000000	-500.000000	

75%	113.000000	-71.000000	-45.000000
max	365243.000000	365243.000000	365243.000000

	NFLAG_INSURED_ON_APPROVAL
count	30839.000000
mean	0.322352
std	0.467384
min	0.000000
25%	0.000000
50%	0.000000
75%	1.000000
max	1.000000

[8 rows x 21 columns]

df_pre.shape

(49999, 37)

df_pre.size

1849963

df_pre.index

RangeIndex(start=0, stop=49999, step=1)

df_pre.columns

Index(['SK_ID_PREV', 'SK_ID_CURR', 'NAME_CONTRACT_TYPE',
'AMT_ANNUITY',
'AMT_APPLICATION', 'AMT_CREDIT', 'AMT_DOWN_PAYMENT',
'AMT_GOODS_PRICE',
'WEEKDAY_APPR_PROCESS_START', 'HOUR_APPR_PROCESS_START',
'FLAG_LAST_APPL_PER_CONTRACT', 'NFLAG_LAST_APPL_IN_DAY',
'RATE_DOWN_PAYMENT', 'RATE_INTEREST_PRIMARY',
'RATE_INTEREST_PRIVILEGED', 'NAME_CASH_LOAN_PURPOSE',
'NAME_CONTRACT_STATUS', 'DAYS_DECISION', 'NAME_PAYMENT_TYPE',
'CODE_REJECT_REASON', 'NAME_TYPE_SUITE', 'NAME_CLIENT_TYPE',
'NAME_GOODS_CATEGORY', 'NAME_PORTFOLIO', 'NAME_PRODUCT_TYPE',
'CHANNEL_TYPE', 'SELLERPLACE_AREA', 'NAME_SELLER_INDUSTRY',
'CNT_PAYMENT', 'NAME_YIELD_GROUP', 'PRODUCT_COMBINATION',
'DAYS_FIRST_DRAWING', 'DAYS_FIRST_DUE',
'DAYS_LAST_DUE_1ST_VERSION',
'DAYS_LAST_DUE', 'DAYS_TERMINATION',
'NFLAG_INSURED_ON_APPROVAL'],
dtype='object')

df_pre.columns.to_list()

['SK_ID_PREV',
'SK_ID_CURR',

```

'NAME_CONTRACT_TYPE',
'AMT_ANNUITY',
'AMT_APPLICATION',
'AMT_CREDIT',
'AMT_DOWN_PAYMENT',
'AMT_GOODS_PRICE',
'WEEKDAY_APPR_PROCESS_START',
'HOUR_APPR_PROCESS_START',
'FLAG_LAST_APPL_PER_CONTRACT',
'NFLAG_LAST_APPL_IN_DAY',
'RATE_DOWN_PAYMENT',
'RATE_INTEREST_PRIMARY',
'RATE_INTEREST_PRIVILEGED',
'NAME_CASH_LOAN_PURPOSE',
'NAME_CONTRACT_STATUS',
'DAYS_DECISION',
'NAME_PAYMENT_TYPE',
'CODE_REJECT_REASON',
'NAME_TYPE_SUITE',
'NAME_CLIENT_TYPE',
'NAME_GOODS_CATEGORY',
'NAME_PORTFOLIO',
'NAME_PRODUCT_TYPE',
'CHANNEL_TYPE',
'SELLERPLACE_AREA',
'NAME_SELLER_INDUSTRY',
'CNT_PAYMENT',
'NAME_YIELD_GROUP',
'PRODUCT_COMBINATION',
'DAYS_FIRST_DRAWING',
'DAYS_FIRST_DUE',
'DAYS_LAST_DUE_1ST_VERSION',
'DAYS_LAST_DUE',
'DAYS_TERMINATION',
'NFLAG_INSURED_ON_APPROVAL']

```

```
df_pre.describe().columns
```

```

Index(['SK_ID_PREV', 'SK_ID_CURR', 'AMT_ANNUITY', 'AMT_APPLICATION',
      'AMT_CREDIT', 'AMT_DOWN_PAYMENT', 'AMT_GOODS_PRICE',
      'HOUR_APPR_PROCESS_START', 'NFLAG_LAST_APPL_IN_DAY',
      'RATE_DOWN_PAYMENT', 'RATE_INTEREST_PRIMARY',
      'RATE_INTEREST_PRIVILEGED', 'DAYS_DECISION',
      'SELLERPLACE_AREA',
      'CNT_PAYMENT', 'DAYS_FIRST_DRAWING', 'DAYS_FIRST_DUE',
      'DAYS_LAST_DUE_1ST_VERSION', 'DAYS_LAST_DUE',
      'DAYS_TERMINATION',
      'NFLAG_INSURED_ON_APPROVAL'],
      dtype='object')

```

```
list(set(df_pre.columns)-set(df_pre.describe().columns))
```

```
['NAME_TYPE_SUITE',
 'NAME_CONTRACT_STATUS',
 'NAME_CONTRACT_TYPE',
 'NAME_PAYMENT_TYPE',
 'WEEKDAY_APPR_PROCESS_START',
 'NAME_CASH_LOAN_PURPOSE',
 'NAME_GOODS_CATEGORY',
 'CODE_REJECT_REASON',
 'PRODUCT_COMBINATION',
 'NAME_CLIENT_TYPE',
 'NAME_SELLER_INDUSTRY',
 'CHANNEL_TYPE',
 'NAME_PORTFOLIO',
 'NAME_PRODUCT_TYPE',
 'FLAG_LAST_APPL_PER_CONTRACT',
 'NAME_YIELD_GROUP']
```

df_pre

	SK_ID_PREV	SK_ID_CURR	NAME_CONTRACT_TYPE	AMT_ANNUITY \
0	2030495	271877	Consumer loans	1730.430
1	2802425	108129	Cash loans	25188.615
2	2523466	122040	Cash loans	15060.735
3	2819243	176158	Cash loans	47041.335
4	1784265	202054	Cash loans	31924.395
...	...	...	...	...
49994	1171956	339569	Cash loans	NaN
49995	1904808	363980	Cash loans	NaN
49996	2331005	231295	Cash loans	22176.405
49997	1960897	346691	Cash loans	NaN
49998	1979352	363244	Cash loans	24909.390

	AMT_APPLICATION	AMT_CREDIT	AMT_DOWN_PAYMENT	AMT_GOODS_PRICE
\				
0	17145.0	17145.0	0.0	17145.0
1	607500.0	679671.0	NaN	607500.0
2	112500.0	136444.5	NaN	112500.0
3	450000.0	470790.0	NaN	450000.0
4	337500.0	404055.0	NaN	337500.0
...	...	...	...	...
49994	0.0	0.0	NaN	NaN
49995	0.0	0.0	NaN	NaN

49996	180000.0	216418.5	NaN	180000.0
49997	0.0	0.0	NaN	NaN
49998	360000.0	409896.0	NaN	360000.0
	WEEKDAY_APPR_PROCESS_START	HOURL_APPR_PROCESS_START	...	\
0	SATURDAY	15	...	
1	THURSDAY	11	...	
2	TUESDAY	11	...	
3	MONDAY	7	...	
4	THURSDAY	9	...	
...	...	...	...	
49994	THURSDAY	11	...	
49995	FRIDAY	10	...	
49996	TUESDAY	12	...	
49997	WEDNESDAY	16	...	
49998	FRIDAY	18	...	
	NAME_SELLER_INDUSTRY	CNT_PAYMENT	NAME_YIELD_GROUP	\
0	Connectivity	12.0	middle	
1	XNA	36.0	low_action	
2	XNA	12.0	high	
3	XNA	12.0	middle	
4	XNA	24.0	high	
...	...	...	...	
49994	XNA	NaN	XNA	
49995	XNA	NaN	XNA	
49996	XNA	12.0	middle	
49997	XNA	NaN	XNA	
49998	XNA	36.0	high	
	PRODUCT_COMBINATION	DAYS_FIRST_DRAWING	DAYS_FIRST_DUE	\
0	POS mobile with interest	365243.0	-42.0	
1	Cash X-Sell: low	365243.0	-134.0	
2	Cash X-Sell: high	365243.0	-271.0	
3	Cash X-Sell: middle	365243.0	-482.0	
4	Cash Street: high	NaN	NaN	
...	...	...	...	
49994	Cash	NaN	NaN	
49995	Cash	NaN	NaN	
49996	Cash X-Sell: middle	365243.0	-670.0	
49997	Cash	NaN	NaN	
49998	Cash X-Sell: high	NaN	NaN	
	DAYS_LAST_DUE_1ST_VERSION	DAYS_LAST_DUE	DAYS_TERMINATION	\
0	300.0	-42.0	-37.0	
1	916.0	365243.0	365243.0	

2	59.0	365243.0	365243.0
3	-152.0	-182.0	-177.0
4	NaN	NaN	NaN
...	...	...	...
49994	NaN	NaN	NaN
49995	NaN	NaN	NaN
49996	-340.0	-340.0	-338.0
49997	NaN	NaN	NaN
49998	NaN	NaN	NaN

	NFLAG_INSURED_ON_APPROVAL
0	0.0
1	1.0
2	1.0
3	1.0
4	NaN
...	...
49994	NaN
49995	NaN
49996	1.0
49997	NaN
49998	NaN

[49999 rows x 37 columns]

missing_value(df_pre)

RATE_INTEREST_PRIVILEGED	99.67
RATE_INTEREST_PRIMARY	99.67
RATE_DOWN_PAYMENT	50.40
AMT_DOWN_PAYMENT	50.40
NAME_TYPE_SUITE	48.49
NFLAG_INSURED_ON_APPROVAL	38.32
DAYS_FIRST_DRAWING	38.32
DAYS_FIRST_DUE	38.32
DAYS_LAST_DUE_1ST_VERSION	38.32
DAYS_LAST_DUE	38.32
DAYS_TERMINATION	38.32
AMT_GOODS_PRICE	21.49
AMT_ANNUITY	21.18
CNT_PAYMENT	21.18
PRODUCT_COMBINATION	0.02
CHANNEL_TYPE	0.00
NAME_PRODUCT_TYPE	0.00
NAME_YIELD_GROUP	0.00
SELLERPLACE_AREA	0.00
NAME_SELLER_INDUSTRY	0.00
NAME_GOODS_CATEGORY	0.00
NAME_PORTFOLIO	0.00
SK_ID_PREV	0.00


```

NAME_CLIENT_TYPE      0.00
CODE_REJECT_REASON    0.00
SK_ID_CURR             0.00
DAYS_DECISION          0.00
NAME_CONTRACT_STATUS  0.00
NAME_CASH_LOAN_PURPOSE 0.00
NFLAG_LAST_APPL_IN_DAY 0.00
FLAG_LAST_APPL_PER_CONTRACT 0.00
HOUR_APPR_PROCESS_START 0.00
WEEKDAY_APPR_PROCESS_START 0.00
AMT_CREDIT             0.00
AMT_APPLICATION        0.00
NAME_CONTRACT_TYPE     0.00
NAME_PAYMENT_TYPE      0.00
dtype: float64

```

```

nullpre=missing_value(df_pre)[missing_value(df_pre)>50]

```

```

nullpre.index

```

```

Index(['RATE_INTEREST_PRIVILEGED', 'RATE_INTEREST_PRIMARY',
      'RATE_DOWN_PAYMENT', 'AMT_DOWN_PAYMENT'],
      dtype='object')

```

```

df_pre.drop(columns=nullpre.index,inplace=True)

```

```

df_pre

```

	SK_ID_PREV	SK_ID_CURR	NAME_CONTRACT_TYPE	AMT_ANNUITY \
0	2030495	271877	Consumer loans	1730.430
1	2802425	108129	Cash loans	25188.615
2	2523466	122040	Cash loans	15060.735
3	2819243	176158	Cash loans	47041.335
4	1784265	202054	Cash loans	31924.395
...	...	...	...	...
49994	1171956	339569	Cash loans	NaN
49995	1904808	363980	Cash loans	NaN
49996	2331005	231295	Cash loans	22176.405
49997	1960897	346691	Cash loans	NaN
49998	1979352	363244	Cash loans	24909.390

	AMT_APPLICATION	AMT_CREDIT	AMT_GOODS_PRICE \
0	17145.0	17145.0	17145.0
1	607500.0	679671.0	607500.0
2	112500.0	136444.5	112500.0
3	450000.0	470790.0	450000.0
4	337500.0	404055.0	337500.0
...	...	...	...
49994	0.0	0.0	NaN
49995	0.0	0.0	NaN
49996	180000.0	216418.5	180000.0

49997	0.0	0.0	NaN
49998	360000.0	409896.0	360000.0
	WEEKDAY_APPR_PROCESS_START	hour_APPR_PROCESS_START	\
0	SATURDAY	15	
1	THURSDAY	11	
2	TUESDAY	11	
3	MONDAY	7	
4	THURSDAY	9	
...	...	...	
49994	THURSDAY	11	
49995	FRIDAY	10	
49996	TUESDAY	12	
49997	WEDNESDAY	16	
49998	FRIDAY	18	
	FLAG_LAST_APPL_PER_CONTRACT	...	NAME_SELLER_INDUSTRY
CNT_PAYMENT	\		
0	Y	...	Connectivity
12.0			
1	Y	...	XNA
36.0			
2	Y	...	XNA
12.0			
3	Y	...	XNA
12.0			
4	Y	...	XNA
24.0			
...	...	...	...
..			
49994	Y	...	XNA
NaN			
49995	Y	...	XNA
NaN			
49996	Y	...	XNA
12.0			
49997	Y	...	XNA
NaN			
49998	Y	...	XNA
36.0			
	NAME_YIELD_GROUP	PRODUCT_COMBINATION	DAYS_FIRST_DRAWING \
0	middle	POS mobile with interest	365243.0
1	low_action	Cash X-Sell: low	365243.0
2	high	Cash X-Sell: high	365243.0
3	middle	Cash X-Sell: middle	365243.0
4	high	Cash Street: high	NaN
...	...	...	...
49994	XNA	Cash	NaN
49995	XNA	Cash	NaN

49996	middle	Cash X-Sell: middle	365243.0
49997	XNA	Cash	NaN
49998	high	Cash X-Sell: high	NaN

	DAYS_FIRST_DUE	DAYS_LAST_DUE_1ST_VERSION	DAYS_LAST_DUE
DAYS_TERMINATION \			
0	-42.0	300.0	-42.0
-37.0			
1	-134.0	916.0	365243.0
365243.0			
2	-271.0	59.0	365243.0
365243.0			
3	-482.0	-152.0	-182.0
-177.0			
4	NaN	NaN	NaN
NaN			
...	...	...	...
...			
49994	NaN	NaN	NaN
NaN			
49995	NaN	NaN	NaN
NaN			
49996	-670.0	-340.0	-340.0
-338.0			
49997	NaN	NaN	NaN
NaN			
49998	NaN	NaN	NaN
NaN			

	NFLAG_INSURED_ON_APPROVAL
0	0.0
1	1.0
2	1.0
3	1.0
4	NaN
...	...
49994	NaN
49995	NaN
49996	1.0
49997	NaN
49998	NaN

[49999 rows x 33 columns]

```
df_pre["NAME_TYPE_SUITE"].isnull().sum()
```

24243

```
df_pre["NAME_TYPE_SUITE"]
```

```

0          NaN
1    Unaccompanied
2    Spouse, partner
3          NaN
4          NaN
...
49994      NaN
49995      NaN
49996      NaN
49997      NaN
49998      NaN
Name: NAME_TYPE_SUITE, Length: 49999, dtype: object

```

```
df_pre["NAME_TYPE_SUITE"]
```

```

0          NaN
1    Unaccompanied
2    Spouse, partner
3          NaN
4          NaN
...
49994      NaN
49995      NaN
49996      NaN
49997      NaN
49998      NaN
Name: NAME_TYPE_SUITE, Length: 49999, dtype: object

```

```
df_pre["NAME_TYPE_SUITE"]=df_pre["NAME_TYPE_SUITE"].fillna(df_pre["NAME_TYPE_SUITE"].mode()[0])
```

```
df_pre["NAME_TYPE_SUITE"].isnull().sum()
```

```
0
```

```
df_pre["AMT_GOODS_PRICE"]
```

```

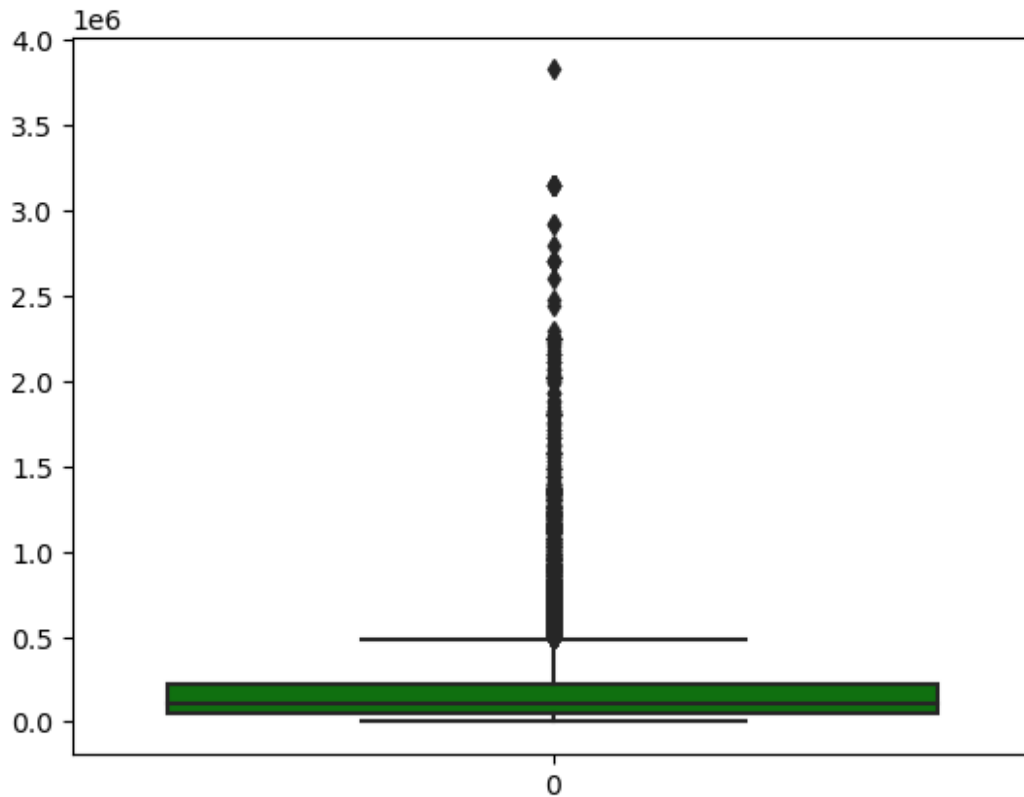
0      17145.0
1     607500.0
2     112500.0
3     450000.0
4     337500.0
...
49994      NaN
49995      NaN
49996     180000.0
49997      NaN
49998     360000.0
Name: AMT_GOODS_PRICE, Length: 49999, dtype: float64

```

```
df_pre["AMT_GOODS_PRICE"].isnull().sum()
```

10744

```
sns.boxplot(df_pre["AMT_GOODS_PRICE"],color="green")  
plt.show()
```



```
df_pre["AMT_GOODS_PRICE"]=df_pre["AMT_GOODS_PRICE"].fillna(df_pre["AMT_GOODS_PRICE"]==df_pre["AMT_CREDIT"])
```

```
df_pre["AMT_GOODS_PRICE"].isnull().sum()
```

0

```
df_pre["AMT_ANNUITY"]
```

0	1730.430
1	25188.615
2	15060.735
3	47041.335
4	31924.395
	...
49994	NaN
49995	NaN
49996	22176.405
49997	NaN

```

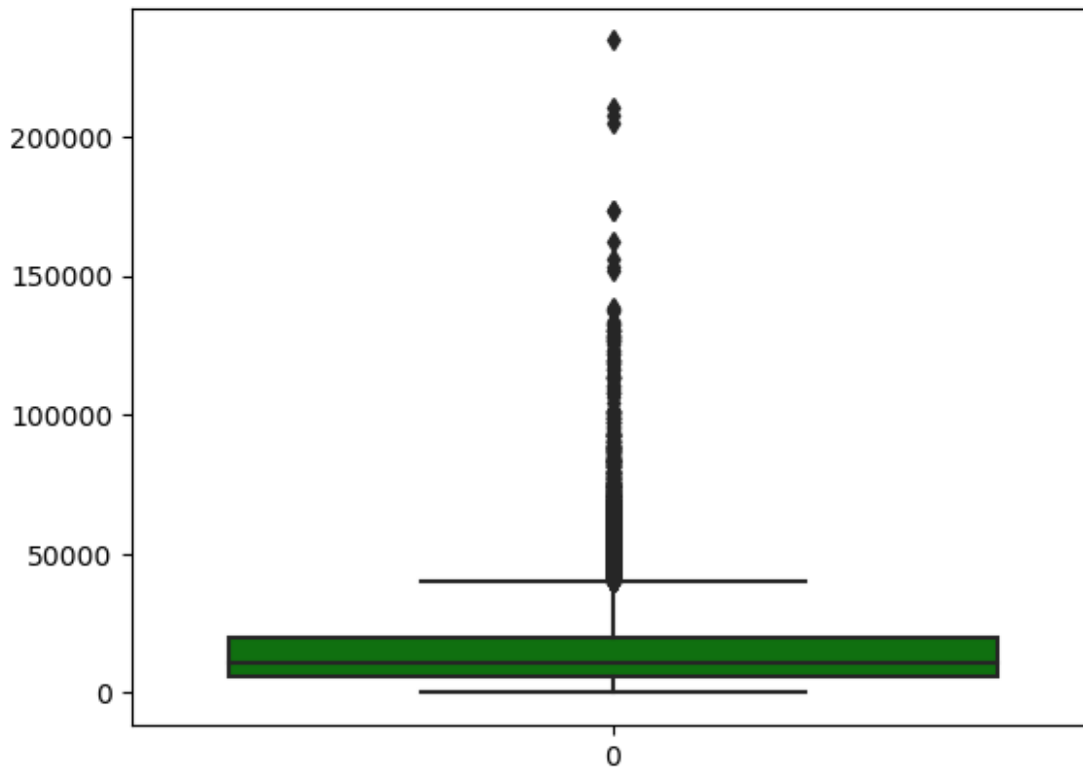
49998      24909.390
Name: AMT_ANNUIITY, Length: 49999, dtype: float64

df_pre["AMT_ANNUIITY"].isnull().sum()

10592

sns.boxplot(df_pre["AMT_ANNUIITY"],color="green")
plt.show()

```



```

df_pre["AMT_ANNUIITY"]=df_pre["AMT_ANNUIITY"].fillna(df_pre["AMT_ANNUIITY"]
).median())

df_pre["AMT_ANNUIITY"].isnull().sum()

0

df_pre["PRODUCT_COMBINATION"]

0      POS mobile with interest
1      Cash X-Sell: low
2      Cash X-Sell: high
3      Cash X-Sell: middle
4      Cash Street: high
...
49994      ...      Cash

```

```
49995          Cash
49996      Cash X-Sell: middle
49997          Cash
49998      Cash X-Sell: high
Name: PRODUCT_COMBINATION, Length: 49999, dtype: object
```

```
df_pre["PRODUCT_COMBINATION"].isnull().sum()
```

```
8
```

```
df_pre["PRODUCT_COMBINATION"].value_counts()
```

```
PRODUCT_COMBINATION
POS household with interest    8510
Cash                          7939
POS mobile with interest      7029
Cash X-Sell: middle           3953
Cash X-Sell: low              3539
Card Street                   3323
POS industry with interest    3231
POS household without interest 2799
Card X-Sell                   2302
Cash Street: high             1752
Cash X-Sell: high             1657
Cash Street: low              1056
Cash Street: middle           960
POS mobile without interest    731
POS other with interest        728
POS industry without interest  390
POS others without interest    92
Name: count, dtype: int64
```

```
df_pre["PRODUCT_COMBINATION"] = df_pre["PRODUCT_COMBINATION"].fillna(df_pre["PRODUCT_COMBINATION"].mode()[0])
```

```
df_pre["PRODUCT_COMBINATION"].isnull().sum()
```

```
0
```

```
df_pre["CNT_PAYMENT"]
```

```
0      12.0
1      36.0
2      12.0
3      12.0
4      24.0
...
49994   NaN
49995   NaN
49996   12.0
49997   NaN
```

```

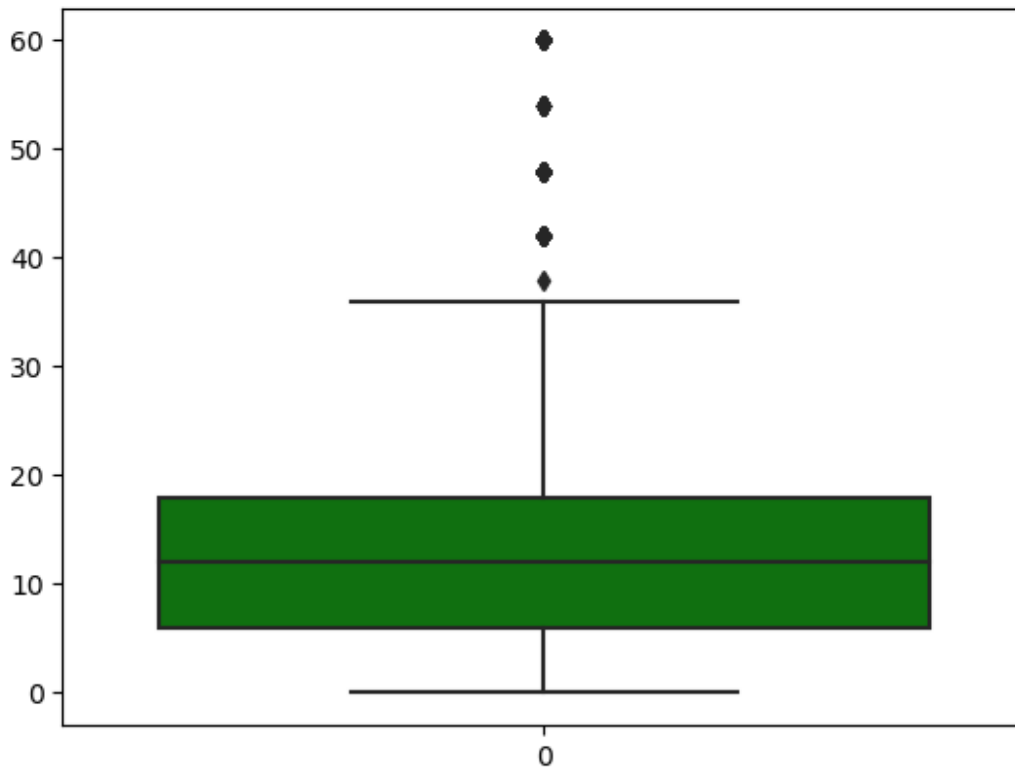
49998      36.0
Name: CNT_PAYMENT, Length: 49999, dtype: float64

df_pre["CNT_PAYMENT"].isnull().sum()

10592

sns.boxplot(df_pre["CNT_PAYMENT"],color="green")
plt.show()

```



```

df_pre["CNT_PAYMENT"]=df_pre["CNT_PAYMENT"].fillna(df_pre["CNT_PAYMENT"]
"].median())

df_pre["CNT_PAYMENT"].isnull().sum()

0

df_pre[["AMT_ANNUITY","AMT_CREDIT","AMT_APPLICATION"]]

```

	AMT_ANNUITY	AMT_CREDIT	AMT_APPLICATION
0	1730.430	17145.0	17145.0
1	25188.615	679671.0	607500.0
2	15060.735	136444.5	112500.0
3	47041.335	470790.0	450000.0
4	31924.395	404055.0	337500.0
...	...	...	...

49994	10879.920	0.0	0.0
49995	10879.920	0.0	0.0
49996	22176.405	216418.5	180000.0
49997	10879.920	0.0	0.0
49998	24909.390	409896.0	360000.0

[49999 rows x 3 columns]

```
df_pre[["AMT_ANNUITY_LAC", "AMT_CREDIT_LAC", "AMT_APPLICATION_LAC"]]=round(df_pre[["AMT_ANNUITY", "AMT_CREDIT", "AMT_APPLICATION"]]/100000,2)
```

```
df_pre[["AMT_ANNUITY_LAC", "AMT_CREDIT_LAC", "AMT_APPLICATION_LAC"]]
```

	AMT_ANNUITY_LAC	AMT_CREDIT_LAC	AMT_APPLICATION_LAC
0	0.02	0.17	0.17
1	0.25	6.80	6.08
2	0.15	1.36	1.12
3	0.47	4.71	4.50
4	0.32	4.04	3.38
...	...	...	...
49994	0.11	0.00	0.00
49995	0.11	0.00	0.00
49996	0.22	2.16	1.80
49997	0.11	0.00	0.00
49998	0.25	4.10	3.60

[49999 rows x 3 columns]

```
list(set(df_pre.columns)-set(df_pre.describe().columns))
```

```
[ 'NAME_TYPE_SUITE',
'NAME_CONTRACT_STATUS',
'NAME_CONTRACT_TYPE',
'NAME_PAYMENT_TYPE',
'WEEKDAY_APPR_PROCESS_START',
'NAME_CASH_LOAN_PURPOSE',
'NAME_GOODS_CATEGORY',
'AMT_GOODS_PRICE',
'CODE_REJECT_REASON',
'PRODUCT_COMBINATION',
'NAME_CLIENT_TYPE',
'NAME_SELLER_INDUSTRY',
'CHANNEL_TYPE',
'NAME_PORTFOLIO',
'NAME_PRODUCT_TYPE',
'FLAG_LAST_APPL_PER_CONTRACT',
'NAME_YIELD_GROUP']
```

```
df_pre.describe().columns
```

```
Index(['SK_ID_PREV', 'SK_ID_CURR', 'AMT_ANNUITY', 'AMT_APPLICATION',
      'AMT_CREDIT', 'HOUR_APPR_PROCESS_START',
      'NFLAG_LAST_APPL_IN_DAY',
      'DAYS_DECISION', 'SELLERPLACE_AREA', 'CNT_PAYMENT',
      'DAYS_FIRST_DRAWING', 'DAYS_FIRST_DUE',
      'DAYS_LAST_DUE_1ST_VERSION',
      'DAYS_LAST_DUE', 'DAYS_TERMINATION',
      'NFLAG_INSURED_ON_APPROVAL',
      'AMT_ANNUITY_LAC', 'AMT_CREDIT_LAC', 'AMT_APPLICATION_LAC'],
      dtype='object')
```

```
df_pre[['DAYS_DECISION', 'DAYS_FIRST_DRAWING', 'DAYS_FIRST_DUE',
        'DAYS_LAST_DUE_1ST_VERSION', 'DAYS_LAST_DUE', 'DAYS_TERMINATION']]
```

	DAYS_DECISION	DAYS_FIRST_DRAWING	DAYS_FIRST_DUE \
0	-73	365243.0	-42.0
1	-164	365243.0	-134.0
2	-301	365243.0	-271.0
3	-512	365243.0	-482.0
4	-781	NaN	NaN
...	...	...	...
49994	-24	NaN	NaN
49995	-425	NaN	NaN
49996	-700	365243.0	-670.0
49997	-13	NaN	NaN
49998	-441	NaN	NaN

	DAYS_LAST_DUE_1ST_VERSION	DAYS_LAST_DUE	DAYS_TERMINATION
0	300.0	-42.0	-37.0
1	916.0	365243.0	365243.0
2	59.0	365243.0	365243.0
3	-152.0	-182.0	-177.0
4	NaN	NaN	NaN
...	...	...	...
49994	NaN	NaN	NaN
49995	NaN	NaN	NaN
49996	-340.0	-340.0	-338.0
49997	NaN	NaN	NaN
49998	NaN	NaN	NaN

```
[49999 rows x 6 columns]
```

```
df_pre[['DAYS_DECISION', 'DAYS_FIRST_DRAWING', 'DAYS_FIRST_DUE',
        'DAYS_LAST_DUE_1ST_VERSION', 'DAYS_LAST_DUE',
        'DAYS_TERMINATION']] = abs(df_pre[['DAYS_DECISION', 'DAYS_FIRST_DRAWING',
        'DAYS_FIRST_DUE', 'DAYS_LAST_DUE_1ST_VERSION', 'DAYS_LAST_DUE',
        'DAYS_TERMINATION']])
```

```
df_pre[['DAYS_DECISION', 'DAYS_FIRST_DRAWING', 'DAYS_FIRST_DUE',
        'DAYS_LAST_DUE_1ST_VERSION', 'DAYS_LAST_DUE', 'DAYS_TERMINATION']]
```

	DAYS_DECISION	DAYS_FIRST_DRAWING	DAYS_FIRST_DUE	\
0	73	365243.0	42.0	
1	164	365243.0	134.0	
2	301	365243.0	271.0	
3	512	365243.0	482.0	
4	781	NaN	NaN	
...	...	...	...	
49994	24	NaN	NaN	
49995	425	NaN	NaN	
49996	700	365243.0	670.0	
49997	13	NaN	NaN	
49998	441	NaN	NaN	

	DAYS_LAST_DUE_1ST_VERSION	DAYS_LAST_DUE	DAYS_TERMINATION
0	300.0	42.0	37.0
1	916.0	365243.0	365243.0
2	59.0	365243.0	365243.0
3	152.0	182.0	177.0
4	NaN	NaN	NaN
...	...	...	...
49994	NaN	NaN	NaN
49995	NaN	NaN	NaN
49996	340.0	340.0	338.0
49997	NaN	NaN	NaN
49998	NaN	NaN	NaN

[49999 rows x 6 columns]

```
df_pre[['DAYS_DECISION', 'DAYS_FIRST_DRAWING', 'DAYS_FIRST_DUE',
'DAYS_LAST_DUE_1ST_VERSION', 'DAYS_LAST_DUE',
'DAYS_TERMINATION']] = round(df_pre[['DAYS_DECISION', 'DAYS_FIRST_DRAWING',
'DAYS_FIRST_DUE', 'DAYS_LAST_DUE_1ST_VERSION', 'DAYS_LAST_DUE',
'DAYS_TERMINATION']] / 365, 2)
```

```
df_pre[['DAYS_DECISION', 'DAYS_FIRST_DRAWING', 'DAYS_FIRST_DUE',
'DAYS_LAST_DUE_1ST_VERSION', 'DAYS_LAST_DUE', 'DAYS_TERMINATION']]
```

	DAYS_DECISION	DAYS_FIRST_DRAWING	DAYS_FIRST_DUE	\
0	0.20	1000.67	0.12	
1	0.45	1000.67	0.37	
2	0.82	1000.67	0.74	
3	1.40	1000.67	1.32	
4	2.14	NaN	NaN	
...	...	...	...	
49994	0.07	NaN	NaN	
49995	1.16	NaN	NaN	
49996	1.92	1000.67	1.84	
49997	0.04	NaN	NaN	
49998	1.21	NaN	NaN	

	DAYS_LAST_DUE_1ST_VERSION	DAYS_LAST_DUE	DAYS_TERMINATION
0	0.82	0.12	0.10
1	2.51	1000.67	1000.67
2	0.16	1000.67	1000.67
3	0.42	0.50	0.48
4	NaN	NaN	NaN
...	...	...	...
49994	NaN	NaN	NaN
49995	NaN	NaN	NaN
49996	0.93	0.93	0.93
49997	NaN	NaN	NaN
49998	NaN	NaN	NaN

[49999 rows x 6 columns]

```
df_pre["AMT_CREDIT_LAC"].value_counts()
```

AMT_CREDIT_LAC

0.00	9435
0.45	1305
1.35	619
0.90	563
4.50	550

...	
6.77	1
9.36	1
5.48	1
8.49	1
5.86	1

Name: count, Length: 1326, dtype: int64

```
df_pre["AMT_CREDIT_LAC"].describe()
```

count	49999.000000
mean	1.885440
std	3.084702
min	0.000000
25%	0.260000
50%	0.790000
75%	1.980000
max	41.040000

Name: AMT_CREDIT_LAC, dtype: float64

```
df_pre["AMT_CREDIT_LAC_RANGE"] = pd.cut(df_pre["AMT_CREDIT_LAC"], bins=[0, 5, 10, 15, 20, 25, 30, 35, 40, 45], labels=["0-5l", "5-10l", "10-15l", "15-20l", "20-25l", "25-30l", "30-35l", "35-40l", "40&more"])
```

```
df_pre["AMT_CREDIT_LAC_RANGE"].value_counts()
```

AMT_CREDIT_LAC_RANGE

0-5l	35521
------	-------

```
5-10l      3278
10-15l     1281
15-20l      298
20-25l     137
25-30l      36
30-35l      10
35-40l       2
40&more      1
Name: count, dtype: int64
```

```
df_pre["AMT_ANNUITY_LAC"].value_counts()
```

```
AMT_ANNUITY_LAC
```

```
0.11      12515
0.07       2455
0.04       2437
0.06       2400
0.05       2289
```

```
...
1.23        1
1.25        1
2.10        1
1.27        1
1.52        1
```

```
Name: count, Length: 141, dtype: int64
```

```
df_pre["AMT_ANNUITY_LAC"].describe()
```

```
count      49999.000000
mean        0.145208
std         0.130373
min         0.000000
25%         0.070000
50%         0.110000
75%         0.160000
max         2.340000
```

```
Name: AMT_ANNUITY_LAC, dtype: float64
```

```
df_pre["AMT_APPLICATION_LAC"].value_counts()
```

```
AMT_APPLICATION_LAC
```

```
0.00      10938
0.45       1704
1.35       1287
2.25       1260
0.90       1136
```

```
...
8.89        1
9.49        1
4.45        1
6.71        1
```

```

5.19      1
Name: count, Length: 804, dtype: int64

df_pre["AMT_APPLICATION_LAC"].describe()
count      49999.000000
mean        1.688925
std         2.822091
min         0.000000
25%         0.220000
50%         0.720000
75%         1.800000
max         38.260000
Name: AMT_APPLICATION_LAC, dtype: float64

df_pre["AMT_APPLICATION_LAC_RANGE"] = pd.cut(df_pre["AMT_APPLICATION_LAC"],
bins=[0,5,10,15,20,25,30,35,40], labels=["0-5l", "5-10l", "10-15l", "15-20l", "20-25l", "25-30l", "30-35l", "35&more"])

df_pre["AMT_APPLICATION_LAC_RANGE"].value_counts()
AMT_APPLICATION_LAC_RANGE
0-5l      35014
5-10l     2586
10-15l    1149
15-20l     171
20-25l     123
25-30l       9
30-35l       8
35&more      1
Name: count, dtype: int64

df_app.groupby(["CODE_GENDER"])["AMT_ANNUITY"].mean()
CODE_GENDER
F      26264.853304
M      28718.742692
XNA    16312.500000
Name: AMT_ANNUITY, dtype: float64

df_app.groupby(["CODE_GENDER", "OCCUPATION_TYPE"])["TARGET"].count()
CODE_GENDER  OCCUPATION_TYPE
F            Accountants      1568
             Cleaning staff    690
             Cooking staff     871
             Core staff       3542
             Drivers          144
             HR staff          97
             High skill tech staff 1165
             IT staff          24

```

	Laborers	3747
	Low-skill Laborers	56
	Managers	1867
	Medicine staff	1365
	Private service staff	427
	Realty agents	109
	Sales staff	4558
	Secretaries	201
	Security staff	320
	Unknown	11876
	Waiters/barmen staff	196
M	Accountants	53
	Cleaning staff	49
	Cooking staff	92
	Core staff	892
	Drivers	2900
	HR staff	4
	High skill tech staff	687
	IT staff	56
	Laborers	5205
	Low-skill Laborers	300
	Managers	1622
	Medicine staff	38
	Private service staff	20
	Realty agents	14
	Sales staff	602
	Secretaries	11
	Security staff	820
	Unknown	3777
	Waiters/barmen staff	32
XNA	Low-skill Laborers	1
	Unknown	1

Name: TARGET, dtype: int64

df_app["TARGET"].value_counts()

TARGET

0 45973

1 4026

Name: count, dtype: int64

df_app.describe().columns.to_list()

```
[ 'SK_ID_CURR',
  'TARGET',
  'CNT_CHILDREN',
  'AMT_INCOME_TOTAL',
  'AMT_CREDIT',
  'AMT_ANNUITY',
  'REGION_POPULATION_RELATIVE',
```

'DAYS_BIRTH',
'DAYS_EMPLOYED',
'DAYS_REGISTRATION',
'DAYS_ID_PUBLISH',
'FLAG_MOBIL',
'FLAG_EMP_PHONE',
'FLAG_WORK_PHONE',
'FLAG_CONT_MOBILE',
'FLAG_PHONE',
'FLAG_EMAIL',
'CNT_FAM_MEMBERS',
'REGION_RATING_CLIENT',
'REGION_RATING_CLIENT_W_CITY',
'HOUR_APPR_PROCESS_START',
'REG_REGION_NOT_LIVE_REGION',
'REG_REGION_NOT_WORK_REGION',
'LIVE_REGION_NOT_WORK_REGION',
'REG_CITY_NOT_LIVE_CITY',
'REG_CITY_NOT_WORK_CITY',
'LIVE_CITY_NOT_WORK_CITY',
'EXT_SOURCE_2',
'EXT_SOURCE_3',
'OBS_30_CNT_SOCIAL_CIRCLE',
'DEF_30_CNT_SOCIAL_CIRCLE',
'OBS_60_CNT_SOCIAL_CIRCLE',
'DEF_60_CNT_SOCIAL_CIRCLE',
'DAYS_LAST_PHONE_CHANGE',
'FLAG_DOCUMENT_2',
'FLAG_DOCUMENT_3',
'FLAG_DOCUMENT_4',
'FLAG_DOCUMENT_5',
'FLAG_DOCUMENT_6',
'FLAG_DOCUMENT_7',
'FLAG_DOCUMENT_8',
'FLAG_DOCUMENT_9',
'FLAG_DOCUMENT_10',
'FLAG_DOCUMENT_11',
'FLAG_DOCUMENT_12',
'FLAG_DOCUMENT_13',
'FLAG_DOCUMENT_14',
'FLAG_DOCUMENT_15',
'FLAG_DOCUMENT_16',
'FLAG_DOCUMENT_17',
'FLAG_DOCUMENT_18',
'FLAG_DOCUMENT_19',
'FLAG_DOCUMENT_20',
'FLAG_DOCUMENT_21',
'AMT_REQ_CREDIT_BUREAU_HOUR',
'AMT_REQ_CREDIT_BUREAU_DAY',


```
'AMT_REQ_CREDIT_BUREAU_WEEK',
'AMT_REQ_CREDIT_BUREAU_MON',
'AMT_REQ_CREDIT_BUREAU_QRT',
'AMT_REQ_CREDIT_BUREAU_YEAR',
'DAYS_BIRTH_IN_YEAR',
'DAYS_EMPLOYED_IN_YEAR',
'AMT_CREDIT_LAC',
'AMT_INCOME_TOTAL_LAC']
```

```
FLAG_COL=['FLAG_MOBIL','FLAG_EMP_PHONE','FLAG_WORK_PHONE','FLAG_CONT_M
OBILE','FLAG_PHONE','FLAG_EMAIL']
```

```
df_app[FLAG_COL]
```

	FLAG_MOBIL	FLAG_EMP_PHONE	FLAG_WORK_PHONE	
FLAG_CONT_MOBILE \				
0	1	1	0	1
1	1	1	0	1
2	1	1	1	1
3	1	1	0	1
4	1	1	0	1
...	...	...	...	...
49994	1	1	0	1
49995	1	1	0	1
49996	1	1	0	1
49997	1	1	0	1
49998	1	0	0	1

	FLAG_PHONE	FLAG_EMAIL
0	1	0
1	1	0
2	1	0
3	0	0
4	0	0
...	...	...
49994	0	0
49995	0	0
49996	0	1
49997	0	0
49998	0	0

```
[49999 rows x 6 columns]
```

```
df_app["FLAG_MOBIL"]=df_app["FLAG_MOBIL"].apply(lambda x:"YES" if x==1  
else "NO")
```

```
df_app["FLAG_MOBIL"]
```

```
0      YES  
1      YES  
2      YES  
3      YES  
4      YES
```

```
...
```

```
49994   YES  
49995   YES  
49996   YES  
49997   YES  
49998   YES
```

```
Name: FLAG_MOBIL, Length: 49999, dtype: object
```

```
df_app['FLAG_EMP_PHONE']=df_app['FLAG_EMP_PHONE'].apply(lambda x:"YES"  
if x==1 else "NO")
```

```
df_app['FLAG_EMP_PHONE']
```

```
0      YES  
1      YES  
2      YES  
3      YES  
4      YES
```

```
...
```

```
49994   YES  
49995   YES  
49996   YES  
49997   YES  
49998   NO
```

```
Name: FLAG_EMP_PHONE, Length: 49999, dtype: object
```

```
df_app['FLAG_WORK_PHONE']=df_app['FLAG_WORK_PHONE'].apply(lambda  
x:"YES" if x==1 else "NO")
```

```
df_app['FLAG_WORK_PHONE']
```

```
0      NO  
1      NO  
2      YES  
3      NO  
4      NO
```

```
...
```

```
49994   NO
```

```
49995    NO
49996    NO
49997    NO
49998    NO
```

```
Name: FLAG_WORK_PHONE, Length: 49999, dtype: object
```

```
df_app['FLAG_CONT_MOBILE']=df_app['FLAG_CONT_MOBILE'].apply(lambda
x:"YES" if x==1 else "NO")
```

```
df_app['FLAG_CONT_MOBILE']
```

```
0      YES
1      YES
2      YES
3      YES
4      YES
```

```
...
```

```
49994    YES
49995    YES
49996    YES
49997    YES
49998    YES
```

```
Name: FLAG_CONT_MOBILE, Length: 49999, dtype: object
```

```
df_app['FLAG_PHONE']=df_app['FLAG_PHONE'].apply(lambda x:"YES" if x==1
else "NO")
```

```
df_app['FLAG_PHONE']
```

```
0      YES
1      YES
2      YES
3      NO
4      NO
```

```
...
```

```
49994    NO
49995    NO
49996    NO
49997    NO
49998    NO
```

```
Name: FLAG_PHONE, Length: 49999, dtype: object
```

```
df_app['FLAG_EMAIL']=df_app['FLAG_EMAIL'].apply(lambda x:"YES" if x==1
else "NO")
```

```
df_app['FLAG_EMAIL']
```

```
0      NO
1      NO
2      NO
3      NO
```

```

4          NO
...
49994      NO
49995      NO
49996      YES
49997      NO
49998      NO
Name: FLAG_EMAIL, Length: 49999, dtype: object

```

```

df_app["CREDIT_RATIO"] = round(df_app["AMT_CREDIT"] /
df_app["AMT_INCOME_TOTAL"], 2)

```

```

df_app["CREDIT_RATIO"]

```

```

0          2.01
1          4.79
2          2.00
3          2.32
4          4.22
...
49994      6.70
49995      8.93
49996      8.00
49997      3.04
49998      2.18
Name: CREDIT_RATIO, Length: 49999, dtype: float64

```

```

#numerical analysis

```

```

df_app.describe().columns

```

```

Index(['SK_ID_CURR', 'TARGET', 'CNT_CHILDREN', 'AMT_INCOME_TOTAL',
      'AMT_CREDIT', 'AMT_ANNUITY', 'REGION_POPULATION_RELATIVE',
      'DAYS_BIRTH',
      'DAYS_EMPLOYED', 'DAYS_REGISTRATION', 'DAYS_ID_PUBLISH',
      'CNT_FAM_MEMBERS', 'REGION_RATING_CLIENT',
      'REGION_RATING_CLIENT_W_CITY', 'HOUR_APPR_PROCESS_START',
      'REG_REGION_NOT_LIVE_REGION', 'REG_REGION_NOT_WORK_REGION',
      'LIVE_REGION_NOT_WORK_REGION', 'REG_CITY_NOT_LIVE_CITY',
      'REG_CITY_NOT_WORK_CITY', 'LIVE_CITY_NOT_WORK_CITY',
      'EXT_SOURCE_2',
      'EXT_SOURCE_3', 'OBS_30_CNT_SOCIAL_CIRCLE',
      'DEF_30_CNT_SOCIAL_CIRCLE',
      'OBS_60_CNT_SOCIAL_CIRCLE', 'DEF_60_CNT_SOCIAL_CIRCLE',
      'DAYS_LAST_PHONE_CHANGE', 'FLAG_DOCUMENT_2', 'FLAG_DOCUMENT_3',
      'FLAG_DOCUMENT_4', 'FLAG_DOCUMENT_5', 'FLAG_DOCUMENT_6',
      'FLAG_DOCUMENT_7', 'FLAG_DOCUMENT_8', 'FLAG_DOCUMENT_9',
      'FLAG_DOCUMENT_10', 'FLAG_DOCUMENT_11', 'FLAG_DOCUMENT_12',
      'FLAG_DOCUMENT_13', 'FLAG_DOCUMENT_14', 'FLAG_DOCUMENT_15',
      'FLAG_DOCUMENT_16', 'FLAG_DOCUMENT_17', 'FLAG_DOCUMENT_18',
      'FLAG_DOCUMENT_19', 'FLAG_DOCUMENT_20', 'FLAG_DOCUMENT_21',

```

```

        'AMT_REQ_CREDIT_BUREAU_HOUR', 'AMT_REQ_CREDIT_BUREAU_DAY',
        'AMT_REQ_CREDIT_BUREAU_WEEK', 'AMT_REQ_CREDIT_BUREAU_MON',
        'AMT_REQ_CREDIT_BUREAU_QRT', 'AMT_REQ_CREDIT_BUREAU_YEAR',
        'DAYS_BIRTH_IN_YEAR', 'DAYS_EMPLOYED_IN_YEAR',
        'AMT_CREDIT_LAC',
        'AMT_INCOME_TOTAL_LAC', 'CREDIT_RATIO'],
        dtype='object')

numerical_columns=['AMT_ANNUITY', 'DAYS_BIRTH_IN_YEAR',
'DAYS_EMPLOYED_IN_YEAR','AMT_CREDIT_LAC','AMT_INCOME_TOTAL_LAC','CREDIT_RATIO','CNT_FAM_MEMBERS']

numerical_columns

['AMT_ANNUITY',
'DAYS_BIRTH_IN_YEAR',
'DAYS_EMPLOYED_IN_YEAR',
'AMT_CREDIT_LAC',
'AMT_INCOME_TOTAL_LAC',
'CREDIT_RATIO',
'CNT_FAM_MEMBERS']

def uni_numerical_analysis(dataframe,columns):
    sns.set(style="darkgrid")
    plt.figure(figsize=(25,10))

    plt.subplot(1,3,1)

    sns.boxplot(data=dataframe,x=columns,orient="v").set(title="boxplot")

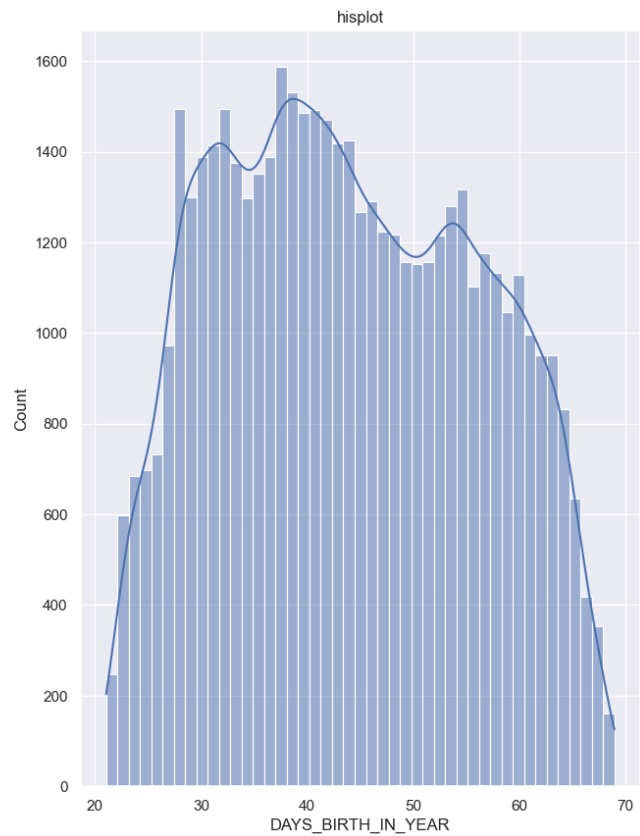
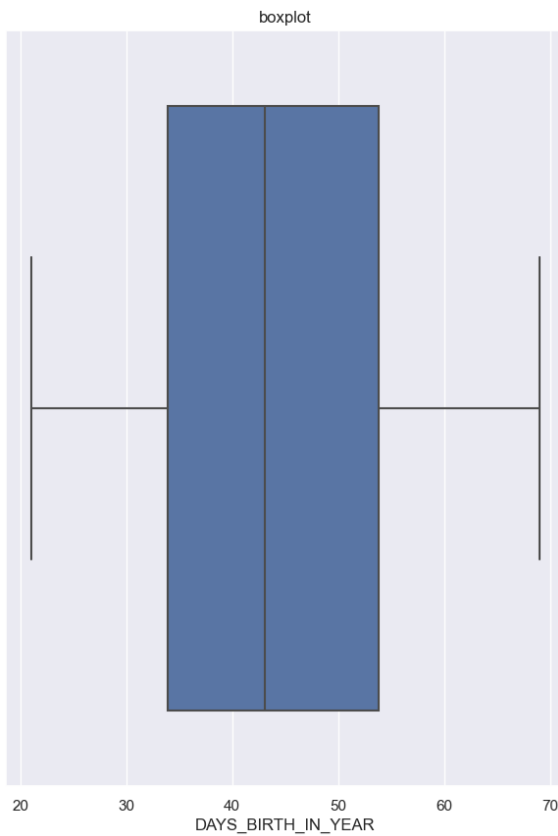
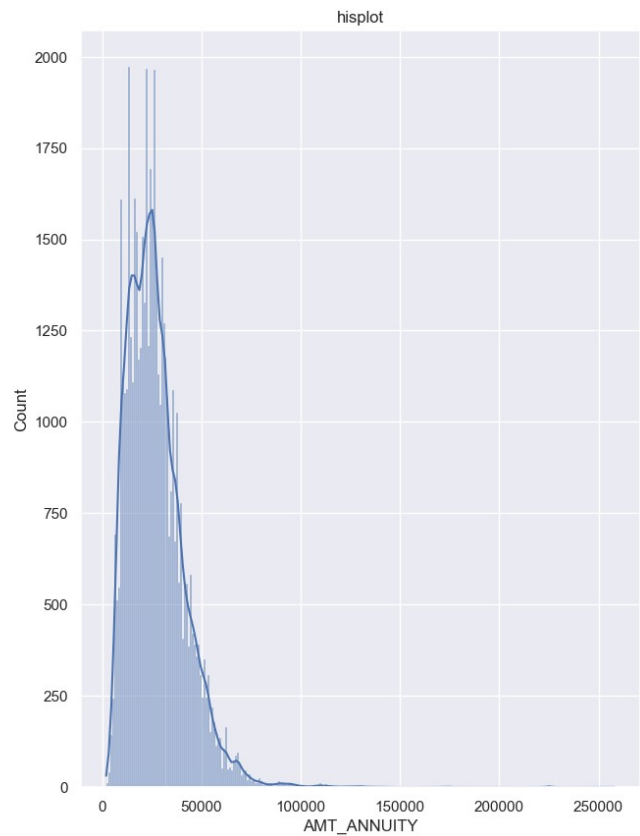
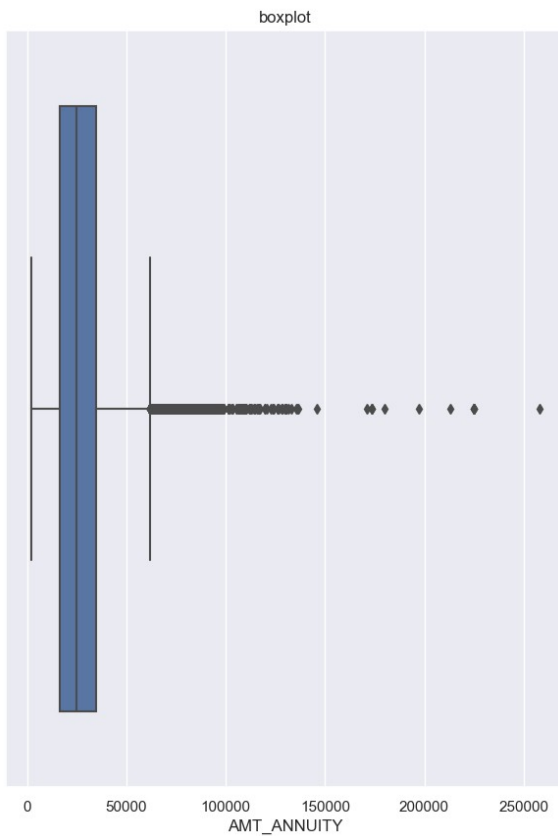
    plt.subplot(1,3,2)
    sns.histplot(dataframe[columns],kde=True).set(title="hisplot")

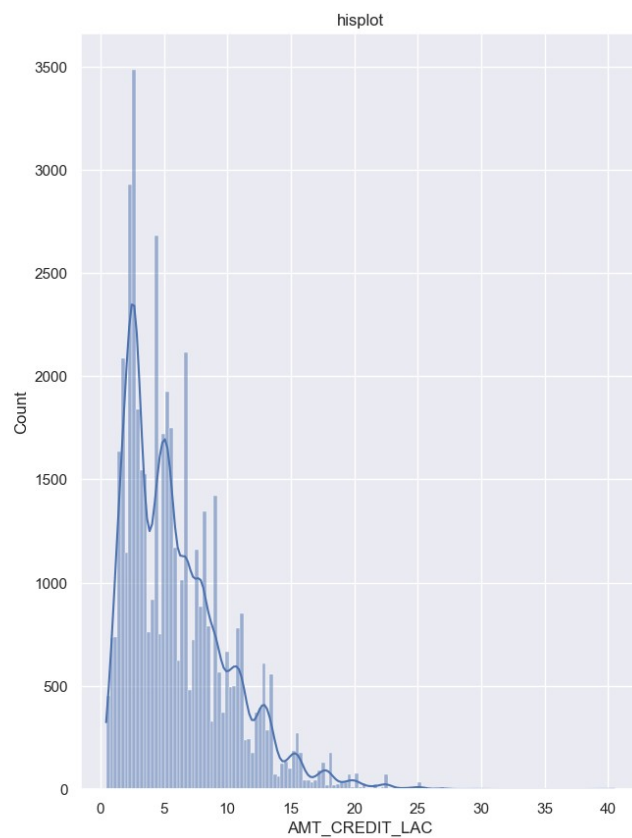
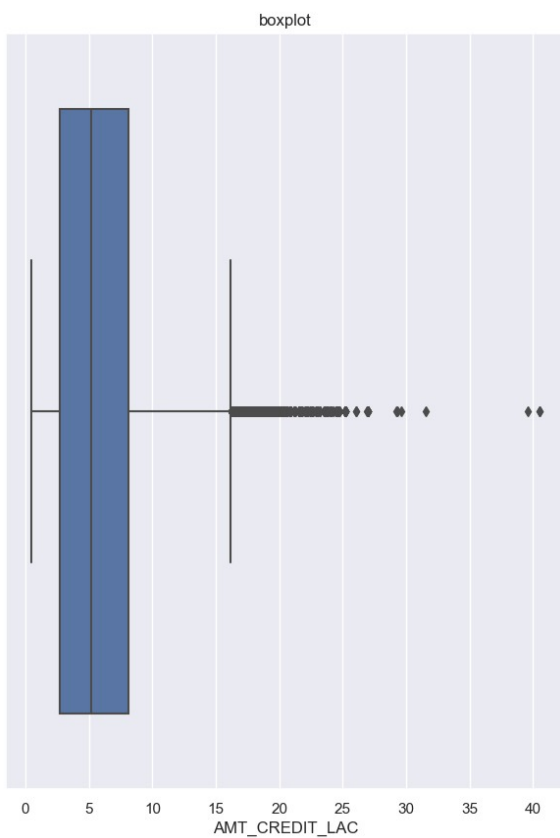
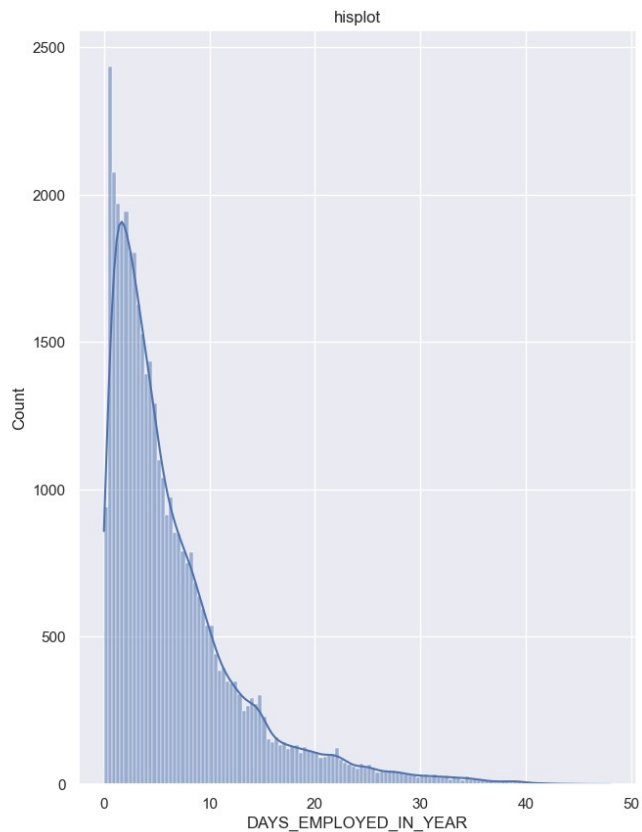
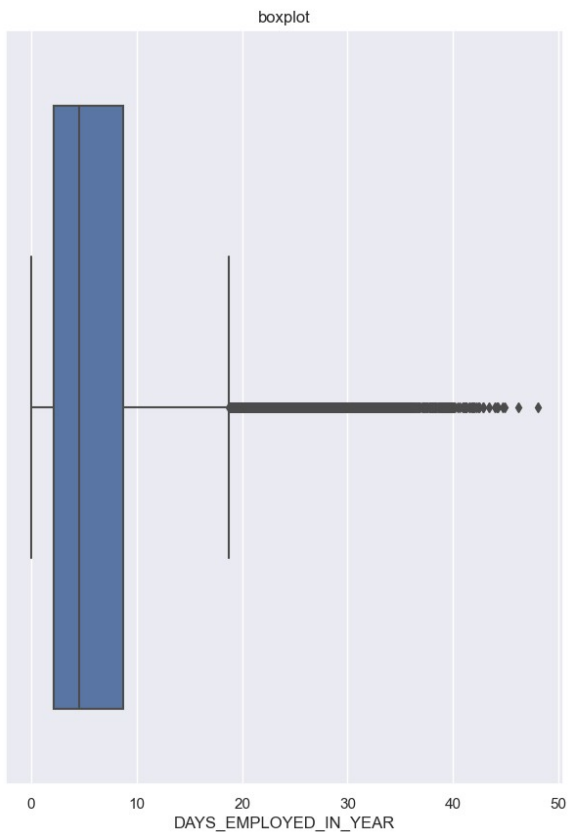
    plt.show()

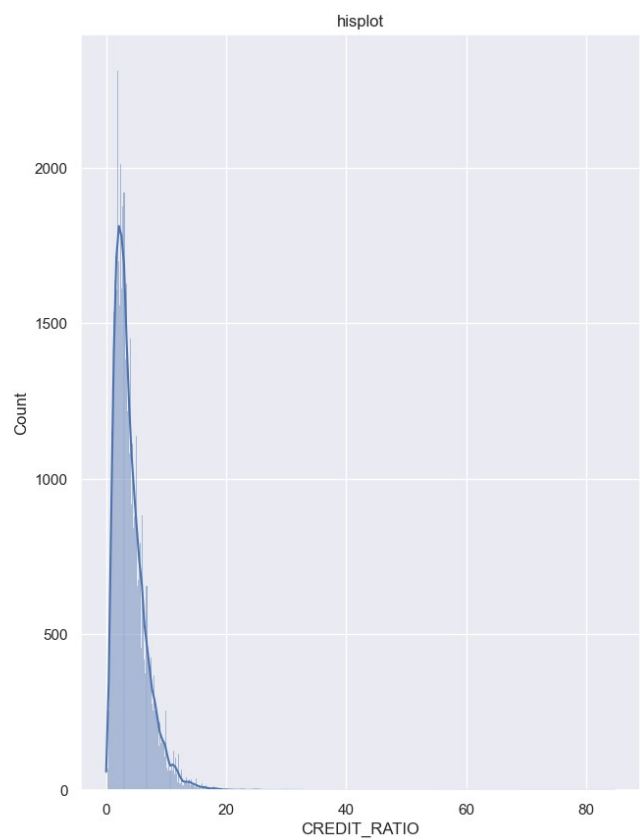
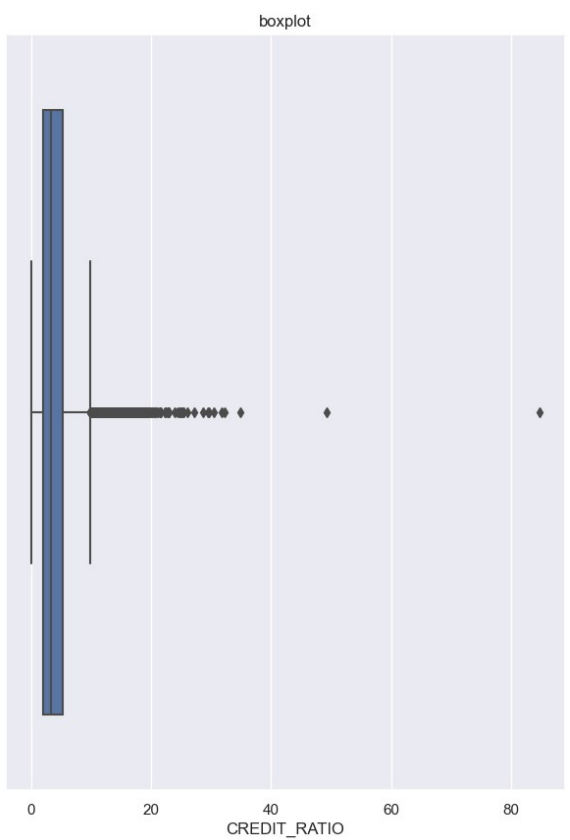
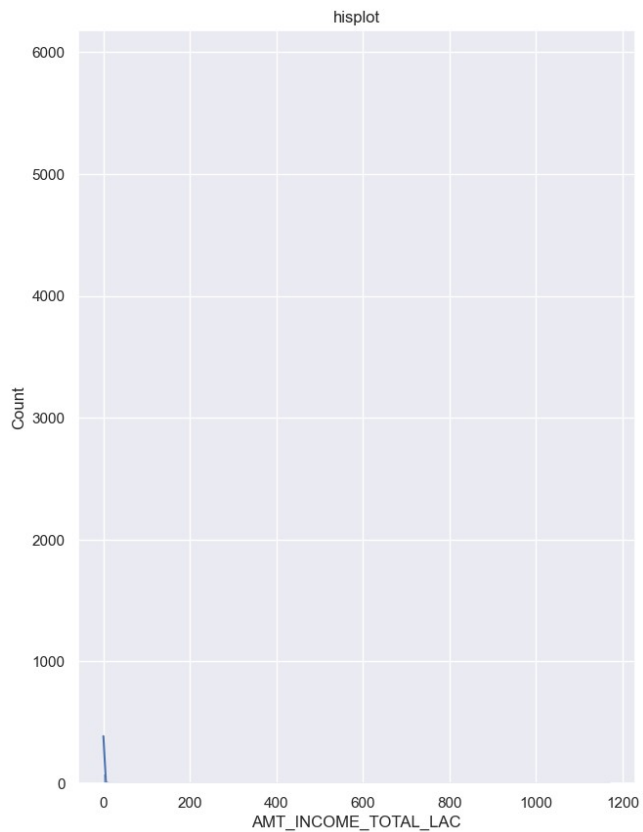
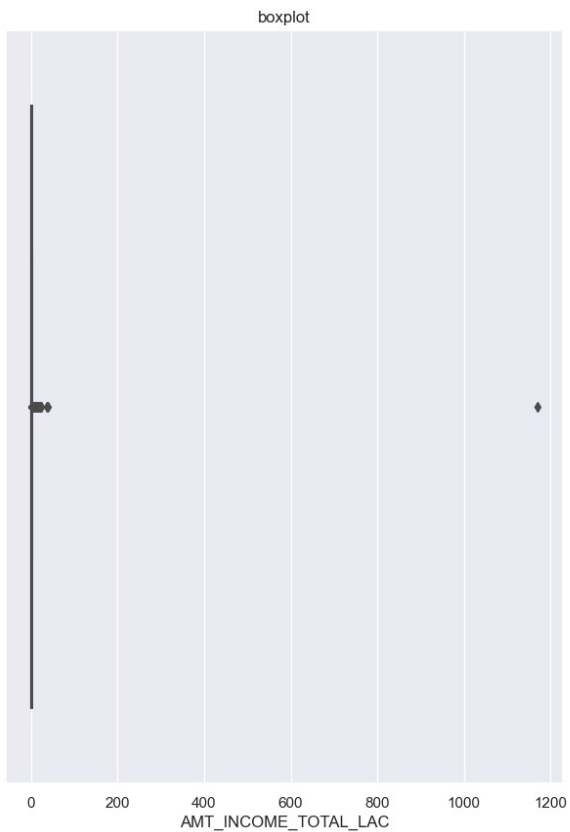
import warnings
warnings.filterwarnings("ignore")

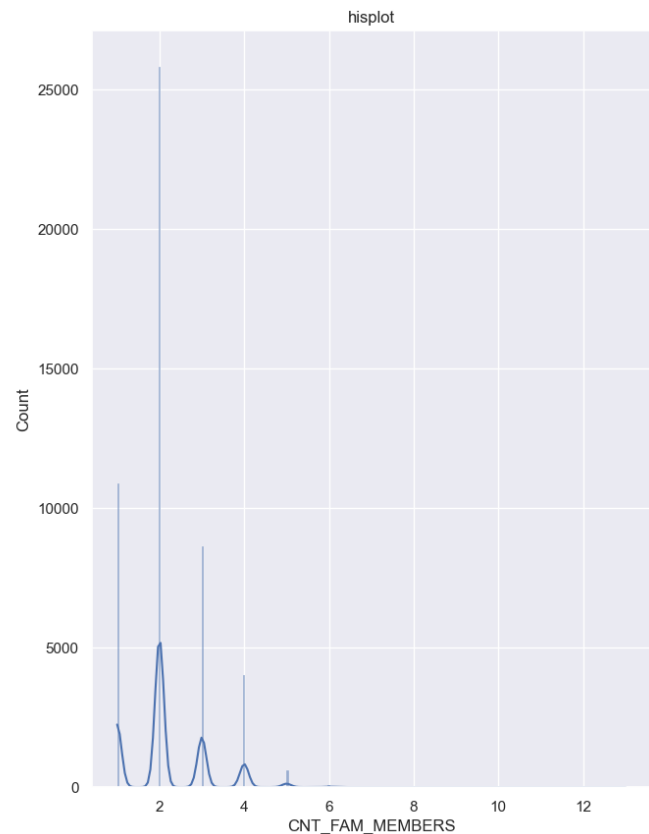
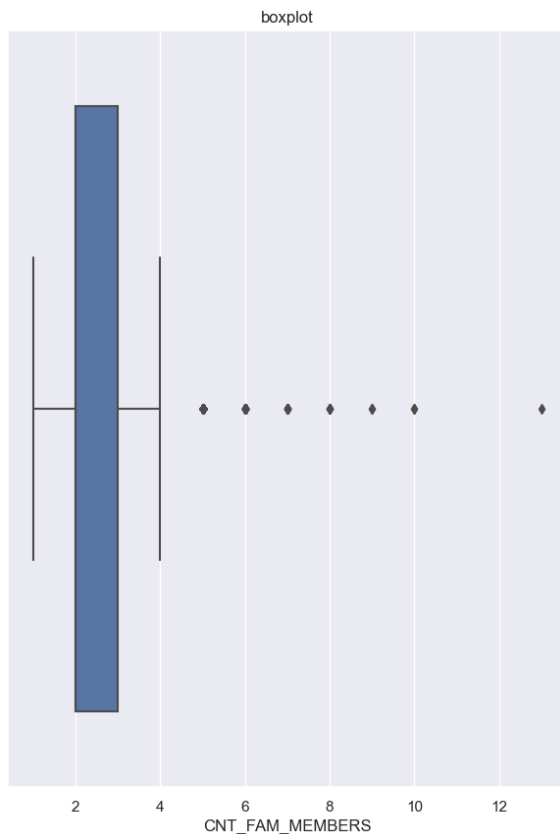
for i in numerical_columns:
    uni_numerical_analysis(df_app,i)

```









```
list(set(df_app.columns)-set(df_app.describe().columns))
```

```
[ 'AMT_CREDIT_LAC_RANGE',
  'FLAG_MOBIL',
  'NAME_HOUSING_TYPE',
  'NAME_TYPE_SUITE',
  'NAME_FAMILY_STATUS',
  'NAME_EDUCATION_TYPE',
  'ORGANIZATION_TYPE',
  'OCCUPATION_TYPE',
  'DAYS_BIRTH_RANGE',
  'FLAG_OWN_CAR',
  'CODE_GENDER',
  'WEEKDAY_APPR_PROCESS_START',
  'NAME_INCOME_TYPE',
  'FLAG_PHONE',
  'FLAG_EMAIL',
  'NAME_CONTRACT_TYPE',
  'AMT_GOODS_PRICE',
  'FLAG_OWN_REALTY',
  'FLAG_CONT_MOBILE',
  'FLAG_WORK_PHONE',
  'FLAG_EMP_PHONE',
  'AMT_INCOME_TOTAL_LAC_RANGE']
```

```

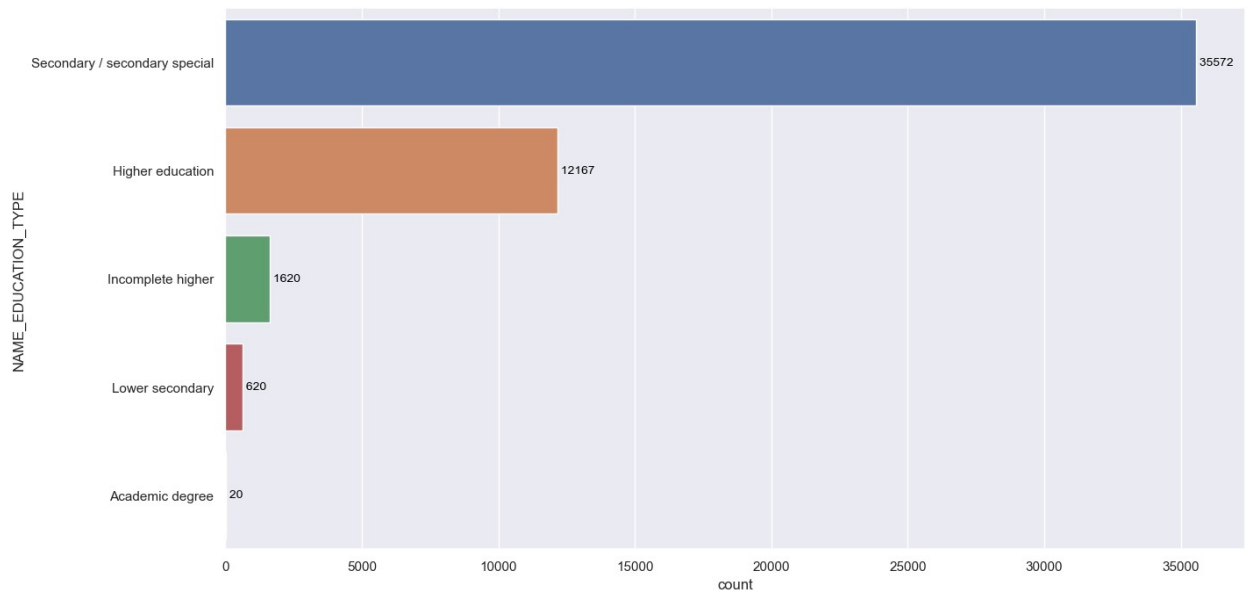
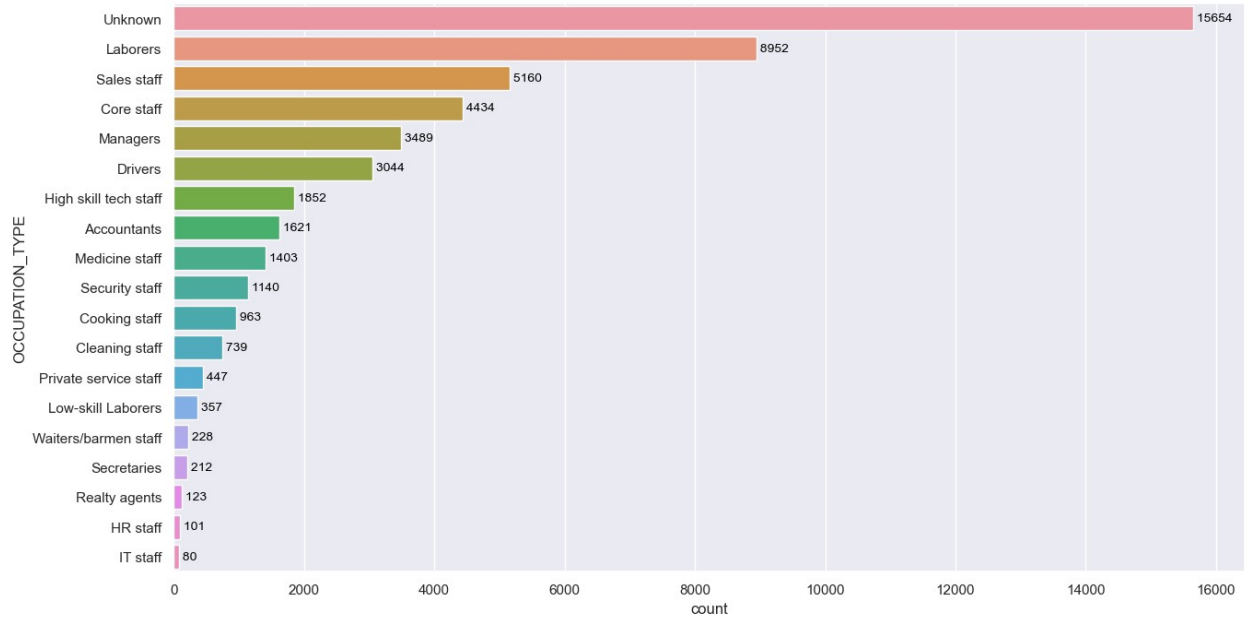
categorical_columns=['OCCUPATION_TYPE',
'AMT_GOODS_PRICE',
'NAME_EDUCATION_TYPE',
'NAME_INCOME_TYPE',
'FLAG_OWN_CAR',
'NAME_HOUSING_TYPE',
'FLAG_WORK_PHONE',
'AMT_INCOME_TOTAL_LAC_RANGE',
'FLAG_MOBIL',
'DAYS_BIRTH_RANGE',
'CODE_GENDER',
'NAME_TYPE_SUITE',
'NAME_FAMILY_STATUS',
'FLAG_EMAIL',
'FLAG_OWN_REALTY',
'FLAG_PHONE',
'FLAG_EMP_PHONE',
'FLAG_CONT_MOBILE',
'ORGANIZATION_TYPE',
'NAME_CONTRACT_TYPE',
'AMT_CREDIT_LAC_RANGE',
'WEEKDAY_APPR_PROCESS_START']

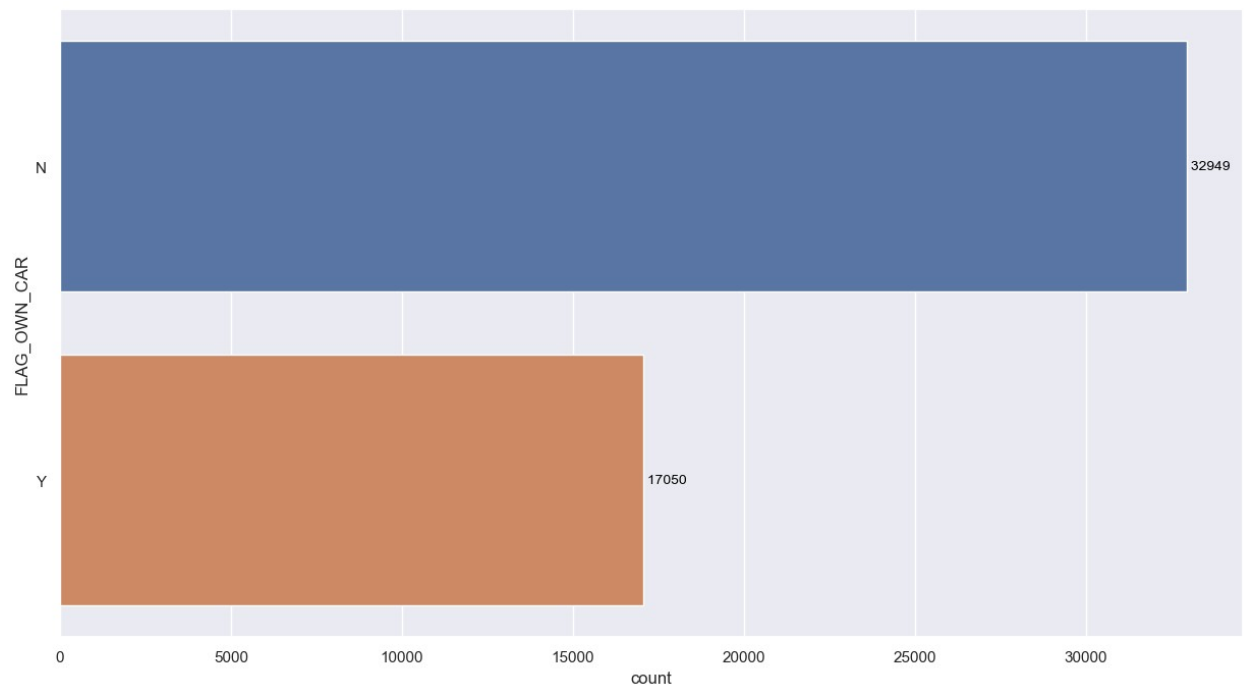
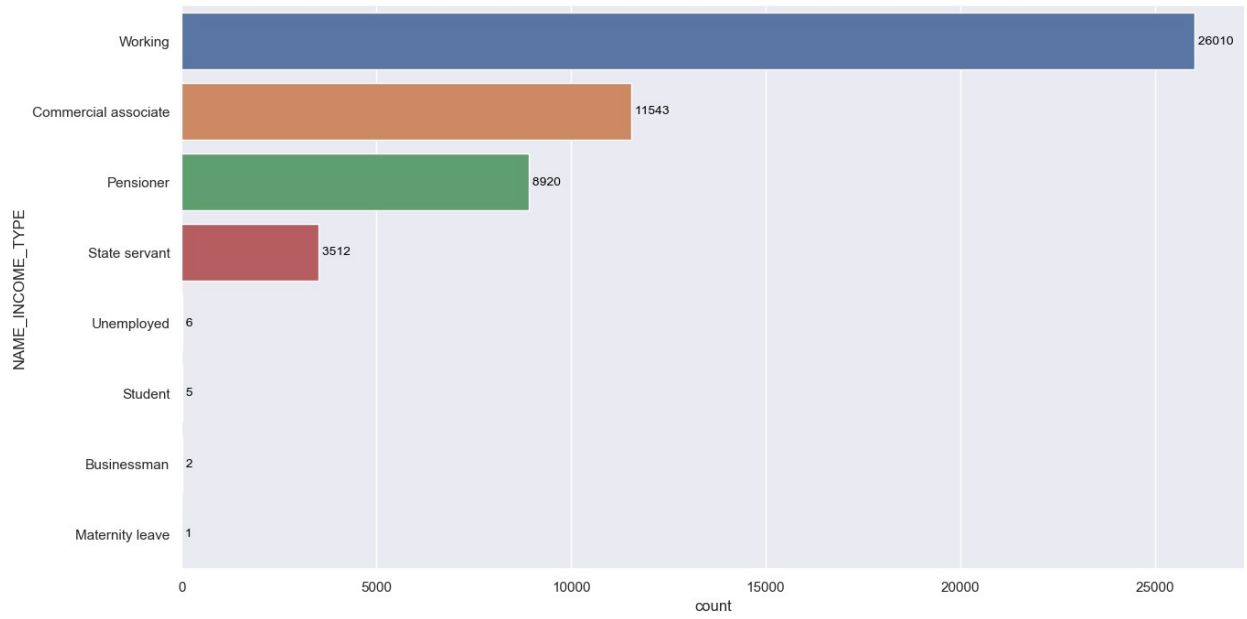
def uni_categorical_analysis(dataframe,columns):
    sns.set(style="darkgrid")
    plt.figure(figsize=(15,8))

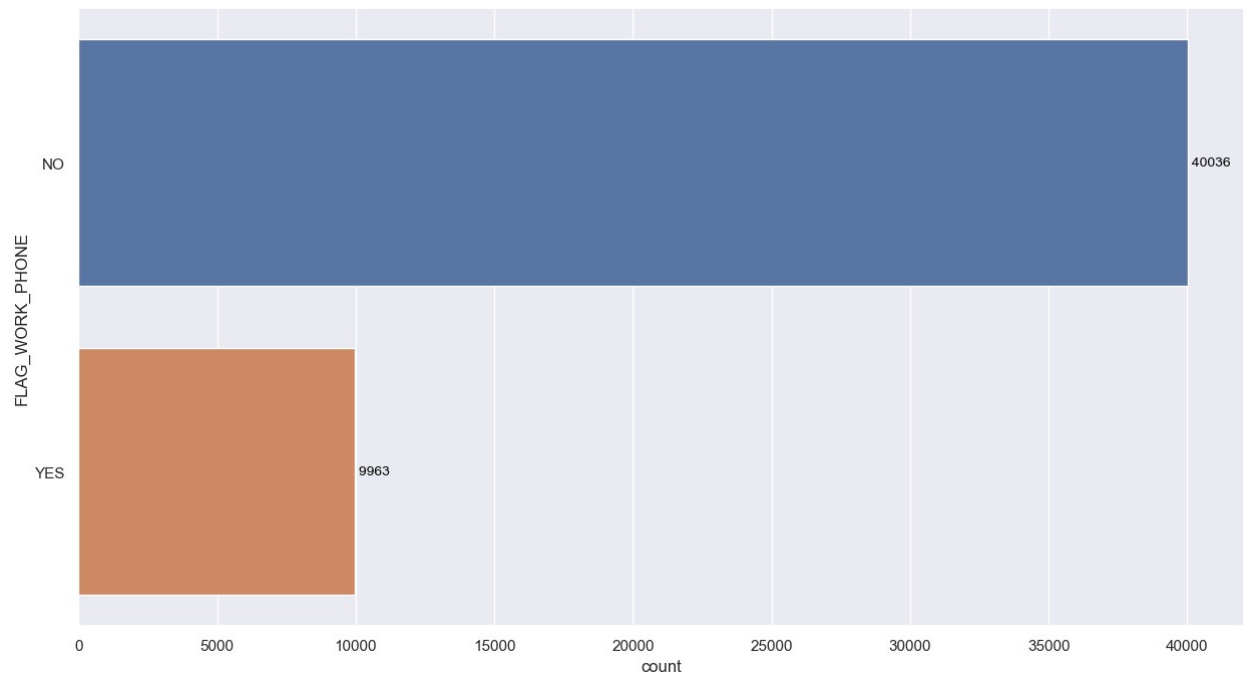
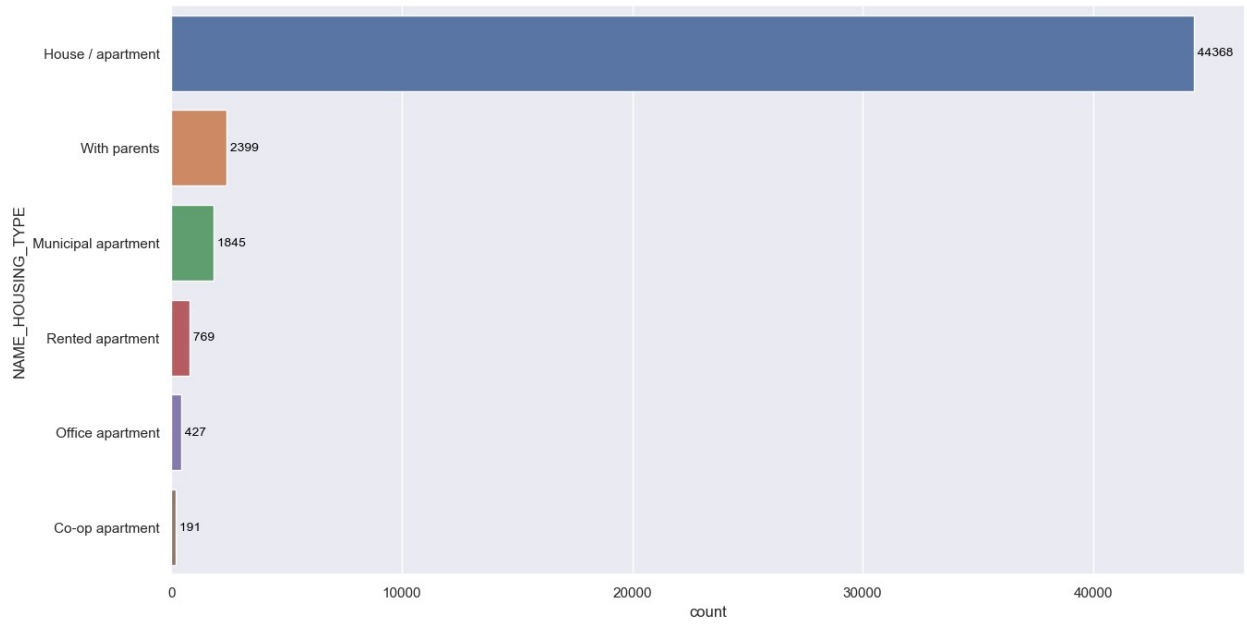
    sns.barplot(x=dataframe[columns].value_counts(),y=dataframe[columns].v
    alue_counts().index)
    for index,value in enumerate(dataframe[columns].value_counts()):
        plt.text(value,index,f"
{value}",ha="left",va="center",color="black",fontsize=10)
    plt.show()

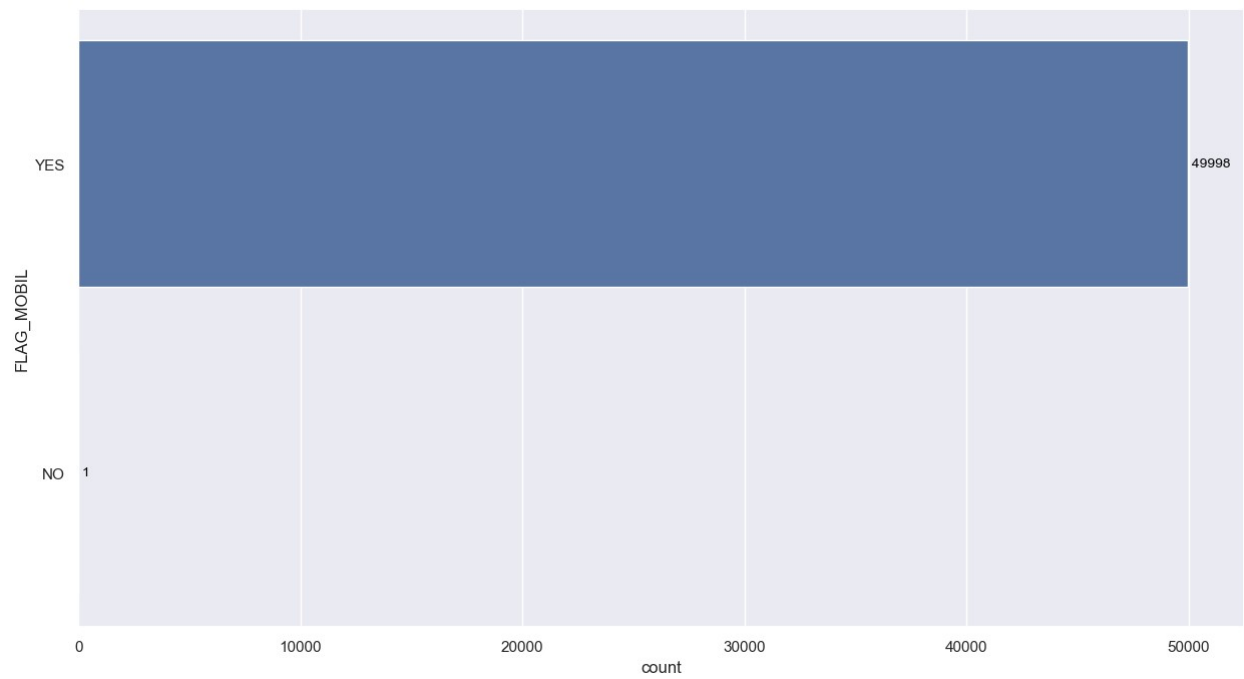
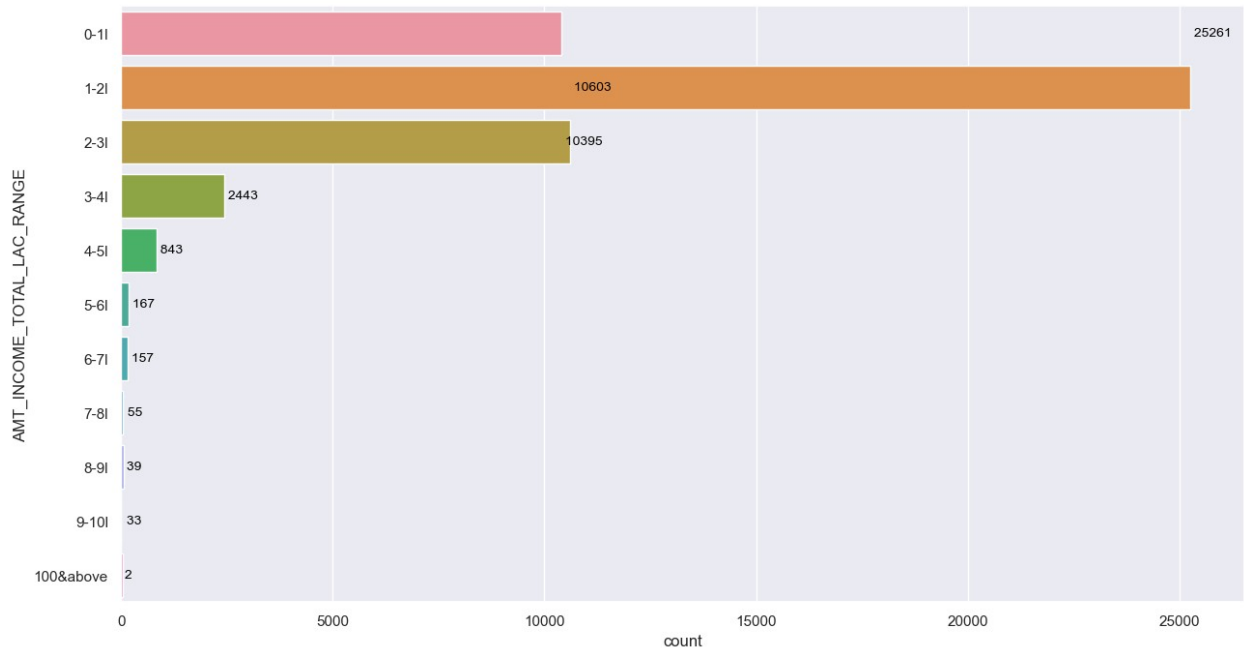
for i in categorical_columns:
    uni_categorical_analysis(df_app,i)

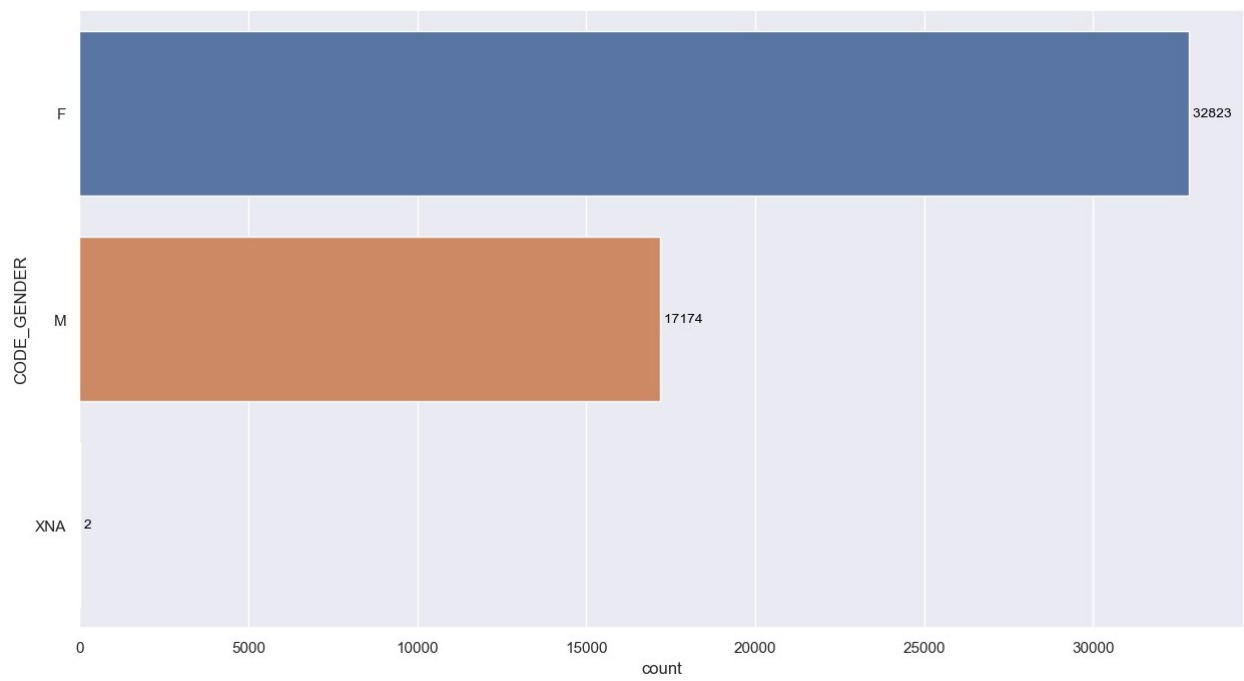
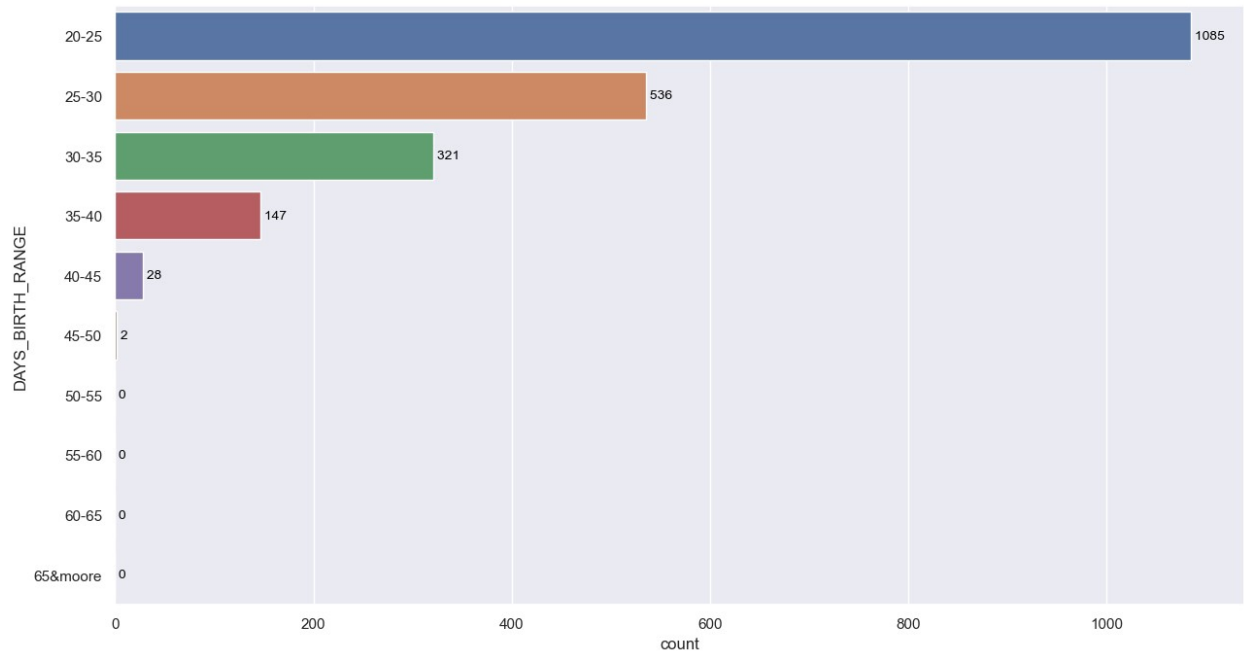
```

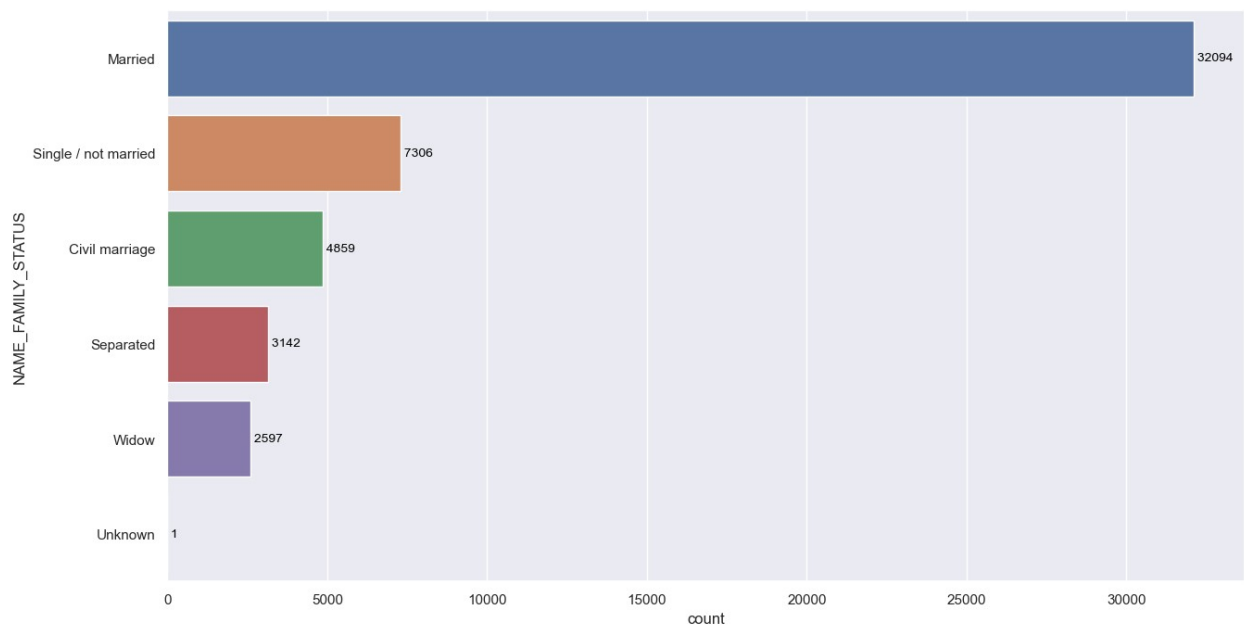
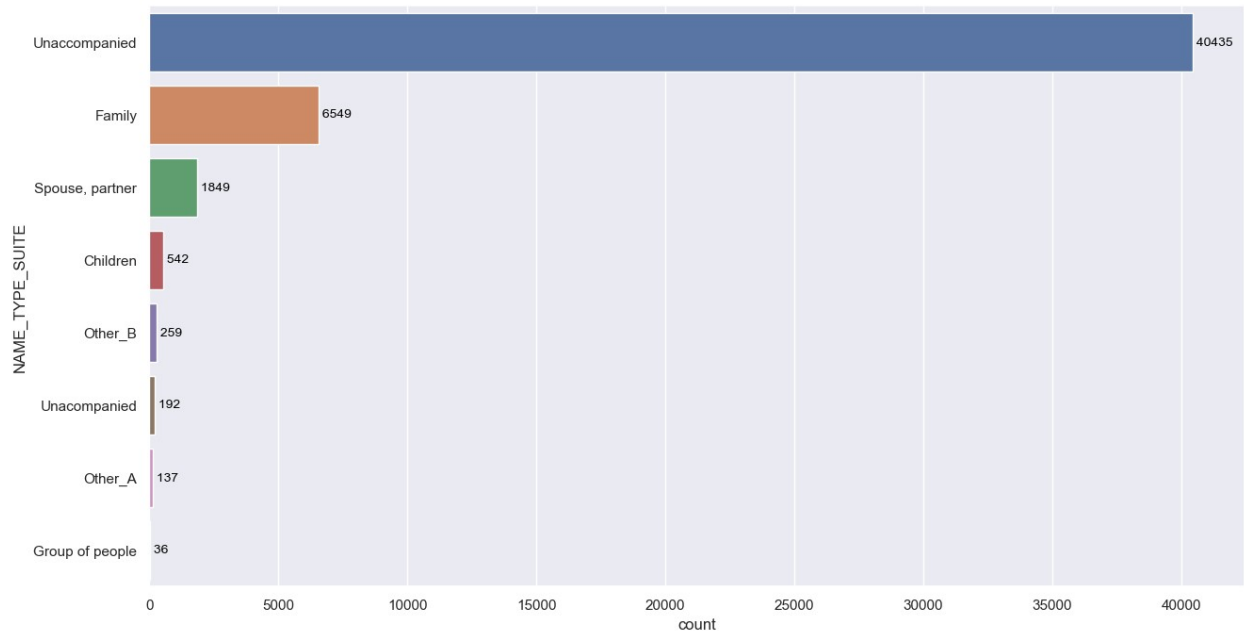


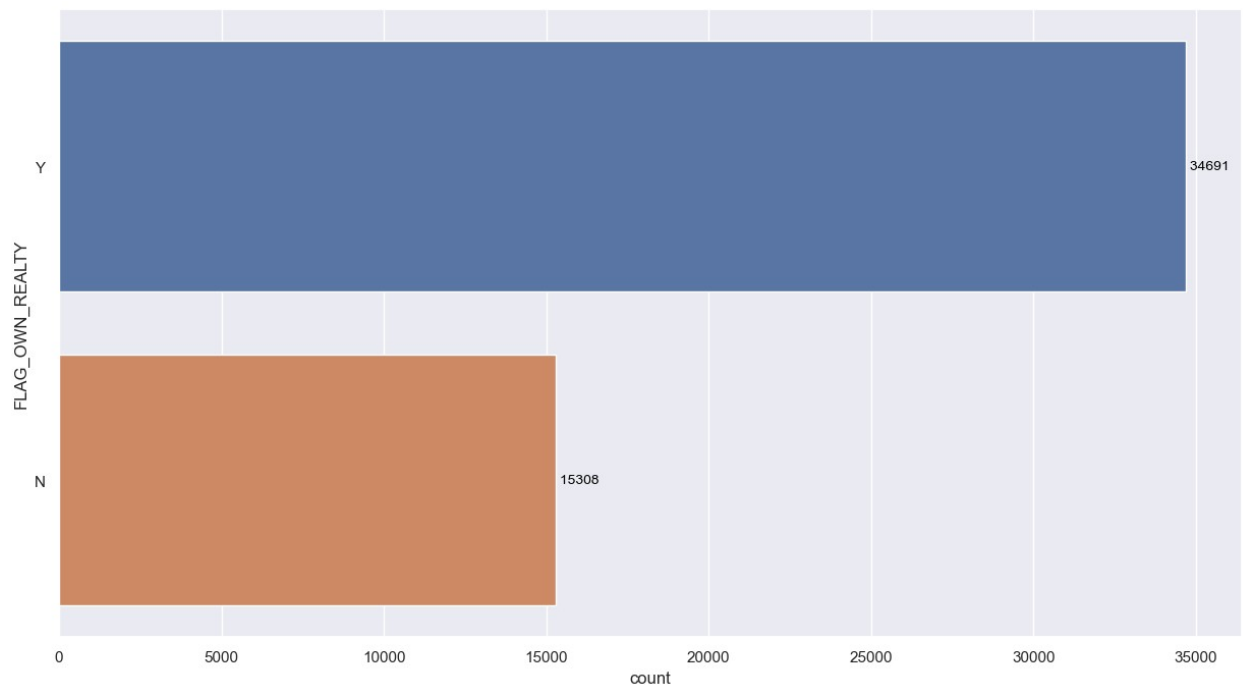
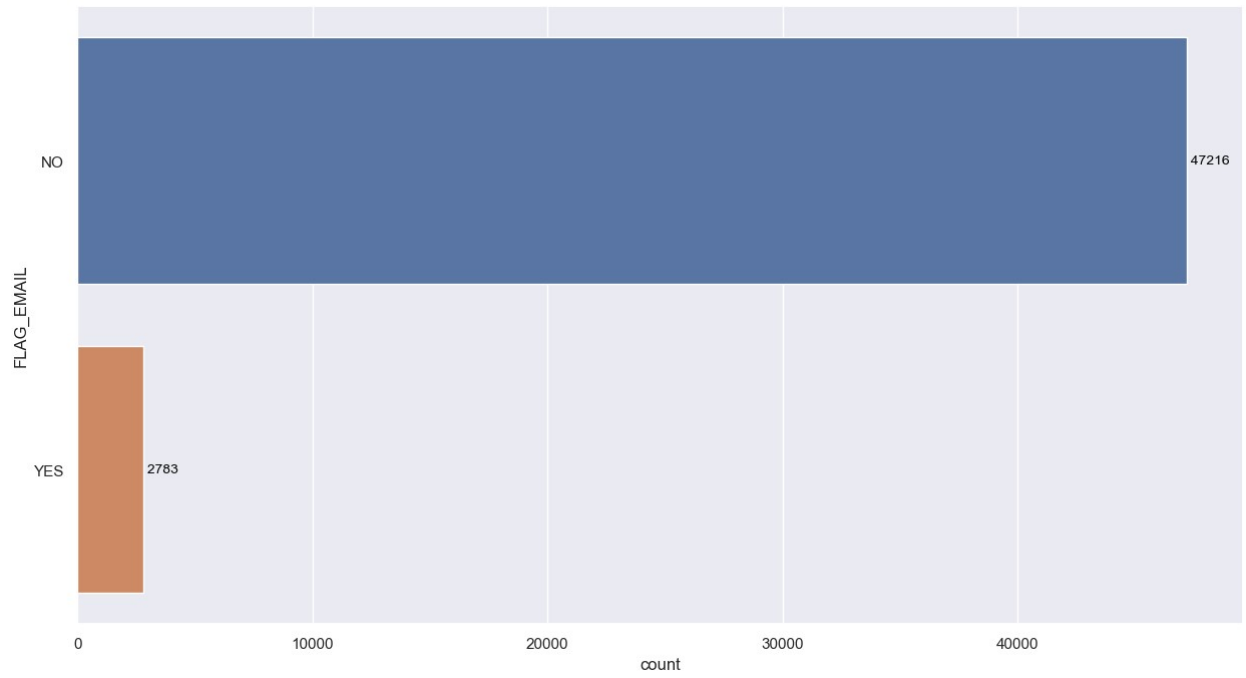


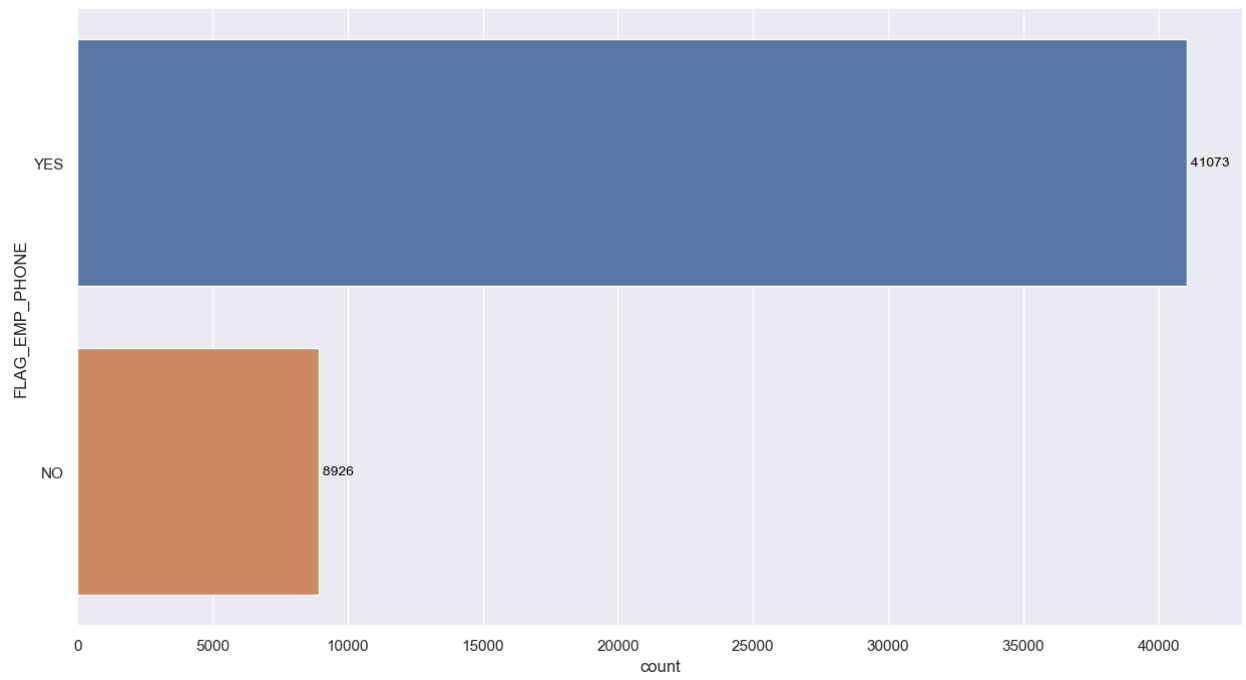
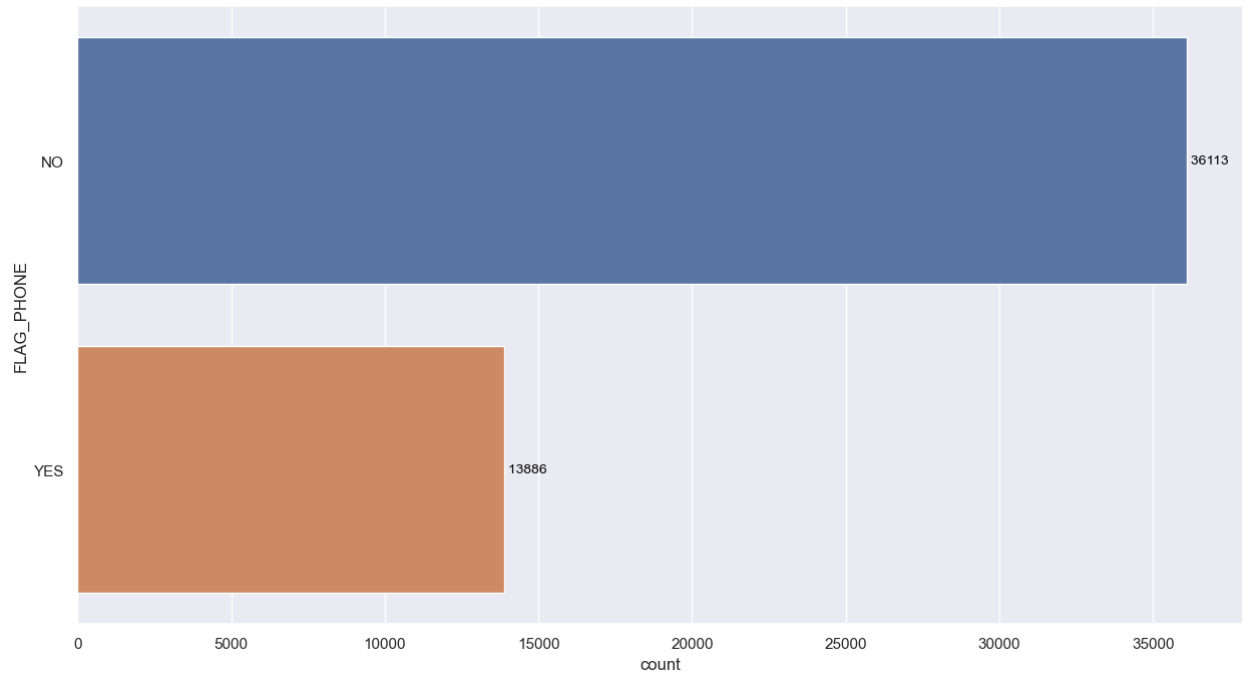


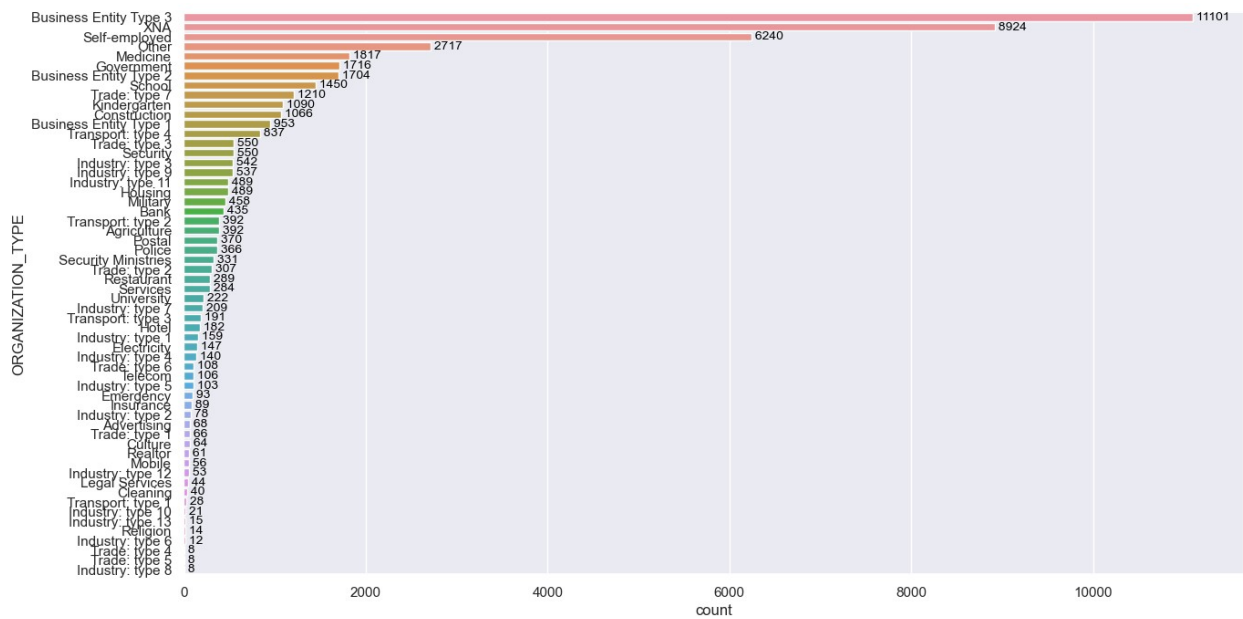
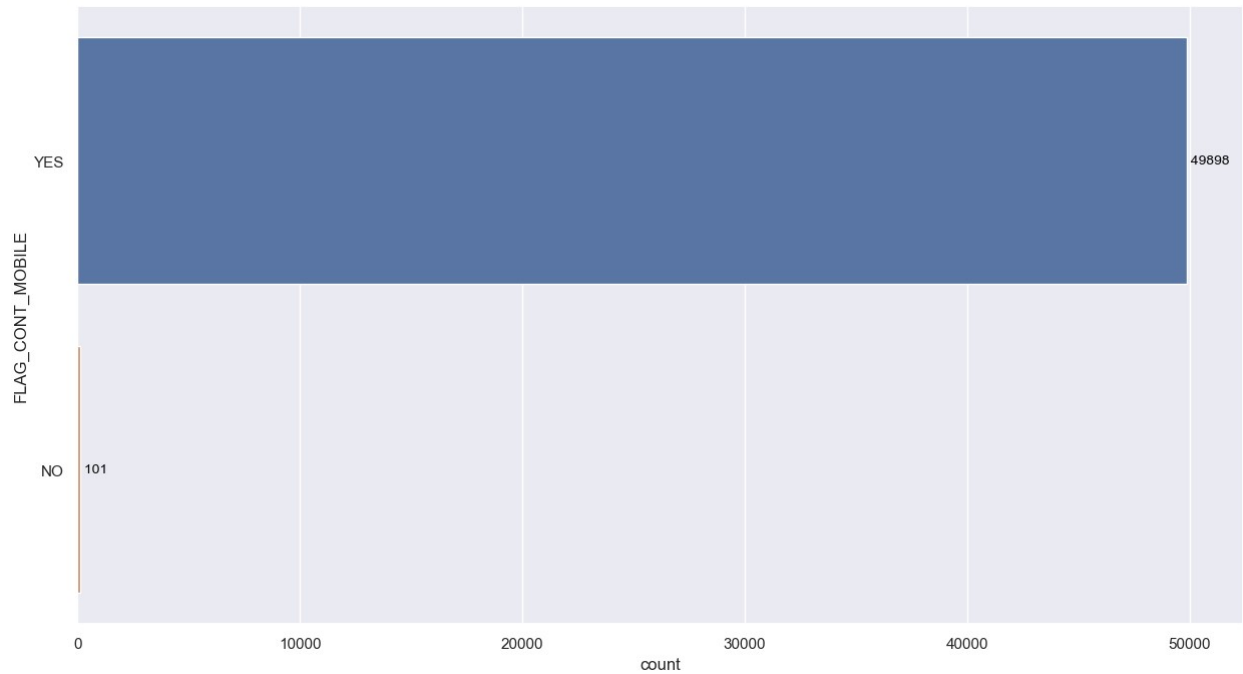


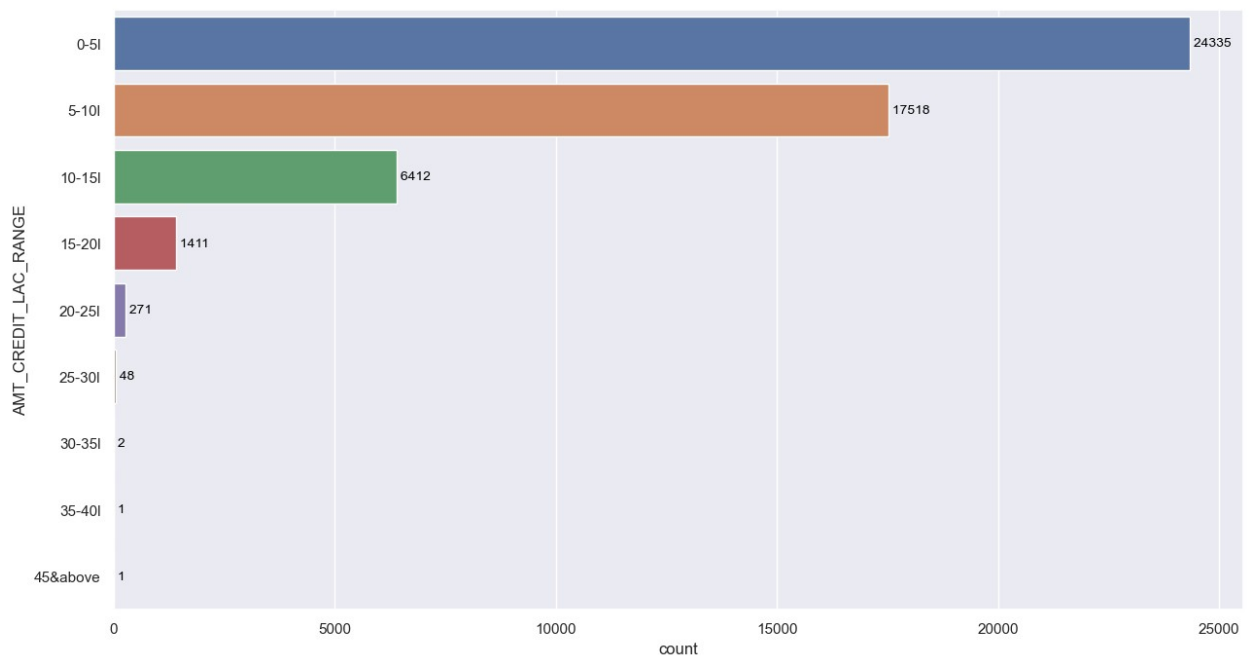
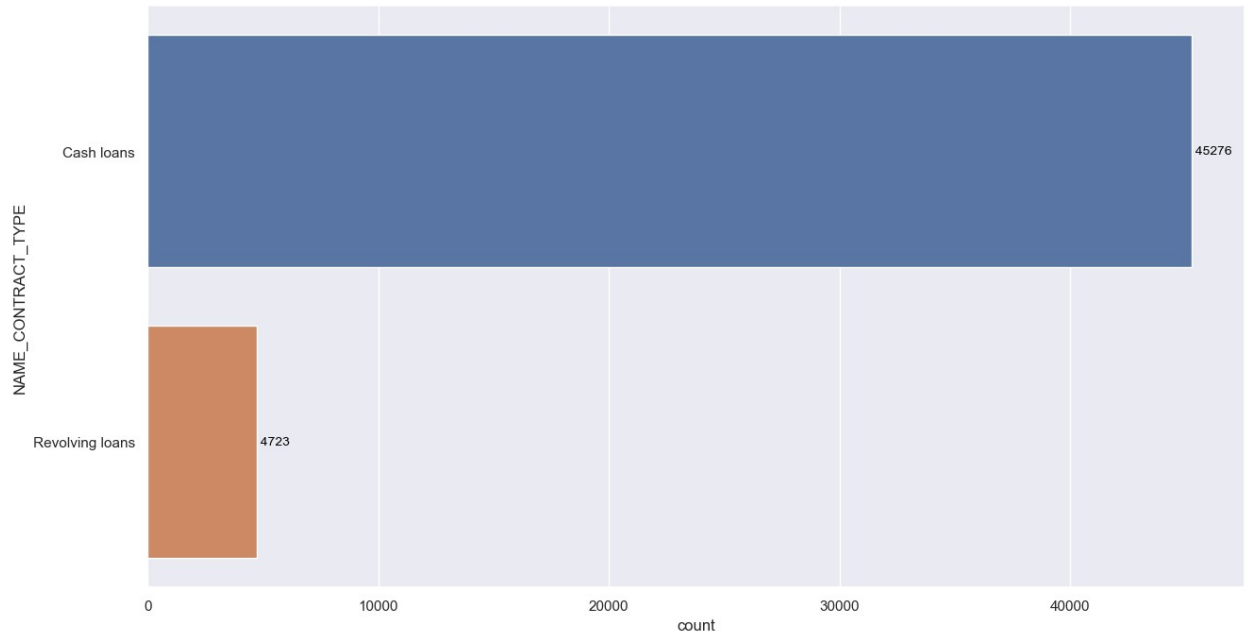


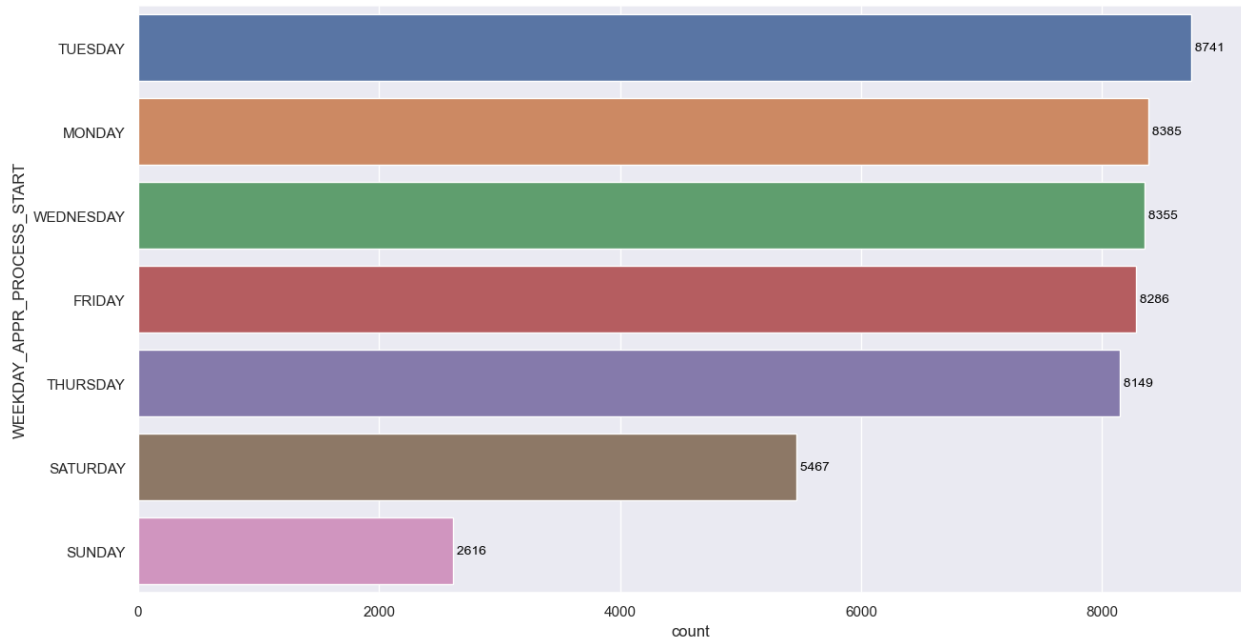












```
df_pre.describe().columns
```

```
Index(['SK_ID_PREV', 'SK_ID_CURR', 'AMT_ANNUITY', 'AMT_APPLICATION',
       'AMT_CREDIT', 'HOUR_APPR_PROCESS_START',
       'NFLAG_LAST_APPL_IN_DAY',
       'DAYS_DECISION', 'SELLERPLACE_AREA', 'CNT_PAYMENT',
       'DAYS_FIRST_DRAWING', 'DAYS_FIRST_DUE',
       'DAYS_LAST_DUE_1ST_VERSION',
       'DAYS_LAST_DUE', 'DAYS_TERMINATION',
       'NFLAG_INSURED_ON_APPROVAL',
       'AMT_ANNUITY_LAC', 'AMT_CREDIT_LAC', 'AMT_APPLICATION_LAC'],
      dtype='object')
```

```
list(set(df_pre.columns)-set(df_pre.describe().columns))
```

```
['NAME_CONTRACT_STATUS',
 'AMT_CREDIT_LAC_RANGE',
 'NAME_PAYMENT_TYPE',
 'CODE_REJECT_REASON',
 'NAME_CLIENT_TYPE',
 'AMT_APPLICATION_LAC_RANGE',
 'NAME_PRODUCT_TYPE',
 'NAME_YIELD_GROUP',
 'NAME_TYPE_SUITE',
 'NAME_CASH_LOAN_PURPOSE',
 'NAME_SELLER_INDUSTRY',
 'NAME_PORTFOLIO',
 'FLAG_LAST_APPL_PER_CONTRACT',
 'WEEKDAY_APPR_PROCESS_START',
 'PRODUCT_COMBINATION',
```

```
'CHANNEL_TYPE',
'NAME_CONTRACT_TYPE',
'NAME_GOODS_CATEGORY',
'AMT_GOODS_PRICE']
```

```
#target analysis
```

```
Target_variable_payment_difficulty=df_app[df_app["TARGET"]==1]
Target_variable_payment_others=df_app[df_app["TARGET"]==0]
```

```
Target_variable_payment_difficulty
```

	SK_ID_CURR	TARGET	NAME_CONTRACT_TYPE	CODE_GENDER	FLAG_OWN_CAR
\					
0	100002	1	Cash loans	M	N
26	100031	1	Cash loans	F	N
40	100047	1	Cash loans	M	N
42	100049	1	Cash loans	F	N
81	100096	1	Cash loans	F	N
...	...	...	...	...	...
49925	157788	1	Cash loans	F	N
49959	157830	1	Cash loans	F	N
49973	157846	1	Cash loans	F	N
49976	157850	1	Cash loans	F	N
49986	157861	1	Cash loans	M	Y

	FLAG_OWN_REALTY	CNT_CHILDREN	AMT_INCOME_TOTAL	AMT_CREDIT	\
0	Y	0	202500.0	406597.5	
26	Y	0	112500.0	979992.0	
40	Y	0	202500.0	1193580.0	
42	N	0	135000.0	288873.0	
81	Y	0	81000.0	252000.0	
...	...	...	...	...	
49925	Y	0	112500.0	717003.0	
49959	Y	0	90000.0	405000.0	
49973	Y	2	157500.0	593010.0	
49976	Y	1	157500.0	679266.0	
49986	N	0	135000.0	270000.0	

```
AMT_ANNUITY ... AMT_REQ_CREDIT_BUREAU_QRT
```

AMT_REQ_CREDIT_BUREAU_YEAR \		
0	24700.5 ...	0.0
1.0		
26	27076.5 ...	2.0
2.0		
40	35028.0 ...	0.0
4.0		
42	16258.5 ...	0.0
2.0		
81	14593.5 ...	0.0
0.0		
...		...
...		
49925	27913.5 ...	0.0
8.0		
49959	19480.5 ...	0.0
5.0		
49973	23107.5 ...	1.0
3.0		
49976	26451.0 ...	0.0
0.0		
49986	21330.0 ...	0.0
1.0		

DAYS_BIRTH_IN_YEAR	DAYS_EMPLOYED_IN_YEAR	DAYS_BIRTH_RANGE \
0	25.92	1.75 NaN
26	51.30	7.20 NaN
40	47.90	3.46 NaN
42	36.67	9.85 NaN
81	67.93	NaN NaN
...	...	...
49925	42.36	4.12 NaN
49959	61.59	NaN NaN
49973	38.04	4.87 NaN
49976	30.74	3.36 NaN
49986	24.33	2.53 NaN

AMT_CREDIT_LAC	AMT_INCOME_TOTAL_LAC	AMT_CREDIT_LAC_RANGE \
0	4.07	2.02 0-5l
26	9.80	1.12 5-10l
40	11.94	2.02 10-15l
42	2.89	1.35 0-5l
81	2.52	0.81 0-5l
...	...	...
49925	7.17	1.12 5-10l
49959	4.05	0.90 0-5l
49973	5.93	1.58 5-10l
49976	6.79	1.58 5-10l
49986	2.70	1.35 0-5l

	AMT_INCOME_TOTAL_LAC_RANGE	CREDIT_RATIO
0	2-3l	2.01
26	1-2l	8.71
40	2-3l	5.89
42	1-2l	2.14
81	0-1l	3.11
...	...	...
49925	1-2l	6.37
49959	0-1l	4.50
49973	1-2l	3.77
49976	1-2l	4.31
49986	1-2l	2.00

[4026 rows x 81 columns]

Target_variable_payment_others

	SK_ID_CURR	TARGET	NAME_CONTRACT_TYPE	CODE_GENDER	FLAG_OWN_CAR
\					
1	100003	0	Cash loans	F	N
2	100004	0	Revolving loans	M	Y
3	100006	0	Cash loans	F	N
4	100007	0	Cash loans	M	N
5	100008	0	Cash loans	M	N
...	...	...	...	...	...
49994	157871	0	Cash loans	F	N
49995	157872	0	Cash loans	M	N
49996	157873	0	Cash loans	M	N
49997	157874	0	Cash loans	F	N
49998	157875	0	Cash loans	F	N

	FLAG_OWN_REALTY	CNT_CHILDREN	AMT_INCOME_TOTAL	AMT_CREDIT	\
1	N	0	270000.0	1293502.5	
2	Y	0	67500.0	135000.0	
3	Y	0	135000.0	312682.5	
4	Y	0	121500.0	513000.0	
5	Y	0	99000.0	490495.5	
...	...	...	...	...	
49994	N	0	180000.0	1206000.0	
49995	N	0	126000.0	1125000.0	

49996	N	1	112500.0	900000.0
49997	Y	0	270000.0	820638.0
49998	Y	0	117000.0	254700.0

	AMT_ANNUIITY	...	AMT_REQ_CREDIT_BUREAU_QRT
AMT_REQ_CREDIT_BUREAU_YEAR	\		

1	35698.5	...	0.0
0.0			
2	6750.0	...	0.0
0.0			
3	29686.5	...	0.0
1.0			
4	21865.5	...	0.0
0.0			
5	27517.5	...	1.0
1.0			
...	...	...	...
...			
49994	45936.0	...	0.0
0.0			
49995	47794.5	...	0.0
0.0			
49996	26316.0	...	0.0
2.0			
49997	34897.5	...	2.0
4.0			
49998	14751.0	...	0.0
0.0			

	DAYS_BIRTH_IN_YEAR	DAYS_EMPLOYED_IN_YEAR	DAYS_BIRTH_RANGE	\
--	--------------------	-----------------------	------------------	---

1	45.93	3.25	NaN
2	52.18	0.62	NaN
3	52.07	8.33	NaN
4	54.61	8.32	NaN
5	46.41	4.35	NaN
...	...	...	...
49994	29.22	0.78	NaN
49995	55.37	12.74	NaN
49996	28.16	3.17	NaN
49997	64.34	5.98	NaN
49998	52.74	NaN	NaN

	AMT_CREDIT_LAC	AMT_INCOME_TOTAL_LAC	AMT_CREDIT_LAC_RANGE	\
--	----------------	----------------------	----------------------	---

1	12.94	2.70	10-15l
2	1.35	0.68	0-5l
3	3.13	1.35	0-5l
4	5.13	1.22	5-10l
5	4.90	0.99	0-5l
...	...	...	...
49994	12.06	1.80	10-15l

49995	11.25	1.26	10-15l
49996	9.00	1.12	5-10l
49997	8.21	2.70	5-10l
49998	2.55	1.17	0-5l

	AMT_INCOME_TOTAL_LAC_RANGE	CREDIT_RATIO
1	2-3l	4.79
2	0-1l	2.00
3	1-2l	2.32
4	1-2l	4.22
5	0-1l	4.95
...	...	...
49994	1-2l	6.70
49995	1-2l	8.93
49996	1-2l	8.00
49997	2-3l	3.04
49998	1-2l	2.18

[45973 rows x 81 columns]

df_app.describe().columns

```
Index(['SK_ID_CURR', 'TARGET', 'CNT_CHILDREN', 'AMT_INCOME_TOTAL',
      'AMT_CREDIT', 'AMT_ANNUITY', 'REGION_POPULATION_RELATIVE',
      'DAYS_BIRTH',
      'DAYS_EMPLOYED', 'DAYS_REGISTRATION', 'DAYS_ID_PUBLISH',
      'CNT_FAM_MEMBERS', 'REGION_RATING_CLIENT',
      'REGION_RATING_CLIENT_W_CITY', 'HOUR_APPR_PROCESS_START',
      'REG_REGION_NOT_LIVE_REGION', 'REG_REGION_NOT_WORK_REGION',
      'LIVE_REGION_NOT_WORK_REGION', 'REG_CITY_NOT_LIVE_CITY',
      'REG_CITY_NOT_WORK_CITY', 'LIVE_CITY_NOT_WORK_CITY',
      'EXT_SOURCE_2',
      'EXT_SOURCE_3', 'OBS_30_CNT_SOCIAL_CIRCLE',
      'DEF_30_CNT_SOCIAL_CIRCLE',
      'OBS_60_CNT_SOCIAL_CIRCLE', 'DEF_60_CNT_SOCIAL_CIRCLE',
      'DAYS_LAST_PHONE_CHANGE', 'FLAG_DOCUMENT_2', 'FLAG_DOCUMENT_3',
      'FLAG_DOCUMENT_4', 'FLAG_DOCUMENT_5', 'FLAG_DOCUMENT_6',
      'FLAG_DOCUMENT_7', 'FLAG_DOCUMENT_8', 'FLAG_DOCUMENT_9',
      'FLAG_DOCUMENT_10', 'FLAG_DOCUMENT_11', 'FLAG_DOCUMENT_12',
      'FLAG_DOCUMENT_13', 'FLAG_DOCUMENT_14', 'FLAG_DOCUMENT_15',
      'FLAG_DOCUMENT_16', 'FLAG_DOCUMENT_17', 'FLAG_DOCUMENT_18',
      'FLAG_DOCUMENT_19', 'FLAG_DOCUMENT_20', 'FLAG_DOCUMENT_21',
      'AMT_REQ_CREDIT_BUREAU_HOUR', 'AMT_REQ_CREDIT_BUREAU_DAY',
      'AMT_REQ_CREDIT_BUREAU_WEEK', 'AMT_REQ_CREDIT_BUREAU_MON',
      'AMT_REQ_CREDIT_BUREAU_QRT', 'AMT_REQ_CREDIT_BUREAU_YEAR',
      'DAYS_BIRTH_IN_YEAR', 'DAYS_EMPLOYED_IN_YEAR',
      'AMT_CREDIT_LAC',
      'AMT_INCOME_TOTAL_LAC', 'CREDIT_RATIO'],
      dtype='object')
```

```
numerical_data_1=['AMT_ANNUITY','AMT_GOODS_PRICE','DAYS_BIRTH_IN_YEAR',  
, 'DAYS_EMPLOYED_IN_YEAR', 'AMT_CREDIT_LAC','AMT_INCOME_TOTAL_LAC',  
'CREDIT_RATIO','CNT_FAM_MEMBERS']
```

```
numerical_data_1
```

```
['AMT_ANNUITY',  
'AMT_GOODS_PRICE',  
'DAYS_BIRTH_IN_YEAR',  
'DAYS_EMPLOYED_IN_YEAR',  
'AMT_CREDIT_LAC',  
'AMT_INCOME_TOTAL_LAC',  
'CREDIT_RATIO',  
'CNT_FAM_MEMBERS']
```

```
list(set(df_app.columns)-set(df_app.describe().columns))
```

```
['AMT_CREDIT_LAC_RANGE',  
'FLAG_MOBIL',  
'NAME_HOUSING_TYPE',  
'NAME_TYPE_SUITE',  
'NAME_FAMILY_STATUS',  
'NAME_EDUCATION_TYPE',  
'ORGANIZATION_TYPE',  
'OCCUPATION_TYPE',  
'DAYS_BIRTH_RANGE',  
'FLAG_OWN_CAR',  
'CODE_GENDER',  
'WEEKDAY_APPR_PROCESS_START',  
'NAME_INCOME_TYPE',  
'FLAG_PHONE',  
'FLAG_EMAIL',  
'NAME_CONTRACT_TYPE',  
'AMT_GOODS_PRICE',  
'FLAG_OWN_REALTY',  
'FLAG_CONT_MOBILE',  
'FLAG_WORK_PHONE',  
'FLAG_EMP_PHONE',  
'AMT_INCOME_TOTAL_LAC_RANGE']
```

```
categorical_data_1=['FLAG_OWN_CAR','NAME_HOUSING_TYPE',  
'NAME_FAMILY_STATUS','AMT_INCOME_TOTAL_LAC_RANGE','DAYS_BIRTH_RANGE',  
'CODE_GENDER','NAME_INCOME_TYPE','NAME_CONTRACT_TYPE','FLAG_OWN_REALTY',  
, 'NAME_EDUCATION_TYPE','NAME_TYPE_SUITE','DAYS_BIRTH_RANGE']
```

```
categorical_data_1
```

```
['FLAG_OWN_CAR',  
'NAME_HOUSING_TYPE',  
'NAME_FAMILY_STATUS',  
'AMT_INCOME_TOTAL_LAC_RANGE',
```

```

'DAYS_BIRTH_RANGE',
'CODE_GENDER',
'NAME_INCOME_TYPE',
'NAME_CONTRACT_TYPE',
'FLAG_OWN_REALTY',
'NAME_EDUCATION_TYPE',
'NAME_TYPE_SUITE',
'DAYS_BIRTH_RANGE']

def target_categorical_uni(dataframe, variable, dataframe_difficulty,
dataframe_others):
    fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(25, 8))

    # Plot for payment difficulty
    sns.countplot(
        x=variable,
        data=dataframe_difficulty,
        linewidth=1,
        ax=ax1,
        edgecolor="black",
        hue=variable,
        palette=sns.color_palette("dark",
n_colors=dataframe_difficulty[variable].nunique())
    )
    ax1.set_ylabel("Total Count")
    ax1.set_title(f"Distribution of {variable} in Payment Difficulty",
fontsize=18)
    ax1.set_xticklabels(ax1.get_xticklabels(), rotation=40,
ha="right")

    for p in ax1.patches:
        ax1.annotate(
            f"{(p.get_height() / len(dataframe_difficulty) * 100):.1f}"
            %,
            (p.get_x() + p.get_width() / 2, p.get_height() + 5),
            ha="center",
            weight="bold"
        )

    # Plot for other payments
    sns.countplot(
        x=variable,
        data=dataframe_others,
        linewidth=1,
        ax=ax2,
        edgecolor="black",
        hue=variable,
        palette=sns.color_palette("dark",
n_colors=dataframe_others[variable].nunique())
    )

```

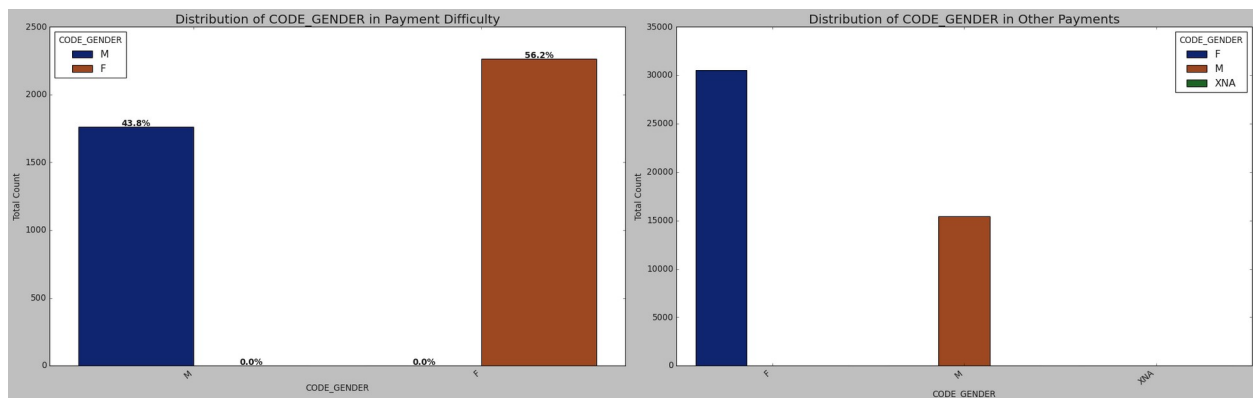
```

ax2.set_ylabel("Total Count")
ax2.set_title(f"Distribution of {variable} in Other Payments",
fontsize=18)
ax2.set_xticklabels(ax2.get_xticklabels(), rotation=40,
ha="right")

plt.tight_layout()
plt.show()

target_categorical_uni(df_app,"CODE_GENDER",Target_variable_payment_difficulty,Target_variable_payment_others)

```

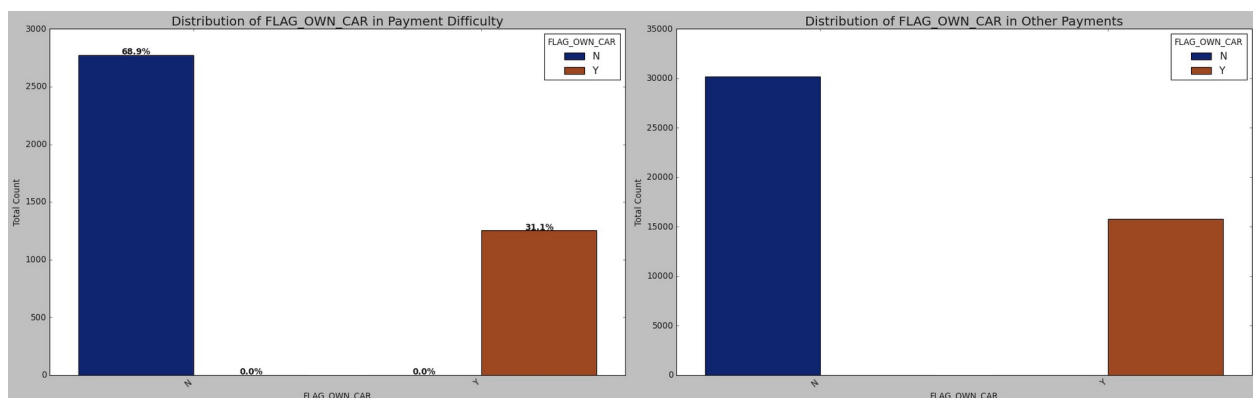


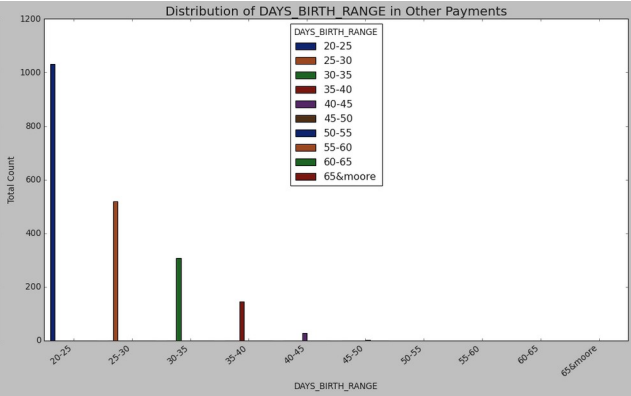
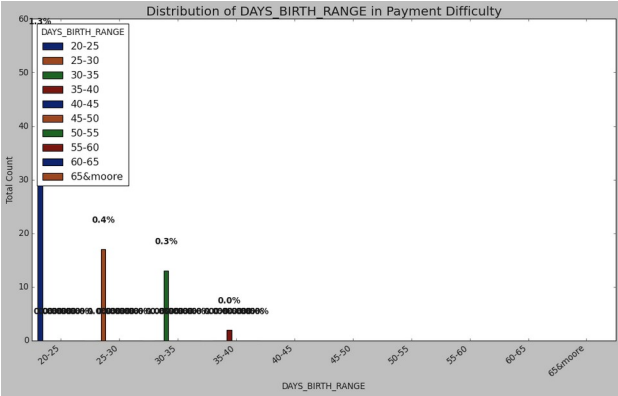
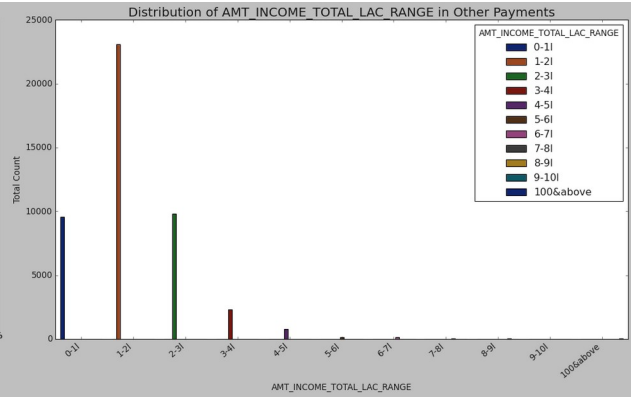
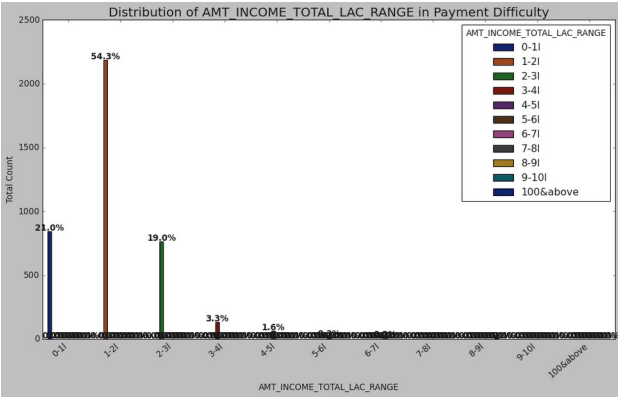
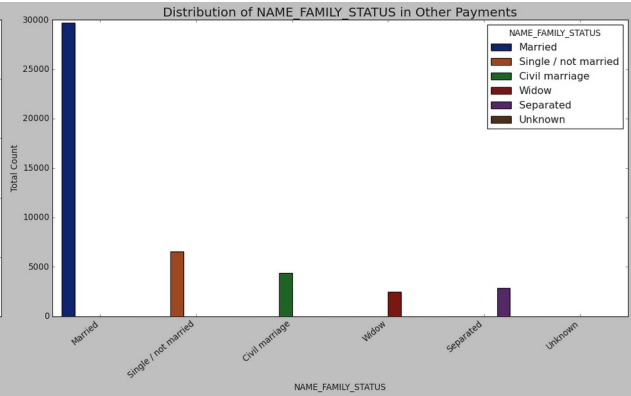
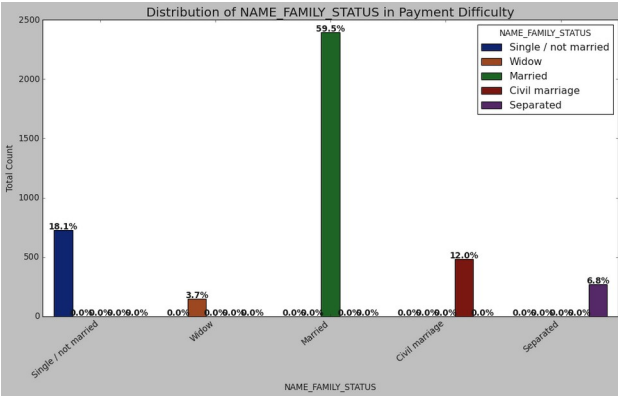
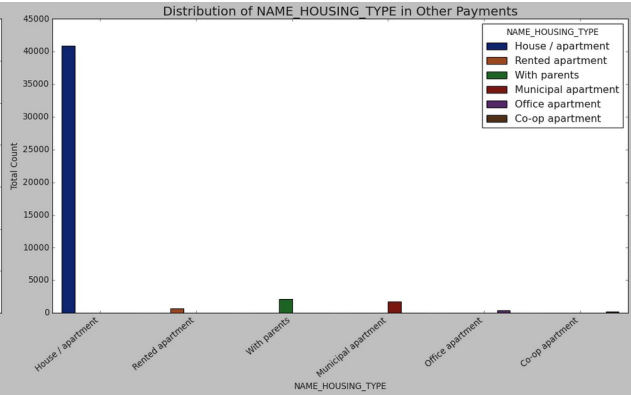
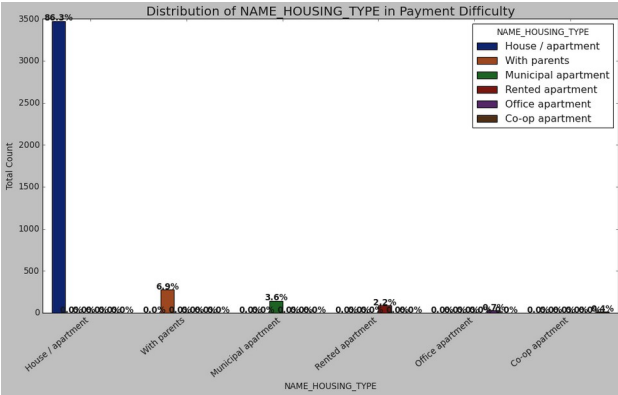
```

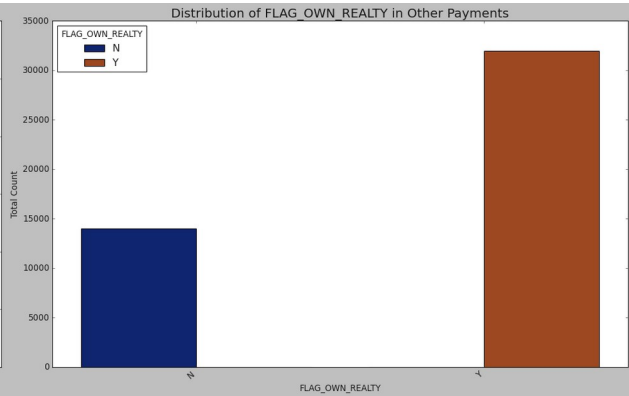
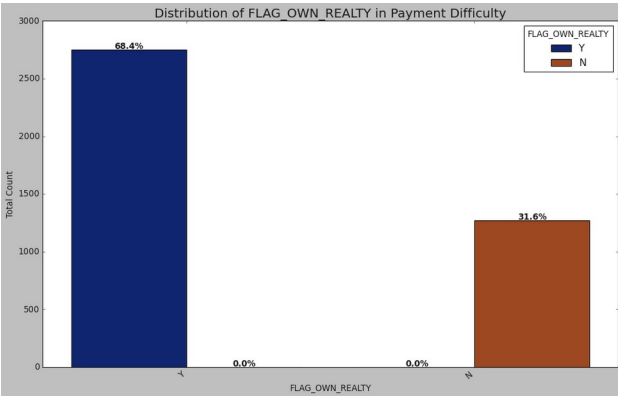
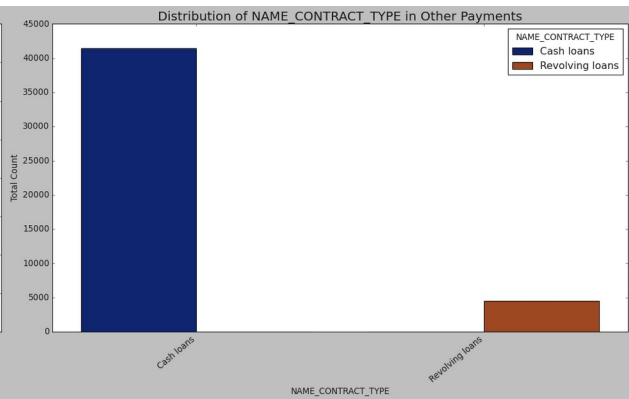
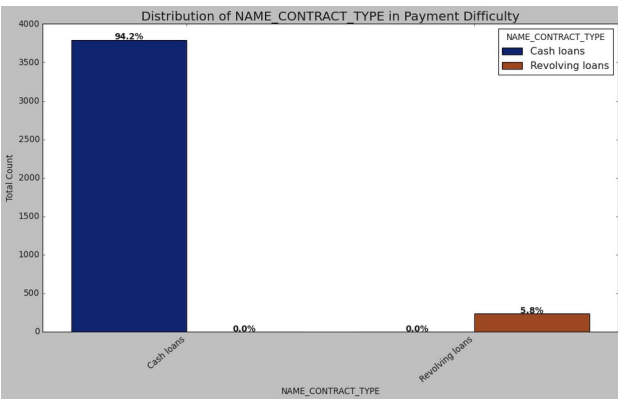
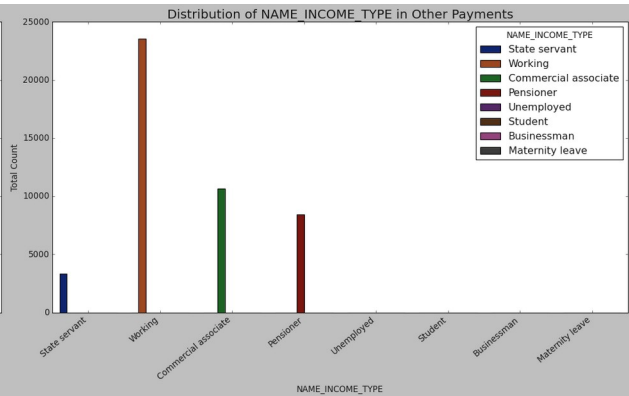
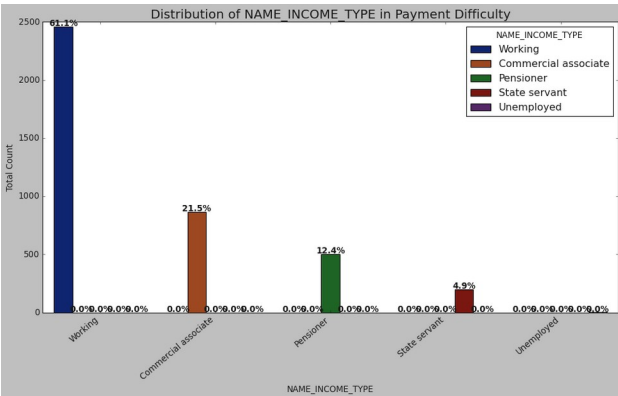
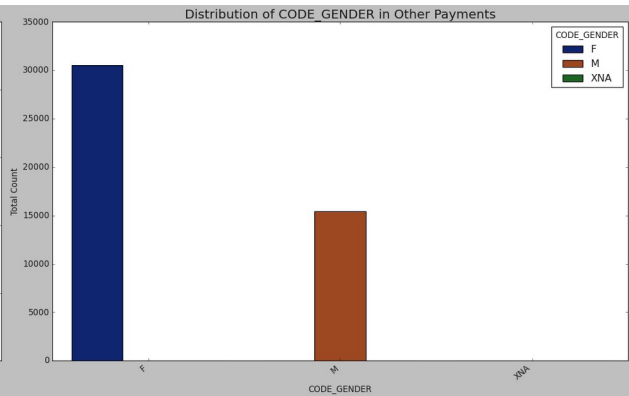
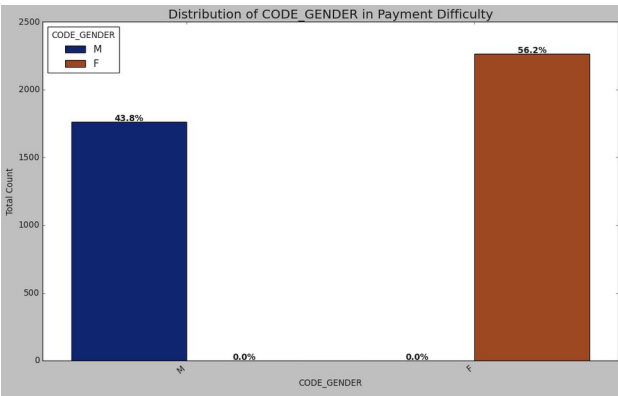
for i in categorical_data_1:

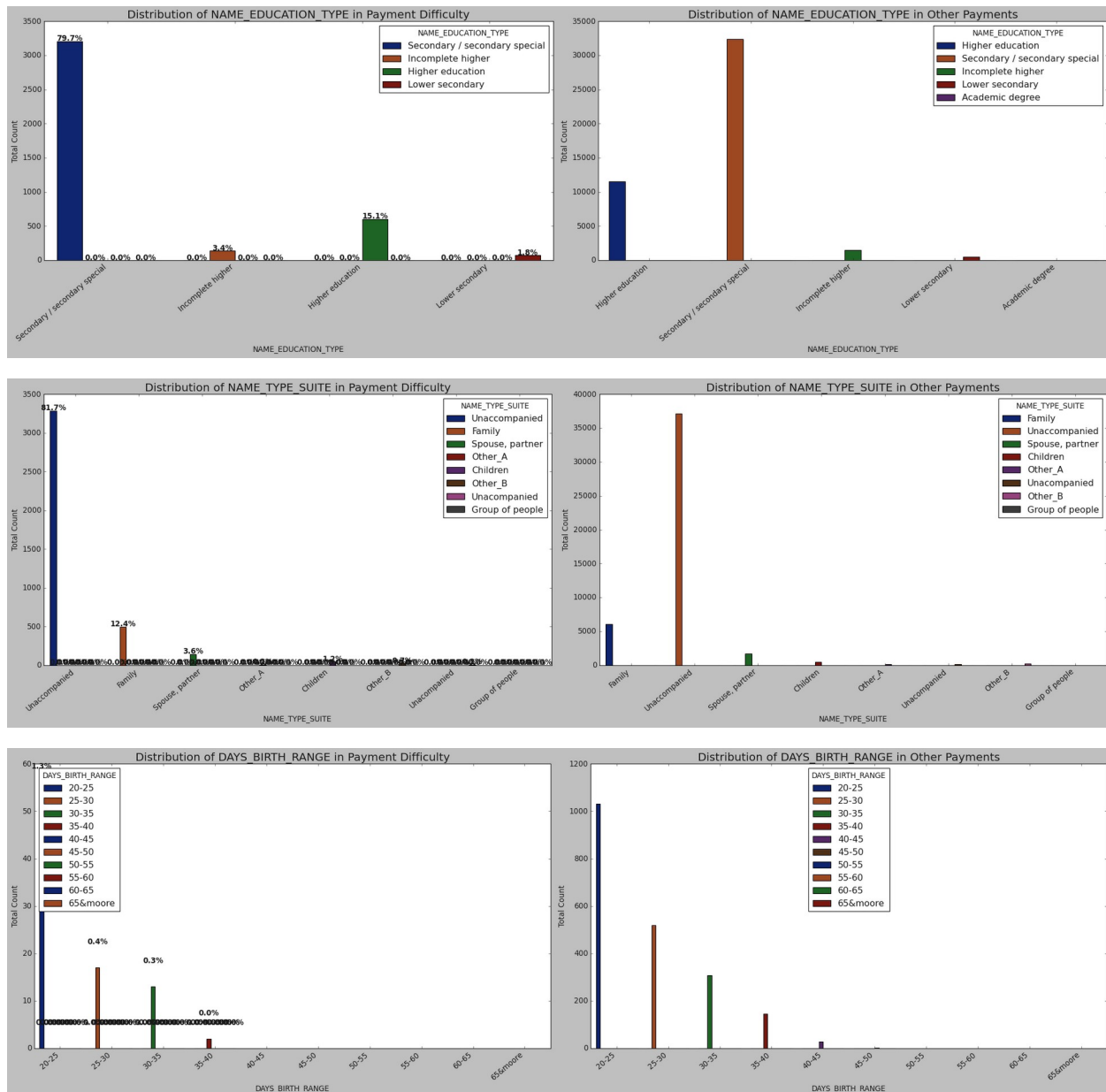
target_categorical_uni(df_app,i,Target_variable_payment_difficulty,Target_variable_payment_others)

```









```
def target_numerical_uni(variable):
    sns.set(style="darkgrid")
    plt.figure(figsize=(15,8))

    plt.subplot(1,2,1)

    sns.histplot(Target_variable_payment_difficulty[variable],kde=True)
    plt.title(f'distribution of {variable} in payment
difficulty', fontsize=15)
    plt.xlabel("variable")
```

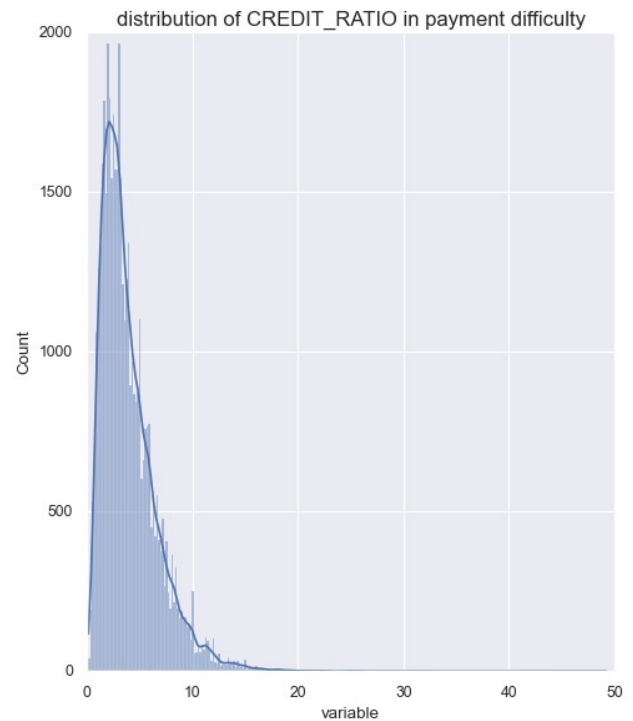
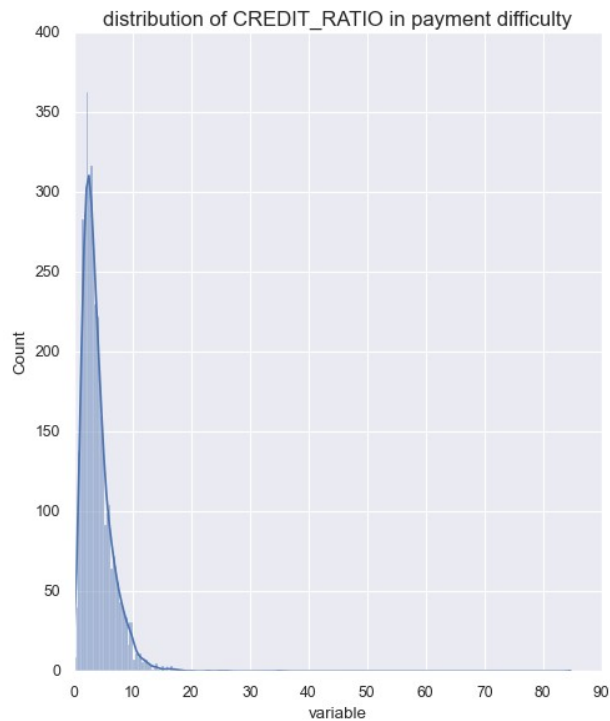


```

plt.subplot(1,2,2)
sns.histplot(Target_variable_payment_others[variable],kde=True)
plt.title(f'distribution of {variable} in payment
difficulty',fontsize=15)
plt.xlabel("variable")
plt.show()

```

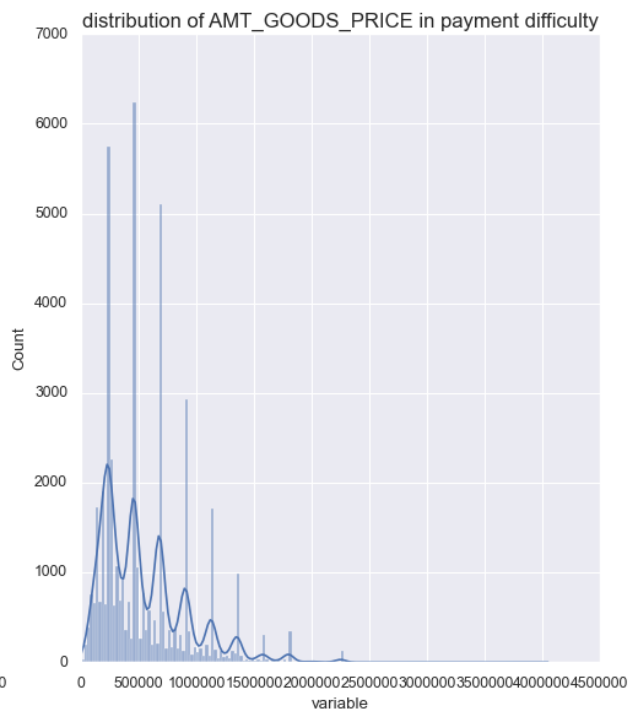
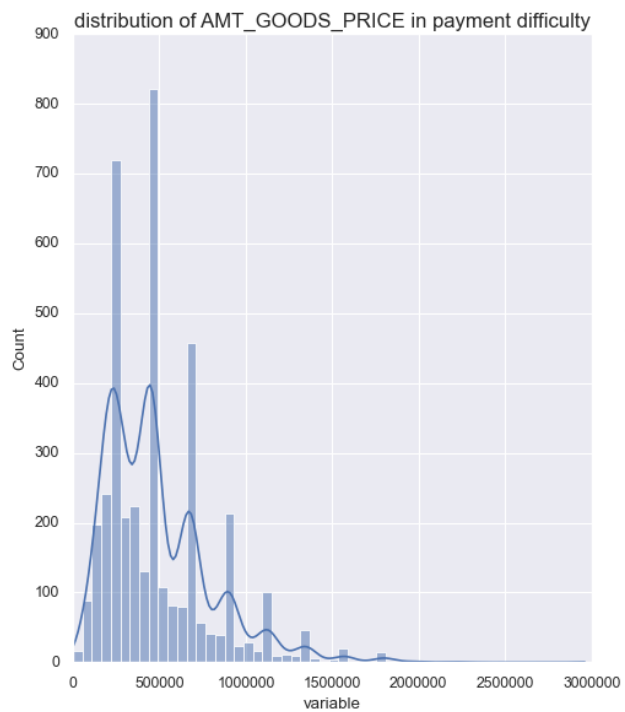
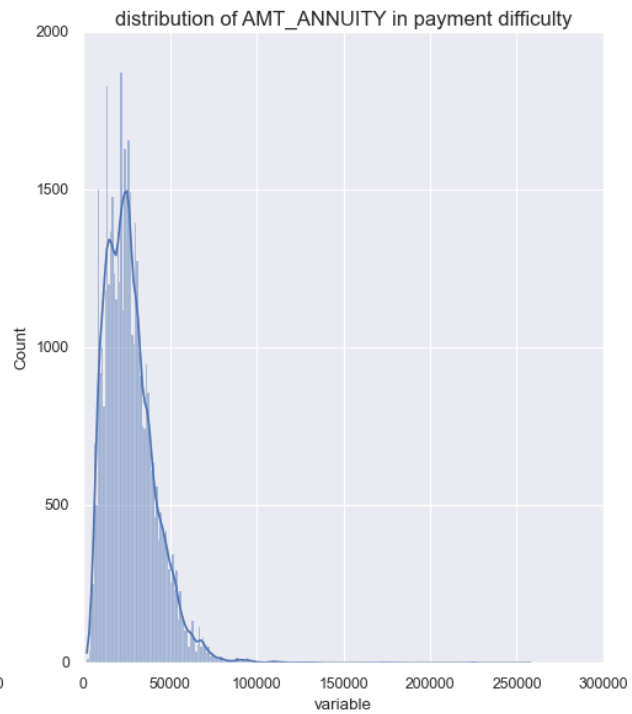
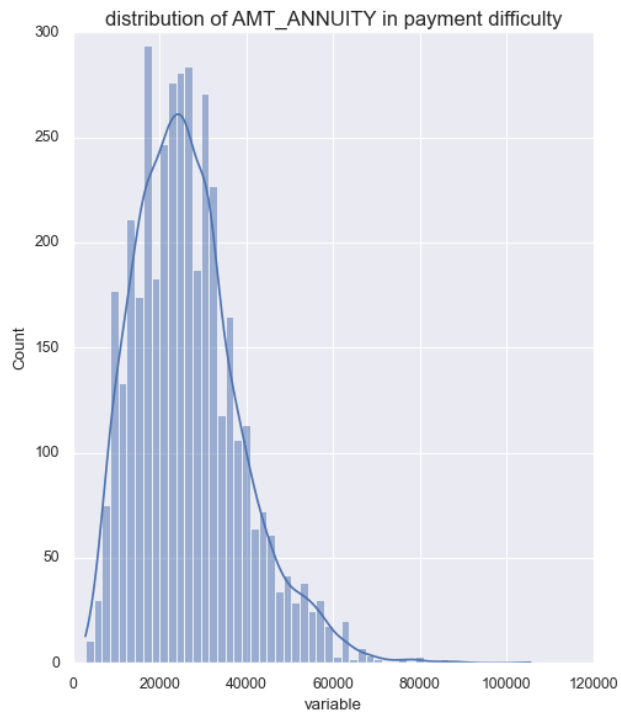
```
target_numerical_uni("CREDIT_RATIO")
```

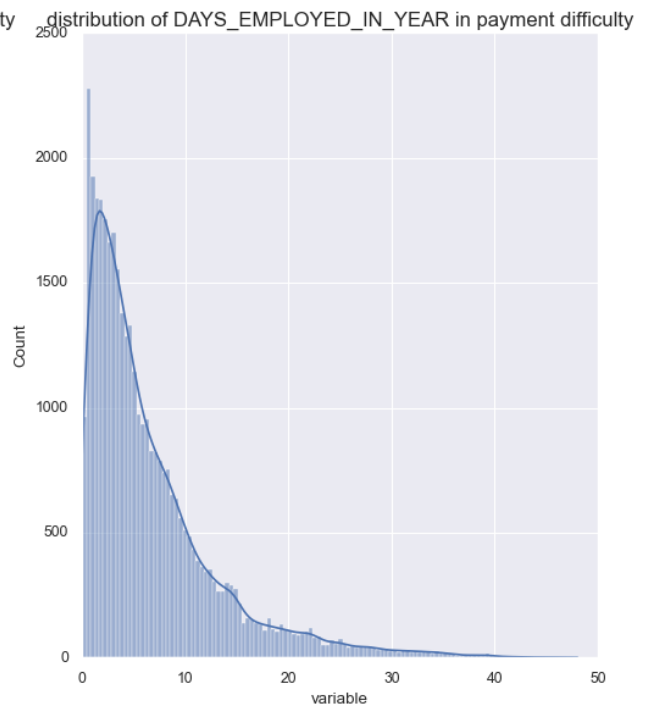
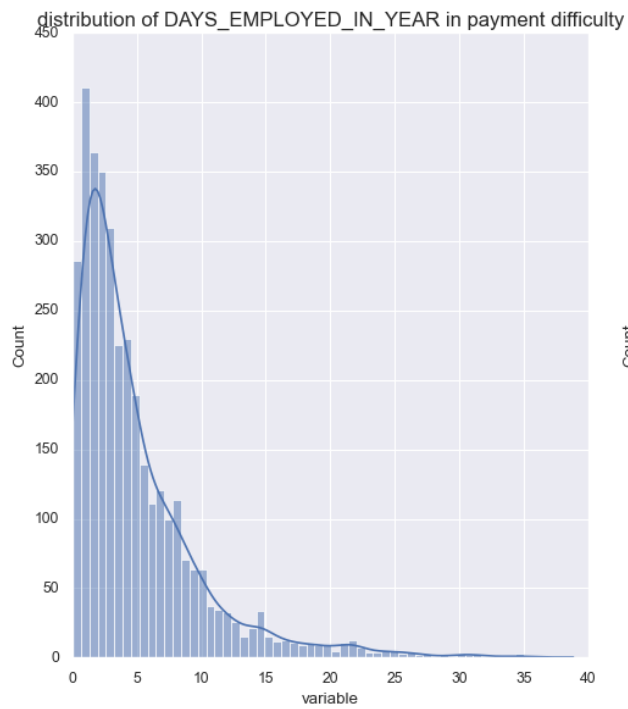
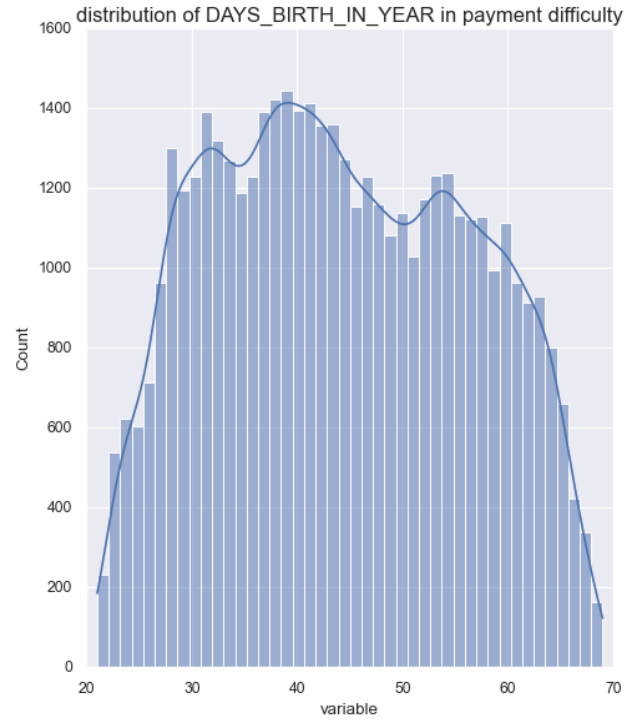
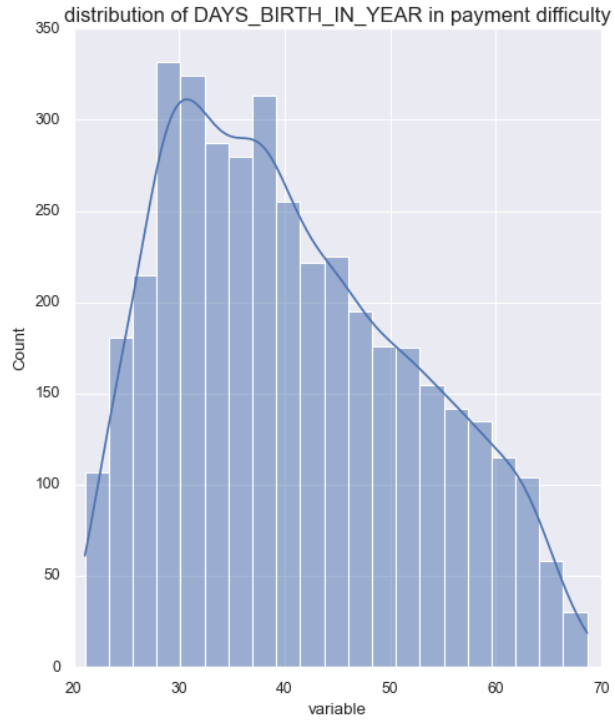


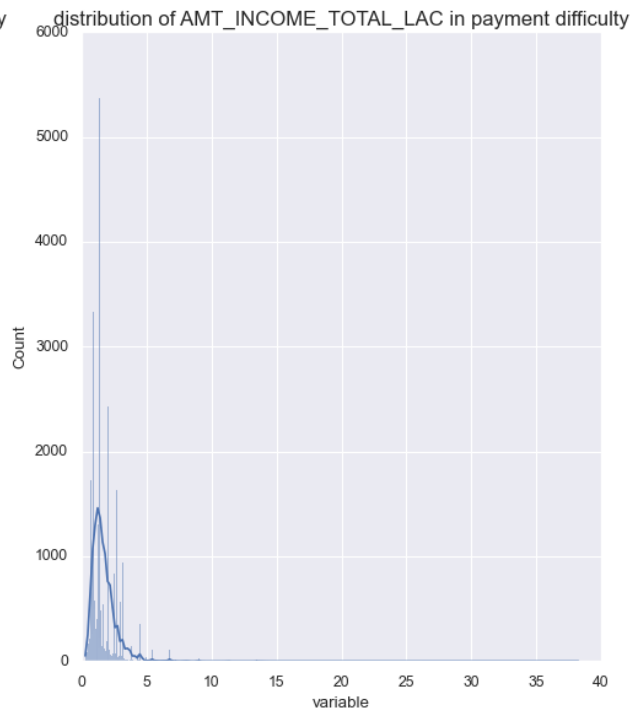
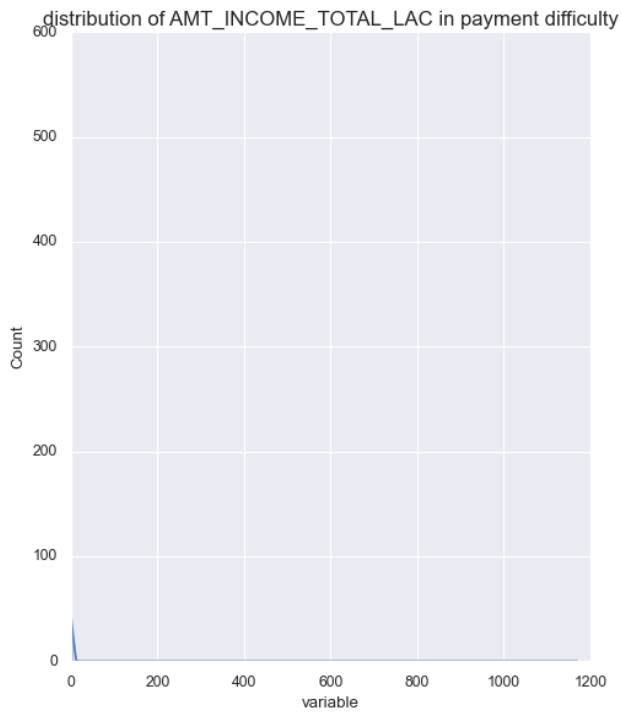
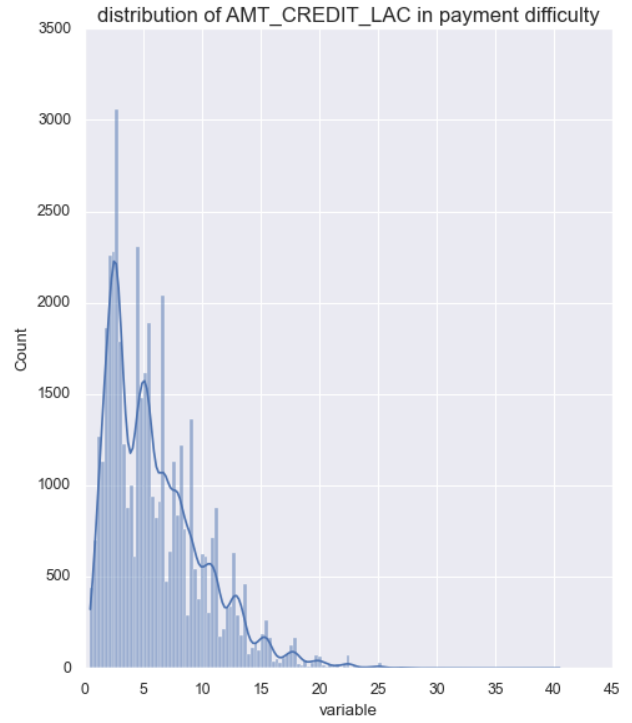
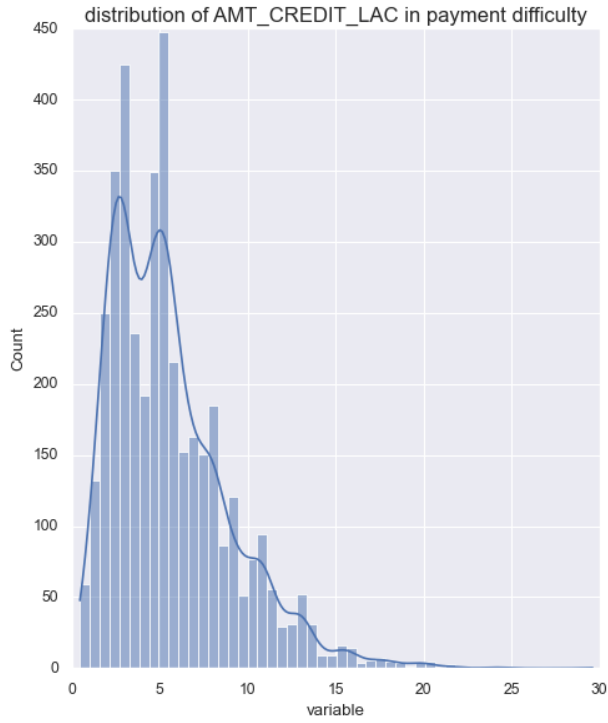
```

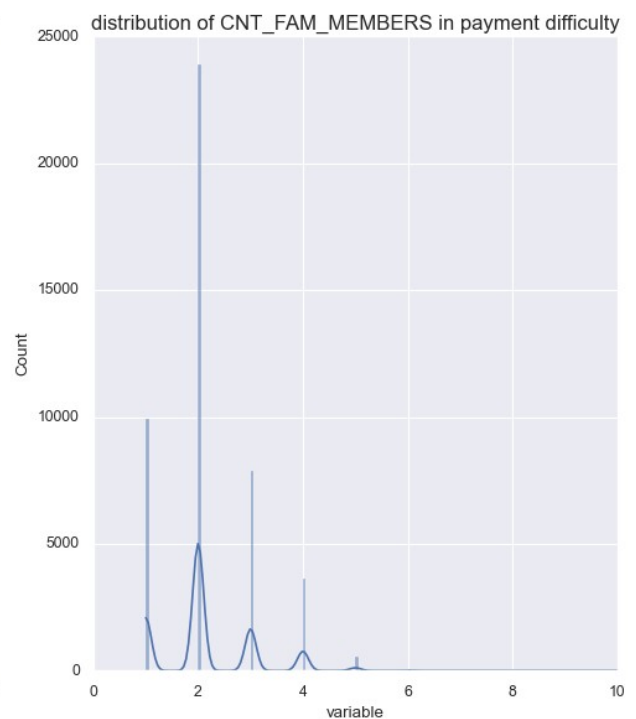
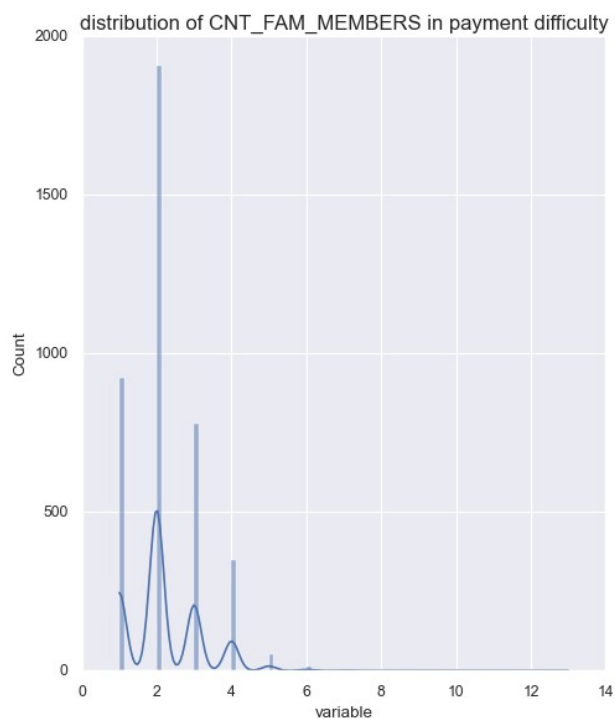
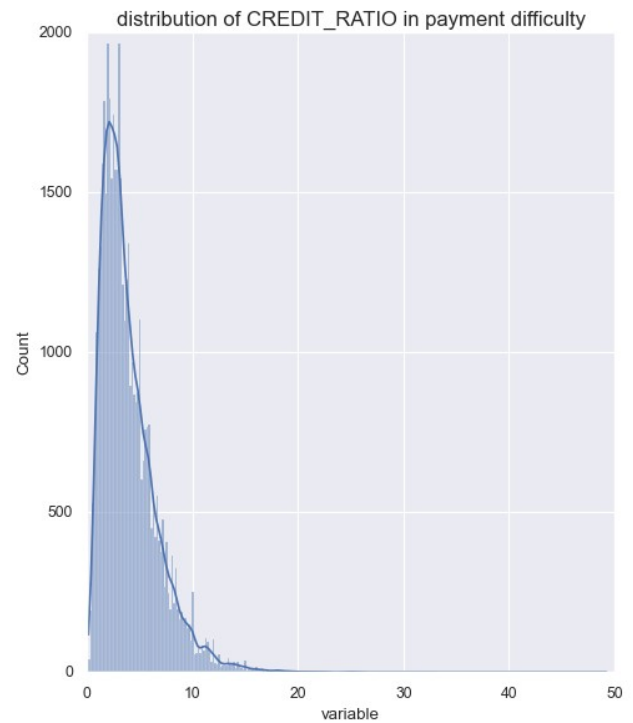
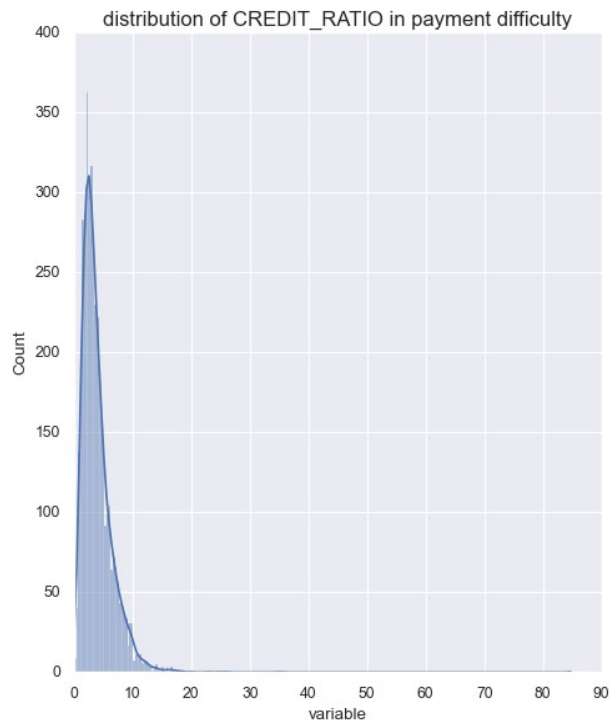
for i in numerical_data_1:
    target_numerical_uni(i)

```









#bi-variate analysis

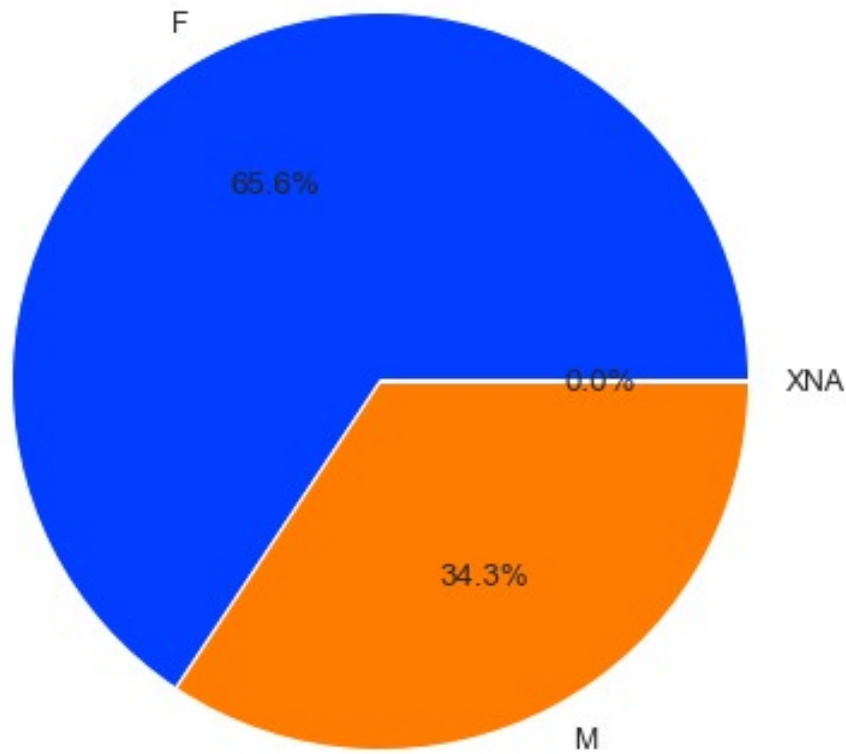
```
sns.scatterplot(x=Target_variable_payment_difficulty["AMT_ANNUITY"],y=
Target_variable_payment_difficulty["AMT_GOODS_PRICE"],data=Target_vari
able_payment_difficulty,hue="TARGET")
plt.show()
```



#pie chart

```
color=sns.color_palette("bright")
plt.pie(x=df_app["CODE_GENDER"].value_counts(),labels=df_app["CODE_GENDER"].value_counts().index,colors=color,autopct="%.1f%%")
plt.title("CODE-GENDER")
plt.show()
```

CODE-GENDER



```
df_app.columns
```

```
Index(['SK_ID_CURR', 'TARGET', 'NAME_CONTRACT_TYPE', 'CODE_GENDER',
      'FLAG_OWN_CAR', 'FLAG_OWN_REALTY', 'CNT_CHILDREN',
      'AMT_INCOME_TOTAL',
      'AMT_CREDIT', 'AMT_ANNUITY', 'AMT_GOODS_PRICE',
      'NAME_TYPE_SUITE',
      'NAME_INCOME_TYPE', 'NAME_EDUCATION_TYPE',
      'NAME_FAMILY_STATUS',
      'NAME_HOUSING_TYPE', 'REGION_POPULATION_RELATIVE',
      'DAYS_BIRTH',
      'DAYS_EMPLOYED', 'DAYS_REGISTRATION', 'DAYS_ID_PUBLISH',
      'FLAG_MOBIL',
      'FLAG_EMP_PHONE', 'FLAG_WORK_PHONE', 'FLAG_CONT_MOBILE',
      'FLAG_PHONE',
      'FLAG_EMAIL', 'OCCUPATION_TYPE', 'CNT_FAM_MEMBERS',
      'REGION_RATING_CLIENT', 'REGION_RATING_CLIENT_W_CITY',
      'WEEKDAY_APPR_PROCESS_START', 'HOUR_APPR_PROCESS_START',
      'REG_REGION_NOT_LIVE_REGION', 'REG_REGION_NOT_WORK_REGION',
      'LIVE_REGION_NOT_WORK_REGION', 'REG_CITY_NOT_LIVE_CITY',
```

```

'REG_CITY_NOT_WORK_CITY', 'LIVE_CITY_NOT_WORK_CITY',
'ORGANIZATION_TYPE', 'EXT_SOURCE_2', 'EXT_SOURCE_3',
'OBS_30_CNT_SOCIAL_CIRCLE', 'DEF_30_CNT_SOCIAL_CIRCLE',
'OBS_60_CNT_SOCIAL_CIRCLE', 'DEF_60_CNT_SOCIAL_CIRCLE',
'DAYS_LAST_PHONE_CHANGE', 'FLAG_DOCUMENT_2', 'FLAG_DOCUMENT_3',
'FLAG_DOCUMENT_4', 'FLAG_DOCUMENT_5', 'FLAG_DOCUMENT_6',
'FLAG_DOCUMENT_7', 'FLAG_DOCUMENT_8', 'FLAG_DOCUMENT_9',
'FLAG_DOCUMENT_10', 'FLAG_DOCUMENT_11', 'FLAG_DOCUMENT_12',
'FLAG_DOCUMENT_13', 'FLAG_DOCUMENT_14', 'FLAG_DOCUMENT_15',
'FLAG_DOCUMENT_16', 'FLAG_DOCUMENT_17', 'FLAG_DOCUMENT_18',
'FLAG_DOCUMENT_19', 'FLAG_DOCUMENT_20', 'FLAG_DOCUMENT_21',
'AMT_REQ_CREDIT_BUREAU_HOUR', 'AMT_REQ_CREDIT_BUREAU_DAY',
'AMT_REQ_CREDIT_BUREAU_WEEK', 'AMT_REQ_CREDIT_BUREAU_MON',
'AMT_REQ_CREDIT_BUREAU_QRT', 'AMT_REQ_CREDIT_BUREAU_YEAR',
'DAYS_BIRTH_IN_YEAR', 'DAYS_EMPLOYED_IN_YEAR',
'DAYS_BIRTH_RANGE',
'AMT_CREDIT_LAC', 'AMT_INCOME_TOTAL_LAC',
'AMT_CREDIT_LAC_RANGE',
'AMT_INCOME_TOTAL_LAC_RANGE', 'CREDIT_RATIO'],
dtype='object')

```

```
df_app.groupby(['CODE_GENDER'])['TARGET'].mean()*100
```

```

CODE_GENDER
F          6.897602
M         10.259695
XNA         0.000000
Name: TARGET, dtype: float64

```

```
df_app.groupby(['AMT_ANNUITY'])['TARGET'].mean()*100
```

```

AMT_ANNUITY
2052.0      0.0
2187.0      0.0
2583.0      0.0
2596.5      0.0
2754.0      0.0
...
180000.0    0.0
197230.5    0.0
213291.0    0.0
225000.0    0.0
258025.5    0.0
Name: TARGET, Length: 8807, dtype: float64

```

#1--> what is default rate of loans based on the gender of the applicant

#2--> how does the income level effect loan aperature.

#3--> how does the amount requested in the previous application effect the credit score in the current application

#4--> what is effect of the contract type in the previous application on the likely hood of default
 #5--> how does the previous loan behavior loan amount, contract type, repayment history
 # influence the approval rate and the loan size in the current application
 # average loan amount approved application.....

#1. what is the default rate of loans based on the gender of the applicant

```
gender_default_rate = (df_app.groupby("CODE_GENDER")
["TARGET"].mean().reset_index().rename(columns={"TARGET":
"DEFAULT_RATE"}))
```

```
gender_default_rate["DEFAULT_RATE"] *= 100
```

```
gender_default_rate
```

	CODE_GENDER	DEFAULT_RATE
0	F	6.897602
1	M	10.259695
2	XNA	0.000000

#questions

#q1 what is default rate of loans based on the gender of the application?

```
target_variable_payment_difficulty=df_app[df_app["TARGET"]==1]
male=1
```

```
male_1 = (df_app["TARGET"]==1)
male_c=df_app["CODE_GENDER"]=="M"
```

```
male_1
```

```
male_c
```

```
plt.figure(figsize=(10,8))
```

```
sns.barplot(x=df_app["CODE_GENDER"],y=df_app["TARGET"],color="blue")
```

```
plt.show()
```

